

1510.06699 — formula report

1063 inline formulas (0 shown: the ones that did not render), 279 display equations, 7 tables, 38 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0241 (A5) conf 0.019	047	■ 0.019 ● S -2	$\bar{\psi}=\frac{\bar{\psi}\left(\rho+\beta i\gamma^5\right)}{\rho^2+\beta^2}$	$\bar{\psi}=\frac{\bar{\psi}\left(\rho+\beta i\gamma^5\right)}{\rho^2+\beta^2}.$	$\bar{\psi}=\frac{\bar{\psi}(\rho+\beta i\gamma^5)\not{I}}{\rho^2+\beta^2}.$
1510.06699_ EQ0248 (B3)	049	■ 0.183 ● N -1	$L_{\mathrm{G}}=-\kappa^{-1}\left(\Lambda+a^0\mathcal{R}\right)+L_{0'}\mathcal{R}^2,$	$L_G=-\kappa^{-1}\left(\Lambda+a^0\mathcal{R}\right)+L_{0'}\mathcal{R}^2,$	$L_G=-\kappa^{-1}(\Lambda+a^0\mathcal{R})+L_{0'}\mathcal{R}^2,$
1510.06699_ EQ0210 (235)	039	■ 0.264 ● C +5	$\begin{aligned} &h\left(s_{\mathrm{mathcal{R}}^2}\right)_{ab}^c=4\alpha_1\left(\delta_{[a}^c\mathcal{D}_{b]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{ab}\right)\mathcal{R}^{\dagger}{}_{[a}^b\delta_{b]}^c+4\alpha_2\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger}{}_{[a}^d\delta_{b]}^e+4\alpha_3\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger d}{}_{[a}\delta_{b]}^e+4\alpha_4\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger}{}_{ab}^{de}+4\alpha_5\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger [d}{}_{[ab]}^{e]}+4\alpha_6\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger de}{}_{ab}, \\ &h\left(s_{\mathcal{H}^2}\right)_{ab}^c=\frac{2}{3}\xi\eta_{f[b}\delta_{a]}^c\left(\mathcal{D}_e^{\dagger}\mathcal{H}^{\dagger ef}-\frac{1}{2}\mathcal{T}^{\dagger f}{}_{de}\mathcal{H}^{\dagger de}\right), \end{aligned}$	$\begin{aligned} h\left(s_{\mathcal{R}^2}\right)_{ab}^c=&4\alpha_1\left(\delta_{[a}^c\mathcal{D}_{b]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{ab}\right)\mathcal{R}^{\dagger}{}_{[a}^b\delta_{b]}^c+4\alpha_2\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger}{}_{[a}^d\delta_{b]}^e \\ &+4\alpha_3\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger d}{}_{[a}\delta_{b]}^e+4\alpha_4\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger}{}_{ab}^{de} \\ &+4\alpha_5\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger [d}{}_{[ab]}^{e]}+4\alpha_6\left(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger}+\frac{1}{2}\mathcal{T}^{\dagger c}{}_{de}\right)\mathcal{R}^{\dagger de}{}_{ab} \\ h\left(s_{\mathcal{H}^2}\right)_{ab}^c=&\frac{2}{3}\xi\eta_{f[b}\delta_{a]}^c\left(\mathcal{D}_e^{\dagger}\mathcal{H}^{\dagger ef}-\frac{1}{2}\mathcal{T}^{\dagger f}{}_{de}\mathcal{H}^{\dagger de}\right), \end{aligned}$	$\begin{aligned} h(s_{\mathcal{R}^2})_{ab}^c &= 4\alpha_1(\delta_{[a}^c\mathcal{D}_{b]}^{\dagger} + \tfrac{1}{2}\mathcal{T}^{\dagger c}{}_{ab})\mathcal{R}^{\dagger} \\ &\quad + 4\alpha_2(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}^{\dagger c}{}_{de})\mathcal{R}^{\dagger}{}_{[a}^d\delta_{b]}^e \\ &\quad + 4\alpha_3(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}^{\dagger c}{}_{de})\mathcal{R}^{\dagger d}{}_{[a}\delta_{b]}^e \\ &\quad + 4\alpha_4(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}^{\dagger c}{}_{de})\mathcal{R}^{\dagger}{}_{ab}^{de} \\ &\quad + 4\alpha_5(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}^{\dagger c}{}_{de})\mathcal{R}^{\dagger [d}{}_{[ab]}^{e]} \\ &\quad + 4\alpha_6(\delta_{[d}^c\mathcal{D}_{e]}^{\dagger} + \tfrac{1}{2}\mathcal{T}^{\dagger c}{}_{de})\mathcal{R}^{\dagger de}{}_{ab}, \\ h(s_{\mathcal{H}^2})_{ab}^c &= \tfrac{2}{3}\xi\eta_{f[b}\delta_{a]}^c(\mathcal{D}_e^{\dagger}\mathcal{H}^{\dagger ef} - \tfrac{1}{2}\mathcal{T}^{\dagger f}{}_{de}\mathcal{H}^{\dagger de}), \end{aligned}$
1510.06699_ EQ0192 (208)	036	■ 0.344 ● N +1	$\begin{aligned} &D_{\mu}^{\dagger}J^a=\partial_{\mu}J^a-w\left(V_{\mu}+\frac{1}{3}T_{\mu}\right)J^a+A_{b\mu}^{\dagger a}J^b \\ &=\partial_{\mu}^{\dagger}J^a+A_{b\mu}^{\dagger a}J^b \end{aligned}$	$\begin{aligned} D_{\mu}^{\dagger}J^a &= \partial_{\mu}J^a - w\left(V_{\mu} + \frac{1}{3}T_{\mu}\right)J^a + A_{b\mu}^{\dagger a}J^b \\ &= \partial_{\mu}^{\dagger}J^a + A_{b\mu}^{\dagger a}J^b \end{aligned}$	$\begin{aligned} D_{\mu}^{\dagger}J^a &= \partial_{\mu}J^a - w(V_{\mu} + \tfrac{1}{3}T_{\mu})J^a + A^{\dagger a}{}_{b\mu}J^b \\ &= \partial_{\mu}^{\dagger}J^a + A^{\dagger a}{}_{b\mu}J^b, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E09067 (71)	015	■ 0.449 ● N +0	$\mathrm{L}_{\mathrm{D}}=h^{-1}\mathrm{L}_{\mathrm{D}}=h^{-1}\left(\frac{1}{2}i\bar{\psi}\gamma^a\dot{\mathcal{D}}_a^*\psi-\mu\phi\bar{\psi}\psi\right)$	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}\left(\frac{1}{2}i\bar{\psi}\gamma^a\dot{\mathcal{D}}_a^*\psi-\mu\phi\bar{\psi}\psi\right),$	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}(\frac{1}{2}i\bar{\psi}\gamma^a\overset{\leftrightarrow}{\mathcal{D}}_a^*\psi-\mu\phi\bar{\psi}\psi),$
1510.06699_ E0105 (113)	021	■ 0.463 ●	$\begin{aligned} \mathrm{L}_{\mathrm{T}}=&\mathrm{L}_{\mathrm{G}}\left(h,\partial h,\partial^2 h,B,\partial B\right)\\ &+\mathcal{L}_{\mathrm{M}}\left(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,\partial^2 h,B,\partial B\right). \end{aligned}$	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}\left(h,\partial h,\partial^2 h,B,\partial B\right)\\ +\mathcal{L}_{\mathrm{M}}\left(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,\partial^2 h,B,\partial B\right).$	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}(h,\partial h,\partial^2 h,B,\partial B)\\ +\mathcal{L}_{\mathrm{M}}(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,\partial^2 h,B,\partial B).$
1510.06699_ E0116 (126)	022	■ 0.469 ● W +0	$\begin{aligned} R_{\sigma\mu\nu}^{\rho} &=2\left(\partial_{[\mu}\Gamma_{ \sigma \nu]}^{\rho}+\Gamma_{\lambda[\mu}^{\rho}\Gamma_{ \sigma \nu]}^{\lambda}\right)-H_{\mu\nu}\delta_{\sigma}^{\rho}, \\ T^{*\lambda}_{\mu\nu} &=2\Gamma_{[\nu\mu]}^{\lambda}, \\ H_{\mu\nu} &=2\partial_{[\mu}B_{\nu]}, \end{aligned}$	$R_{\sigma\mu\nu}^{\rho}=2\left(\partial_{[\mu}\Gamma_{ \sigma \nu]}^{\rho}+\Gamma_{\lambda[\mu}^{\rho}\Gamma_{ \sigma \nu]}^{\lambda}\right)-H_{\mu\nu}\delta_{\sigma}^{\rho},$ $T^{*\lambda}_{\mu\nu}=2\Gamma_{[\nu\mu]}^{\lambda},$ $H_{\mu\nu}=2\partial_{[\mu}B_{\nu]},$	$R^{\rho}_{\sigma\mu\nu}=2(\partial_{[\mu}\Gamma^{\rho}_{ \sigma \nu]}+\Gamma^{\rho}_{\lambda[\mu}\Gamma^{\lambda}_{ \sigma \nu]})-H_{\mu\nu}\delta^{\rho}_{\sigma},$ $T^{*\lambda}_{\mu\nu}=2\Gamma^{\lambda}_{[\nu\mu]},$ $H_{\mu\nu}=2\partial_{[\mu}B_{\nu]},$
1510.06699_ E09046 (48)	012	■ 0.479 ● N +0	$\{ \}^{\{0\}}\mathrm{D}_{\mu}^{\{*\}}\varphi\equiv\left(\partial_{\mu}^{\{*\}}+\frac{1}{2}\{ \}^{\{0\}}A^{\{*ab}\}_{\mu}\Sigma_{ab}\right)\varphi,$	${}^0D_{\mu}^*\varphi\equiv\left(\partial_{\mu}^*+\frac{1}{2}{}^0A^{*ab}_{\mu}\Sigma_{ab}\right)\varphi,$	${}^0D_{\mu}^*\varphi\equiv(\partial_{\mu}^*+\frac{1}{2}A^{*ab}_{\mu}\Sigma_{ab})\varphi,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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1510.06699_ EQ0207 (229)	038	■0.487	$\begin{aligned} & \backslash \text{begin}\{\text{aligned}\} \text{ h}\backslash \text{left}\{\text{t}_{\backslash \text{mathcal}\{\text{R}\}^{\{2\}}\}^{\{\backslash \text{dagger}\}}\} \\ & \backslash \text{right}\}\{\text{a}\}\{\text{ }_{\{b\}} = \& \backslash \alpha_{\{1\}} \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \\ & \backslash \text{left}\{4 \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{a}\}\{\text{ }_{\{b\}} - \\ & \backslash \text{delta}_{\{b\}}^{\{a\}} \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \backslash \text{right}\} \backslash \& + \\ & \backslash \alpha_{\{2\}} \backslash \text{left}[2 \backslash \text{left}\{\backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{c a}\} \\ & \backslash \text{mathcal}\{\text{R}\}_{\{c b\}}^{\{\backslash \text{dagger}\}} - \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{c d}\} \\ & \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{a}\}\{\text{ }_{\{c d b\}} \backslash \text{right}\} - \backslash \text{delta}_{\{b\}}^{\{a\}} \\ & \backslash \text{mathcal}\{\text{R}\}_{\{c d\}}^{\{\backslash \text{dagger}\}} \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{c d}\} \\ & \backslash \text{right}] \backslash \& + \backslash \alpha_{\{3\}} \backslash \text{left}[2 \backslash \text{left}\{\backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \\ & \backslash \text{dagger a c}\} \backslash \text{mathcal}\{\text{R}\}_{\{c b\}}^{\{\backslash \text{dagger}\}} - \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \\ & \backslash \text{dagger d c}\} \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{a}\}\{\text{ }_{\{c d b\}} \\ & \backslash \text{right}\} - \backslash \text{delta}_{\{b\}}^{\{a\}} \backslash \text{mathcal}\{\text{R}\}_{\{c d\}}^{\{\backslash \text{dagger}\}} \\ & \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{d c}\} \backslash \text{right}] \backslash \& + \backslash \alpha_{\{4\}} \\ & \backslash \text{left}[4 \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{c d e a}\} \backslash \text{mathcal}\{\text{R}\}_{\{c d e b\}}^{\{\backslash \text{dagger}\}} - \backslash \text{delta}_{\{b\}}^{\{a\}} \\ & \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{c d e f}\} \backslash \text{mathcal}\{\text{R}\}_{\{c d e f\}}^{\{\backslash \text{dagger}\}} \backslash \text{right}] \backslash \& + \\ & \backslash \alpha_{\{5\}} \backslash \text{left}[2 \backslash \text{left}\{\backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{a c d e}\} - \\ & \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{e c d a}\} \backslash \text{right}\} \backslash \text{mathcal}\{\text{R}\}_{\{c d e b\}}^{\{\backslash \text{dagger}\}} - \backslash \text{delta}_{\{b\}}^{\{a\}} \\ & \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{c e d f}\} \backslash \text{mathcal}\{\text{R}\}_{\{c d e f\}}^{\{\backslash \text{dagger}\}} \backslash \text{right}] \backslash \& + \backslash \alpha_{\{6\}} \\ & \backslash \text{left}[4 \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{e a c d}\} \backslash \text{mathcal}\{\text{R}\}_{\{c d e b\}}^{\{\backslash \text{dagger}\}} - \backslash \text{delta}_{\{b\}}^{\{a\}} \\ & \backslash \text{mathcal}\{\text{R}\}^{\{\backslash \text{dagger}\}} \text{e f c d}\} \backslash \text{mathcal}\{\text{R}\}_{\{c d e f\}}^{\{\backslash \text{dagger}\}} \backslash \text{right}] \backslash \& \text{h} \\ & \backslash \text{left}\{\text{t}_{\backslash \text{mathcal}\{\text{H}\}^{\{2\}}\}^{\{\backslash \text{dagger}\}} \backslash \text{right}\}^{\{a\}}\{\text{ }_{\{b\}} = \\ & \& \backslash \text{frac}\{1\}{\{2\}} \backslash \xi \backslash \text{left}[4 \backslash \text{mathcal}\{\text{H}\}_{\{b c\}}^{\{\backslash \text{dagger}\}} \\ & \backslash \text{mathcal}\{\text{H}\}^{\{\backslash \text{dagger}\}} \text{a c}\} - \backslash \text{delta}_{\{b\}}^{\{a\}} \backslash \text{mathcal}\{\text{H}\}_{\{c d\}}^{\{\backslash \text{dagger}\}} \\ & \backslash \text{mathcal}\{\text{H}\}^{\{\backslash \text{dagger}\}} \text{c d}\} + \backslash \text{frac}\{4\}{\{3\}} \\ & \backslash \text{mathcal}\{\text{D}\}_{\{b\}}^{\{\backslash \text{dagger}\}} \backslash \text{left}\{\backslash \text{mathcal}\{\text{D}\}_{\{c\}}^{\{\backslash \text{dagger}\}} \\ & \backslash \text{mathcal}\{\text{H}\}^{\{\backslash \text{dagger}\}} \text{c a}\} - \backslash \text{frac}\{1\}{\{2\}} \backslash \text{mathcal}\{\text{T}\}^{\{\backslash \text{dagger}\}} \\ & \backslash \text{dagger a}\}\{\text{ }_{\{c d\}} \backslash \text{mathcal}\{\text{H}\}^{\{\backslash \text{dagger}\}} \text{c d}\} \backslash \text{right}\} \backslash \text{right}], \backslash \text{end}\{\text{aligned}\} \end{aligned}$	$\begin{aligned} h\left(t_{\mathcal{R}^2}^{\dagger}\right)_b^a = & \alpha_1 \mathcal{R}^{\dagger}\left(4 \mathcal{R}_{ b}^{\dagger a}-\delta_b^a \mathcal{R}^{\dagger}\right) \\ & +\alpha_2\left[2\left(\mathcal{R}_{ c b}^{\dagger c a}-\mathcal{R}_{ c d b}^{\dagger c d} \mathcal{R}_{ c d b}^{\dagger a}\right)-\delta_b^a \mathcal{R}_{ c d}^{\dagger} \mathcal{R}_{ c d}^{\dagger c d}\right] \\ & +\alpha_3\left[2\left(\mathcal{R}_{ c b}^{\dagger a c}-\mathcal{R}_{ c d b}^{\dagger d c} \mathcal{R}_{ c d b}^{\dagger a}\right)-\delta_b^a \mathcal{R}_{ c d}^{\dagger} \mathcal{R}_{ c d}^{\dagger d c}\right] \\ & +\alpha_4\left[4 \mathcal{R}_{ c d e b}^{\dagger c d e a}-\delta_b^a \mathcal{R}_{ c d e f}^{\dagger c d e f} \mathcal{R}_{ c d e f}^{\dagger}\right] \\ & +\alpha_5\left[2\left(\mathcal{R}_{ c d e b}^{\dagger a c d e}-\mathcal{R}_{ c d e b}^{\dagger e c d a}\right) \mathcal{R}_{ c d e b}^{\dagger}-\delta_b^a \mathcal{R}_{ c d e f}^{\dagger c e d f} \mathcal{R}_{ c d e f}^{\dagger}\right] \\ & +\alpha_6\left[4 \mathcal{R}_{ c d e b}^{\dagger e a c d}-\delta_b^a \mathcal{R}_{ c d e f}^{\dagger e f c d} \mathcal{R}_{ c d e f}^{\dagger}\right], \\ h\left(t_{\mathcal{H}^2}^{\dagger}\right)_b^a = & \frac{1}{2} \xi\left[4 \mathcal{H}_{ b c}^{\dagger} \mathcal{H}^{\dagger a c}-\delta_b^a \mathcal{H}_{ c d}^{\dagger} \mathcal{H}^{\dagger c d}+\frac{4}{3} \mathcal{D}_b^{\dagger}\left(\mathcal{D}_c^{\dagger} \mathcal{H}^{\dagger c a}-\frac{1}{2} \mathcal{T}_{ c d}^{\dagger a} \mathcal{H}^{\dagger c d}\right)\right], \end{aligned}$	
1510.06699_ EQ0117 (128)	022	■0.497 ● N +0	$\widetilde{R}_{\sigma \mu \nu}^{\rho} \equiv R_{\sigma \mu \nu}^{\rho}+H_{\mu \nu} \delta_{\sigma}^{\rho} .$	$\widetilde{R}_{\sigma \mu \nu}^{\rho} \equiv R_{\sigma \mu \nu}^{\rho}+H_{\mu \nu} \delta_{\sigma}^{\rho} .$	
1510.06699_ EQ0249 (B4)	049	■0.500 ● N +0	$L_0 \mathcal{R}^2 = \alpha_1 {}^0 \mathcal{R}^2 + \alpha_2 {}^0 \mathcal{R}_{ab} {}^0 \mathcal{R}^{ab} + \alpha_3 {}^0 \mathcal{R}_{abcd} {}^0 \mathcal{R}^{abcd},$	$L_0 \mathcal{R}^2 = \alpha_1 {}^0 \mathcal{R}^2 + \alpha_2 {}^0 \mathcal{R}_{ab} {}^0 \mathcal{R}^{ab} + \alpha_3 {}^0 \mathcal{R}_{abcd} {}^0 \mathcal{R}^{abcd},$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E09141 (155b)	027	■ 0.505	$\begin{aligned} & \mathcal{R}^{\dagger ab}_{cd} = \mathcal{R}^{ab}_{cd} + 2\delta_d^{[b}(\mathcal{D}_c + \mathcal{V}_c)\mathcal{V}^a] - 2\delta_c^{[b}(\mathcal{D}_d + \mathcal{V}_d)\mathcal{V}^a] - 2\mathcal{V}^e\mathcal{V}_e\delta_c^{[a}\delta_d^{b]} + 2\mathcal{V}^{[a}\mathcal{T}^{b]}_{cd}, \\ & \mathcal{R}^{\dagger a}_c = \mathcal{R}^a_c + 2(\mathcal{D}_c + \tfrac{1}{2}\mathcal{T}_c + \mathcal{V}_c)\mathcal{V}^a + \delta_c^a(\mathcal{D}_b - 2\mathcal{V}_b)\mathcal{V}^b - \mathcal{T}^a_{cb}\mathcal{V}^b, \\ & \mathcal{R}^{\dagger} = \mathcal{R} + 6(\mathcal{D}_a + \tfrac{1}{3}\mathcal{T}_a - \mathcal{V}_a)\mathcal{V}^a, \end{aligned}$	$\mathcal{R}^{\dagger ab}_{cd} = \mathcal{R}^{ab}_{cd} + 2\delta_d^{[b}(\mathcal{D}_c + \mathcal{V}_c)\mathcal{V}^a] - 2\delta_c^{[b}(\mathcal{D}_d + \mathcal{V}_d)\mathcal{V}^a] - 2\mathcal{V}^e\mathcal{V}_e\delta_c^{[a}\delta_d^{b]} + 2\mathcal{V}^{[a}\mathcal{T}^{b]}_{cd}$ $\mathcal{R}^{\dagger a}_c = \mathcal{R}^a_c + 2\left(\mathcal{D}_c + \frac{1}{2}\mathcal{T}_c + \mathcal{V}_c\right)\mathcal{V}^a + \delta_c^a(\mathcal{D}_b - 2\mathcal{V}_b)\mathcal{V}^b - \mathcal{T}^a_{cb}\mathcal{V}^b$ $\mathcal{R}^{\dagger} = \mathcal{R} + 6\left(\mathcal{D}_a + \frac{1}{3}\mathcal{T}_a - \mathcal{V}_a\right)\mathcal{V}^a,$	$\mathcal{R}^{\dagger ab}_{cd} = \mathcal{R}^{ab}_{cd} + 2\delta_d^{[b}(\mathcal{D}_c + \mathcal{V}_c)\mathcal{V}^a] - 2\delta_c^{[b}(\mathcal{D}_d + \mathcal{V}_d)\mathcal{V}^a] - 2\mathcal{V}^e\mathcal{V}_e\delta_c^{[a}\delta_d^{b]} + 2\mathcal{V}^{[a}\mathcal{T}^{b]}_{cd},$ $\mathcal{R}^{\dagger a}_c = \mathcal{R}^a_c + 2(\mathcal{D}_c + \tfrac{1}{2}\mathcal{T}_c + \mathcal{V}_c)\mathcal{V}^a + \delta_c^a(\mathcal{D}_b - 2\mathcal{V}_b)\mathcal{V}^b - \mathcal{T}^a_{cb}\mathcal{V}^b,$ $\mathcal{R}^{\dagger} = \mathcal{R} + 6(\mathcal{D}_a + \tfrac{1}{3}\mathcal{T}_a - \mathcal{V}_a)\mathcal{V}^a,$
1510.06699_ E09235 (269)	044	■ 0.527 ● N +1	$\begin{aligned} & \mathcal{L}'_{\mathcal{M}} = \mathcal{L}_{\mathcal{M}} - h^{-1} \left[\frac{1}{2} \nu \mathcal{P}^{a0} \mathcal{D}_a \phi^2 - 6a \phi^{20} \mathcal{D}_a \mathcal{P}^a \right. \\ & \quad \left. + \left(\frac{1}{2} \nu + 6a \right) \phi^2 \mathcal{P}_a \mathcal{P}^a \right] \end{aligned}$	$\mathcal{L}'_{\mathcal{M}} = \mathcal{L}_{\mathcal{M}} - h^{-1} \left[\frac{1}{2} \nu \mathcal{P}^{a0} \mathcal{D}_a \phi^2 - 6a \phi^{20} \mathcal{D}_a \mathcal{P}^a + \left(\frac{1}{2} \nu + 6a \right) \phi^2 \mathcal{P}_a \mathcal{P}^a \right]$	$\mathcal{L}'_{\mathcal{M}} = \mathcal{L}_{\mathcal{M}} - h^{-1} \left[\frac{1}{2} \nu \mathcal{P}^{a0} \mathcal{D}_a \phi^2 - 6a \phi^{20} \mathcal{D}_a \mathcal{P}^a + \left(\frac{1}{2} \nu + 6a \right) \phi^2 \mathcal{P}_a \mathcal{P}^a \right].$
1510.06699_ E09187 (203)	035	■ 0.544 ● K +0	$\mathcal{L}_{\{\mathrm{G}\}} = \mathcal{L}_{\{0_{\{\mathrm{mathcal{R}}^{\dagger 2}\}}\}} + \mathcal{L}_{\{\mathrm{mathcal{H}}^{\dagger 2}\}}$	$L_G = L_{0_{\mathcal{R}^{\dagger 2}}} + L_{\mathcal{H}^{\dagger 2}},$	$L_G = L_{0_{\mathcal{R}^{\dagger 2}}} + L_{\mathcal{H}^{\dagger 2}},$
1510.06699_ E09244 (A8)	047	■ 0.551 ● K +0	$S = \int d\lambda \left[\Re(i\bar{R}\dot{R}) - p_\mu (\dot{x}^\mu - me\bar{R}\gamma^\mu R) - m^2 e \right].$	$S = \int d\lambda \left[\Re(i\bar{R}\dot{R}) - p_\mu (\dot{x}^\mu - me\bar{R}\gamma^\mu R) - m^2 e \right].$	$S = \int d\lambda \left[\Re(i\bar{R}\dot{R}) - p_\mu (\dot{x}^\mu - me\bar{R}\gamma^\mu R) - m^2 e \right].$
1510.06699_ E09293 (222)	037	■ 0.560 ● N +0	$\begin{aligned} & \mathcal{L}_{\{\mathrm{mathcal{R}}^{\dagger 2}\}} = \alpha_1 \mathcal{L}_{\{\mathrm{mathcal{R}}^{\dagger 2} + \alpha_2 \mathcal{R}^{\dagger}_{ab} \mathcal{R}^{\dagger ab} + \alpha_3 \mathcal{R}^{\dagger}_{ab} \mathcal{R}^{\dagger ba} + \alpha_4 \mathcal{R}^{\dagger}_{abcd} \mathcal{R}^{\dagger abcd} + \alpha_5 \mathcal{R}^{\dagger}_{abcd} \mathcal{R}^{\dagger acbd} + \alpha_6 \mathcal{R}^{\dagger}_{abcd} \mathcal{R}^{\dagger cdab}, \\ & \mathcal{L}_{\mathcal{H}^{\dagger 2}} = \frac{1}{2} \xi \mathcal{H}^{\dagger}_{ab} \mathcal{H}^{\dagger ab}, \end{aligned}$	$L_{\mathcal{R}^{\dagger 2}} = \alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2 \mathcal{R}^{\dagger}_{ab} \mathcal{R}^{\dagger ab} + \alpha_3 \mathcal{R}^{\dagger}_{ab} \mathcal{R}^{\dagger ba} + \alpha_4 \mathcal{R}^{\dagger}_{abcd} \mathcal{R}^{\dagger abcd} + \alpha_5 \mathcal{R}^{\dagger}_{abcd} \mathcal{R}^{\dagger acbd} + \alpha_6 \mathcal{R}^{\dagger}_{abcd} \mathcal{R}^{\dagger cdab},$ $L_{\mathcal{H}^{\dagger 2}} = \frac{1}{2} \xi \mathcal{H}^{\dagger}_{ab} \mathcal{H}^{\dagger ab},$	$L_{\mathcal{R}^{\dagger 2}} = \alpha_1 \mathcal{R}^{\dagger 2} + \alpha_2 \mathcal{R}^{\dagger}_{ab} \mathcal{R}^{\dagger ab} + \alpha_3 \mathcal{R}^{\dagger}_{ab} \mathcal{R}^{\dagger ba} + \alpha_4 \mathcal{R}^{\dagger}_{abcd} \mathcal{R}^{\dagger abcd} + \alpha_5 \mathcal{R}^{\dagger}_{abcd} \mathcal{R}^{\dagger acbd} + \alpha_6 \mathcal{R}^{\dagger}_{abcd} \mathcal{R}^{\dagger cdab},$ $L_{\mathcal{H}^{\dagger 2}} = \frac{1}{2} \xi \mathcal{H}^{\dagger}_{ab} \mathcal{H}^{\dagger ab},$
1510.06699_ E09091 (98)	019	■ 0.575 ● W +1	$\begin{aligned} & v^a = ep^a, \\ & \dot{p}_a = c^c_{ab} v^b p_c + e \mu^2 \phi \partial_a \phi, \\ & p^2 = \mu^2 \phi^2. \end{aligned}$	$v^a = ep^a,$ $\dot{p}_a = c^c_{ab} v^b p_c + e \mu^2 \phi \partial_a \phi,$ $p^2 = \mu^2 \phi^2.$	$v^a = ep^a,$ $\dot{p}_a = c^c_{ab} v^b p_c + e \mu^2 \phi \partial_a \phi,$ $p^2 = \mu^2 \phi^2.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E09261 (C12)	050	■0.600	$\begin{aligned} R_{\mu\nu}-\frac{1}{2}g_{\mu\nu}R&=\kappa\frac{\tau_{\mu\nu}}{\sqrt{-g}}\\ T^{\lambda}{}_{\mu\nu}&=\frac{\kappa}{2\sqrt{-g}}\left(2\sigma_{\mu\nu}{}^{\lambda}+\delta_{\mu}^{\lambda}\sigma_{\nu\rho}{}^{\rho}-\delta_{\nu}^{\lambda}\sigma_{\mu\rho}{}^{\rho}\right),\end{aligned}$	$R_{\mu\nu}-\frac{1}{2}g_{\mu\nu}R=\kappa\frac{\tau_{\mu\nu}}{\sqrt{-g}}\\ T^{\lambda}{}_{\mu\nu}=\frac{\kappa}{2\sqrt{-g}}\left(2\sigma_{\mu\nu}{}^{\lambda}+\delta_{\mu}^{\lambda}\sigma_{\nu\rho}{}^{\rho}-\delta_{\nu}^{\lambda}\sigma_{\mu\rho}{}^{\rho}\right),$	$R_{\mu\nu}-\frac{1}{2}g_{\mu\nu}R=\kappa\frac{\tau_{\mu\nu}}{\sqrt{-g}},\\ T^{\lambda}{}_{\mu\nu}=\frac{\kappa}{2\sqrt{-g}}(2\sigma_{\mu\nu}{}^{\lambda}+\delta_{\mu}^{\lambda}\sigma_{\nu\rho}{}^{\rho}-\delta_{\nu}^{\lambda}\sigma_{\mu\rho}{}^{\rho}),$
1510.06699_ E09011 (12)	008	■0.614 C +4	$L_{\mathrm{D}}=\frac{1}{2}i\left[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-\left(\partial_{\mu}\bar{\psi}\right)\gamma^{\mu}\psi\right]-m\bar{\psi}\psi\equiv\frac{1}{2}i\bar{\psi}\gamma^{\mu}\overleftrightarrow{\partial}_{\mu}\psi-m\bar{\psi}\psi.$	$L_{\mathrm{D}}=\frac{1}{2}i\left[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-\left(\partial_{\mu}\bar{\psi}\right)\gamma^{\mu}\psi\right]-m\bar{\psi}\psi\equiv\frac{1}{2}i\bar{\psi}\gamma^{\mu}\overleftrightarrow{\partial}_{\mu}\psi-m\bar{\psi}\psi.$	$L_{\mathrm{D}}=\frac{1}{2}i[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-(\partial_{\mu}\bar{\psi})\gamma^{\mu}\psi]-m\bar{\psi}\psi\equiv\frac{1}{2}i\bar{\psi}\gamma^{\mu}\overleftrightarrow{\partial}_{\mu}\psi-m\bar{\psi}\psi.$
1510.06699_ E09183 (199b)	034	■0.615 C +4	$\begin{aligned}\widehat{\varphi}&\equiv\left(\frac{\phi}{\phi_0}\right)^{-w}\varphi\\ \widehat{h}_a{}^{\mu}&\equiv\left(\frac{\phi}{\phi_0}\right)^{-1}h_a{}^{\mu}\\ \widehat{A}^{\dagger ab}{}_{\mu}&\equiv A^{ab}{}_{\mu}+(\mathcal{V}^ab_{\mu}^b-\mathcal{V}^bb_{\mu}^a)\\ \widehat{V}_{\mu}&\equiv V_{\mu}+\frac{1}{3}T_{\mu}+\partial_{\mu}\ln\left(\frac{\phi}{\phi_0}\right)\end{aligned}$	$\widehat{\varphi}\equiv\left(\frac{\phi}{\phi_0}\right)^{-w}\varphi\\ \widehat{h}_a{}^{\mu}\equiv\left(\frac{\phi}{\phi_0}\right)^{-1}h_a{}^{\mu}\\ \widehat{A}^{\dagger ab}{}_{\mu}\equiv A^{ab}{}_{\mu}+(\mathcal{V}^ab_{\mu}^b-\mathcal{V}^bb_{\mu}^a)\\ \widehat{V}_{\mu}\equiv V_{\mu}+\frac{1}{3}T_{\mu}+\partial_{\mu}\ln\left(\frac{\phi}{\phi_0}\right)$	$\widehat{\varphi}\equiv\left(\frac{\phi}{\phi_0}\right)^{-w}\varphi,\\ \widehat{h}_a{}^{\mu}\equiv\left(\frac{\phi}{\phi_0}\right)^{-1}h_a{}^{\mu},\\ \widehat{A}^{\dagger ab}{}_{\mu}\equiv A^{ab}{}_{\mu}+(\mathcal{V}^ab_{\mu}^b-\mathcal{V}^bb_{\mu}^a),\\ \widehat{V}_{\mu}\equiv V_{\mu}+\frac{1}{3}T_{\mu}+\partial_{\mu}\ln\left(\frac{\phi}{\phi_0}\right),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0097 (105b)	020	■ 0.649 ● W +2	$\begin{aligned} \widehat{\lambda} &\equiv \left(\frac{\phi}{\phi_0}\right) \lambda \\ \widehat{p}_a &\equiv \widehat{h}_a{}^\mu p_\mu = \left(\frac{\phi}{\phi_0}\right)^{-1} p_a \\ \widehat{v}^a &\equiv \widehat{b}^a{}_\mu \frac{dx^\mu}{d\widehat{\lambda}} = v^a, \\ \widehat{e} &\equiv \left(\frac{\phi}{\phi_0}\right) e, \end{aligned}$	$\begin{aligned} \widehat{\lambda} &\equiv \left(\frac{\phi}{\phi_0}\right) \lambda \\ \widehat{p}_a &\equiv \widehat{h}_a{}^\mu p_\mu = \left(\frac{\phi}{\phi_0}\right)^{-1} p_a \\ \widehat{v}^a &\equiv \widehat{b}^a{}_\mu \frac{dx^\mu}{d\widehat{\lambda}} = v^a, \\ \widehat{e} &\equiv \left(\frac{\phi}{\phi_0}\right) e, \end{aligned}$	$\begin{aligned} \widehat{\lambda} &\equiv \left(\frac{\phi}{\phi_0}\right) \lambda, \\ \widehat{p}_a &\equiv \widehat{h}_a{}^\mu p_\mu = \left(\frac{\phi}{\phi_0}\right)^{-1} p_a, \\ \widehat{v}^a &\equiv \widehat{b}^a{}_\mu \frac{dx^\mu}{d\widehat{\lambda}} = v^a, \\ \widehat{e} &\equiv \left(\frac{\phi}{\phi_0}\right) e, \end{aligned}$
1510.06699_ EQ0085 (91b)	018	■ 0.651 ●	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi \\ \widehat{h}_a{}^\mu &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^\mu \\ \widehat{A}_\mu^{ab} &\equiv A_\mu^{ab} \\ \widehat{B}_\mu &\equiv B_\mu - \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right) \end{aligned}$	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi \\ \widehat{h}_a{}^\mu &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^\mu \\ \widehat{A}_\mu^{ab} &\equiv A_\mu^{ab} \\ \widehat{B}_\mu &\equiv B_\mu - \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right) \end{aligned}$	$\begin{aligned} \widehat{\varphi} &\equiv \left(\frac{\phi}{\phi_0}\right)^{-w} \varphi, \\ \widehat{h}_a{}^\mu &\equiv \left(\frac{\phi}{\phi_0}\right)^{-1} h_a{}^\mu, \\ \widehat{A}^{ab}{}_\mu &\equiv A^{ab}{}_\mu, \\ \widehat{B}_\mu &\equiv B_\mu - \partial_\mu \ln \left(\frac{\phi}{\phi_0}\right), \end{aligned}$
1510.06699_ EQ0063 (67a)	014	■ 0.657 ● N +0	$\begin{aligned} &\frac{\delta L_M}{\delta \varphi} = \frac{\bar{\partial} L_M}{\partial \varphi} - (\mathcal{D}_a^* + \mathcal{T}_a^*) \left(\frac{\partial L_M}{\partial (\mathcal{D}_a^* \varphi)} \right) = 0, \\ &\frac{\delta L_M}{\delta \phi} = \frac{\bar{\partial} L_M}{\partial \phi} - (\mathcal{D}_a^* + \mathcal{T}_a^*) \left(\frac{\partial L_M}{\partial (\mathcal{D}_a^* \phi)} \right) = 0. \end{aligned}$	$\begin{aligned} \frac{\delta L_M}{\delta \varphi} &= \frac{\bar{\partial} L_M}{\partial \varphi} - (\mathcal{D}_a^* + \mathcal{T}_a^*) \left(\frac{\partial L_M}{\partial (\mathcal{D}_a^* \varphi)} \right) = 0, \\ \frac{\delta L_M}{\delta \phi} &= \frac{\bar{\partial} L_M}{\partial \phi} - (\mathcal{D}_a^* + \mathcal{T}_a^*) \left(\frac{\partial L_M}{\partial (\mathcal{D}_a^* \phi)} \right) = 0. \end{aligned}$	$\begin{aligned} \frac{\delta L_M}{\delta \varphi} &= \frac{\bar{\partial} L_M}{\partial \varphi} - (\mathcal{D}_a^* + \mathcal{T}_a^*) \left(\frac{\partial L_M}{\partial (\mathcal{D}_a^* \varphi)} \right) = 0, \\ \frac{\delta L_M}{\delta \phi} &= \frac{\bar{\partial} L_M}{\partial \phi} - (\mathcal{D}_a^* + \mathcal{T}_a^*) \left(\frac{\partial L_M}{\partial (\mathcal{D}_a^* \phi)} \right) = 0. \end{aligned}$
1510.06699_ EQ0227 (261)	042	■ 0.664 ● C +6	$\begin{aligned} \mathcal{L}_M &= h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^\dagger \phi) (\mathcal{D}^{\dagger a} \phi) - \lambda \phi^4 + \frac{1}{12} \nu \phi^2 \mathcal{R}^\dagger + \phi^2 L_{\mathcal{T}^{\dagger 2}} \right] - \frac{1}{2} \nu h^{-1} (\mathcal{D}_a + \mathcal{T}_a) (\phi^2 \mathcal{V}^a), \\ \mathcal{L}_M &= h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^\dagger \phi) (\mathcal{D}^{\dagger a} \phi) - \lambda \phi^4 + \frac{1}{12} \nu \phi^2 \mathcal{R}^\dagger + \phi^2 L_{\mathcal{T}^{\dagger 2}} \right] - \frac{1}{2} \nu h^{-1} (\mathcal{D}_a + \mathcal{T}_a) (\phi^2 \mathcal{V}^a), \end{aligned}$	$\begin{aligned} \mathcal{L}_M &= h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^\dagger \phi) (\mathcal{D}^{\dagger a} \phi) - \lambda \phi^4 + \frac{1}{12} \nu \phi^2 \mathcal{R}^\dagger + \phi^2 L_{\mathcal{T}^{\dagger 2}} \right] - \frac{1}{2} \nu h^{-1} (\mathcal{D}_a + \mathcal{T}_a) (\phi^2 \mathcal{V}^a), \\ \mathcal{L}_M &= h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^\dagger \phi) (\mathcal{D}^{\dagger a} \phi) - \lambda \phi^4 + \frac{1}{12} \nu \phi^2 \mathcal{R}^\dagger + \phi^2 L_{\mathcal{T}^{\dagger 2}} \right] - \frac{1}{2} \nu h^{-1} (\mathcal{D}_a + \mathcal{T}_a) (\phi^2 \mathcal{V}^a), \end{aligned}$	$\begin{aligned} \mathcal{L}_M &= h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^\dagger \phi) (\mathcal{D}^{\dagger a} \phi) - \lambda \phi^4 + \frac{1}{12} \nu \phi^2 \mathcal{R}^\dagger + \phi^2 L_{\mathcal{T}^{\dagger 2}} \right] - \frac{1}{2} \nu h^{-1} (\mathcal{D}_a + \mathcal{T}_a) (\phi^2 \mathcal{V}^a), \\ \mathcal{L}_M &= h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^\dagger \phi) (\mathcal{D}^{\dagger a} \phi) - \lambda \phi^4 + \frac{1}{12} \nu \phi^2 \mathcal{R}^\dagger + \phi^2 L_{\mathcal{T}^{\dagger 2}} \right] - \frac{1}{2} \nu h^{-1} (\mathcal{D}_a + \mathcal{T}_a) (\phi^2 \mathcal{V}^a), \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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1510.06699_ E00074 (79)	016	■ 0.671 ● C +5	$\begin{aligned} \mathrm{mathcal{L}}_{\mathrm{M}} = & h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^* \phi) (\mathcal{D}^{*a} \phi) - \lambda \phi^4 \right. \\ & \left. - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}} \right] \end{aligned}$	$\mathcal{L}_{\mathrm{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^* \phi) (\mathcal{D}^{*a} \phi) - \lambda \phi^4 - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}} \right]$	$\mathcal{L}_{\mathrm{M}} = h^{-1} [\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \frac{1}{2} \nu (\mathcal{D}_a^* \phi) (\mathcal{D}^{*a} \phi) - \lambda \phi^4 - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}}], \quad (79)$
1510.06699_ E00211 (238)	039	■ 0.675 ●	$\begin{aligned} & h \left(\sigma_{\phi} \right)_{ab}{}^c = -\frac{1}{3} \nu \phi \delta_{[a}^c \mathcal{D}_{b]}^{\dagger} \phi \\ & h \left(\sigma_{\mathcal{R}} \right)_{ab}{}^c = -a \left(\delta_{[a}^c \mathcal{D}_{b]}^{\dagger} + \frac{1}{2} \mathcal{T}^{\dagger c}{}_{ab} \right) \phi^2 \\ & h \left(\sigma_{\mathcal{T}^2} \right)_{ab}{}^c = -4 \beta_1 \phi^2 \mathcal{T}^{\dagger}{}_{[ab]}{}^c + 2 \beta_2 \phi^2 \left(\mathcal{T}^{\dagger c}{}_{ab} + \mathcal{T}^{\dagger}{}_{[ab]}{}^c \right) \phi \end{aligned}$	$\begin{aligned} h(\sigma_{\phi})_{ab}{}^c &= -\frac{1}{3} \nu \phi \delta_{[a}^c \mathcal{D}_{b]}^{\dagger} \phi \\ h(\sigma_{\mathcal{R}})_{ab}{}^c &= -a \left(\delta_{[a}^c \mathcal{D}_{b]}^{\dagger} + \frac{1}{2} \mathcal{T}^{\dagger c}{}_{ab} \right) \phi^2 \\ h(\sigma_{\mathcal{T}^2})_{ab}{}^c &= -4 \beta_1 \phi^2 \mathcal{T}^{\dagger}{}_{[ab]}{}^c + 2 \beta_2 \phi^2 \left(\mathcal{T}^{\dagger c}{}_{ab} + \mathcal{T}^{\dagger}{}_{[ab]}{}^c \right) \phi \end{aligned}$	$\begin{aligned} h(\sigma_{\phi})_{ab}{}^c &= -\frac{1}{3} \nu \phi \delta_{[a}^c \mathcal{D}_{b]}^{\dagger} \phi, \\ h(\sigma_{\mathcal{R}})_{ab}{}^c &= -a (\delta_{[a}^c \mathcal{D}_{b]}^{\dagger} + \frac{1}{2} \mathcal{T}^{\dagger c}{}_{ab}) \phi^2, \\ h(\sigma_{\mathcal{T}^2})_{ab}{}^c &= -4 \beta_1 \phi^2 \mathcal{T}^{\dagger}{}_{[ab]}{}^c + 2 \beta_2 \phi^2 (\mathcal{T}^{\dagger c}{}_{ab} + \mathcal{T}^{\dagger}{}_{[ab]}{}^c) \phi \end{aligned}$
1510.06699_ E00216 (248)	039	■ 0.686 ● N +0	$\begin{aligned} & \partial_{\phi} L_{\mathcal{R}^2} = 0, \\ & \partial_{\phi} L_{\mathcal{H}^2} = 0, \end{aligned}$	$\begin{aligned} \partial_{\phi} L_{\mathcal{R}^2} &= 0, \\ \partial_{\phi} L_{\mathcal{H}^2} &= 0, \end{aligned}$	$\begin{aligned} \partial_{\phi} L_{\mathcal{R}^2} &= 0, \\ \partial_{\phi} L_{\mathcal{H}^2} &= 0, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00213 (239)	039	0.687 C +16	$\begin{aligned} & \hleft(j_{\mathcal{R}^2})_a = & -12\alpha_1\mathcal{D}_a^\dagger\mathcal{R}^\dagger \\ & -4\alpha_2\left[\mathcal{D}_b^\dagger\left(\mathcal{R}_a^{\dagger b}+\frac{1}{2}\delta_a^b\mathcal{R}^\dagger\right)-\frac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}_b^{\dagger c}\right] \\ & -4\alpha_3\left[\mathcal{D}_b^\dagger\left(\mathcal{R}^{\dagger b}{}_a+\frac{1}{2}\delta_a^b\mathcal{R}^\dagger\right)-\frac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}^{\dagger c}{}_b\right] \\ & -8\alpha_4\left[\mathcal{D}_b^\dagger\mathcal{R}_a^{\dagger b}-\frac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{abcd}^\dagger\right] \\ & -2\alpha_5\left[\mathcal{D}_b^\dagger\left(\mathcal{R}_a^{\dagger b}+\mathcal{R}^{\dagger b}{}_a\right)-\mathcal{T}^{\dagger bcd}\left(\mathcal{R}_{cbad}^\dagger+\mathcal{R}_{cabd}^\dagger\right)\right] \\ & -8\alpha_6\left[\mathcal{D}_b^\dagger\mathcal{R}^{\dagger b}{}_a-\frac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{cdab}^\dagger\right], \end{aligned} \tag{241}$ $h(j_{\mathcal{H}^2})_a = -2\xi\left[\mathcal{D}_b^\dagger\mathcal{H}^{\dagger b}{}_a-\frac{1}{2}\mathcal{T}^{\dagger}{}_{acd}\mathcal{H}^{\dagger cd}\right], \tag{242}$	$\begin{aligned} h(j_{\mathcal{R}^2})_a = & -12\alpha_1\mathcal{D}_a^\dagger\mathcal{R}^\dagger \\ & -4\alpha_2\left[\mathcal{D}_b^\dagger\left(\mathcal{R}_a^{\dagger b}+\frac{1}{2}\delta_a^b\mathcal{R}^\dagger\right)-\frac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}_b^{\dagger c}\right] \\ & -4\alpha_3\left[\mathcal{D}_b^\dagger\left(\mathcal{R}^{\dagger b}{}_a+\frac{1}{2}\delta_a^b\mathcal{R}^\dagger\right)-\frac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}^{\dagger c}{}_b\right] \\ & -8\alpha_4\left[\mathcal{D}_b^\dagger\mathcal{R}_a^{\dagger b}-\frac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{abcd}^\dagger\right] \\ & -2\alpha_5\left[\mathcal{D}_b^\dagger\left(\mathcal{R}_a^{\dagger b}+\mathcal{R}^{\dagger b}{}_a\right)-\mathcal{T}^{\dagger bcd}\left(\mathcal{R}_{cbad}^\dagger+\mathcal{R}_{cabd}^\dagger\right)\right] \\ & -8\alpha_6\left[\mathcal{D}_b^\dagger\mathcal{R}^{\dagger b}{}_a-\frac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{cdab}^\dagger\right] \end{aligned}$ $h(j_{\mathcal{H}^2})_a = -2\xi\left[\mathcal{D}_b^\dagger\mathcal{H}^{\dagger b}{}_a-\frac{1}{2}\mathcal{T}^{\dagger}{}_{acd}\mathcal{H}^{\dagger cd}\right]$	$\begin{aligned} h(j_{\mathcal{R}^2})_a = & -12\alpha_1\mathcal{D}_a^\dagger\mathcal{R}^\dagger \\ & -4\alpha_2[\mathcal{D}_b^\dagger(\mathcal{R}_a^{\dagger b} + \tfrac{1}{2}\delta_a^b\mathcal{R}^\dagger) - \tfrac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}_b^{\dagger c}] \\ & -4\alpha_3[\mathcal{D}_b^\dagger(\mathcal{R}^{\dagger b}{}_a + \tfrac{1}{2}\delta_a^b\mathcal{R}^\dagger) - \tfrac{1}{2}\mathcal{T}^{\dagger b}{}_{ac}\mathcal{R}^{\dagger c}{}_b] \\ & -8\alpha_4[\mathcal{D}_b^\dagger\mathcal{R}_a^{\dagger b} - \tfrac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{abcd}^\dagger] \\ & -2\alpha_5[\mathcal{D}_b^\dagger(\mathcal{R}_a^{\dagger b} + \mathcal{R}^{\dagger b}{}_a) - \mathcal{T}^{\dagger bcd}(\mathcal{R}_{cbad}^\dagger + \mathcal{R}_{cabd}^\dagger)] \\ & -8\alpha_6[\mathcal{D}_b^\dagger\mathcal{R}^{\dagger b}{}_a - \tfrac{1}{2}\mathcal{T}^{\dagger bcd}\mathcal{R}_{cdab}^\dagger], \end{aligned} \tag{241}$ $h(j_{\mathcal{H}^2})_a = -2\xi[\mathcal{D}_b^\dagger\mathcal{H}^{\dagger b}{}_a - \tfrac{1}{2}\mathcal{T}^{\dagger}{}_{acd}\mathcal{H}^{\dagger cd}], \tag{242}$
1510.06699_ E00201 (219)	037	0.691	$\begin{aligned} & \{ \}^{\{0\}} \Gamma^{\dagger} \lambda \{ \} _ \mu \nu = & \frac{1}{2} g^{\lambda \rho} \left(\partial_\mu^\dagger g_{\nu \rho} + \partial_\nu^\dagger g_{\mu \rho} - \partial_\rho^\dagger g_{\mu \nu} \right) \\ & = {}^0 \Gamma^\lambda{}_{\mu \nu} - \delta_\nu^\lambda \left(V_\mu + \frac{1}{3} T_\mu \right) - \delta_\mu^\lambda \left(V_\nu + \frac{1}{3} T_\nu \right) \\ & + g_{\mu \nu} \left(V^\lambda + \frac{1}{3} T^\lambda \right) \end{aligned}$	$\begin{aligned} {}^0 \Gamma^{\dagger \lambda}{}_{\mu \nu} = & \frac{1}{2} g^{\lambda \rho} \left(\partial_\mu^\dagger g_{\nu \rho} + \partial_\nu^\dagger g_{\mu \rho} - \partial_\rho^\dagger g_{\mu \nu} \right) \\ = {}^0 \Gamma^\lambda{}_{\mu \nu} - \delta_\nu^\lambda \left(V_\mu + \frac{1}{3} T_\mu \right) - \delta_\mu^\lambda \left(V_\nu + \frac{1}{3} T_\nu \right) \\ & + g_{\mu \nu} \left(V^\lambda + \frac{1}{3} T^\lambda \right), \end{aligned}$	$\begin{aligned} {}^0 \Gamma^{\dagger \lambda}{}_{\mu \nu} = & \frac{1}{2} g^{\lambda \rho} \left(\partial_\mu^\dagger g_{\nu \rho} + \partial_\nu^\dagger g_{\mu \rho} - \partial_\rho^\dagger g_{\mu \nu} \right) \\ = {}^0 \Gamma^\lambda{}_{\mu \nu} - \delta_\nu^\lambda (V_\mu + \tfrac{1}{3} T_\mu) - \delta_\mu^\lambda (V_\nu + \tfrac{1}{3} T_\nu) \\ & + g_{\mu \nu} (V^\lambda + \tfrac{1}{3} T^\lambda), \end{aligned}$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0147 (161b)	028	0.692 C +4	$\begin{aligned} & \mathcal{D}_{[a}^{\dagger} \mathcal{R}^{\dagger de}_{bc]} - \mathcal{T}^{\dagger f}_{[ab} \mathcal{R}^{\dagger de}_{c]f} = 0, \\ & \mathcal{D}_{[a}^{\dagger} \mathcal{T}^{\dagger d}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{T}^{\dagger d}_{c]e} - \mathcal{R}^{\dagger d}_{[abc]} + \mathcal{H}^{\dagger}_{[ab} \delta^d_{c]} = 0, \\ & \mathcal{D}_{[a}^{\dagger} \mathcal{H}^{\dagger}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{H}^{\dagger}_{c]e} = 0. \end{aligned}$	$\mathcal{D}_{[a}^{\dagger} \mathcal{R}^{\dagger de}_{bc]} - \mathcal{T}^{\dagger f}_{[ab} \mathcal{R}^{\dagger de}_{c]f} = 0, \\ \mathcal{D}_{[a}^{\dagger} \mathcal{T}^{\dagger d}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{T}^{\dagger d}_{c]e} - \mathcal{R}^{\dagger d}_{[abc]} + \mathcal{H}^{\dagger}_{[ab} \delta^d_{c]} = 0, \\ \mathcal{D}_{[a}^{\dagger} \mathcal{H}^{\dagger}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{H}^{\dagger}_{c]e} = 0.$	$\mathcal{D}_{[a}^{\dagger} \mathcal{R}^{\dagger de}_{bc]} - \mathcal{T}^{\dagger f}_{[ab} \mathcal{R}^{\dagger de}_{c]f} = 0, (\\ \mathcal{D}_{[a}^{\dagger} \mathcal{T}^{\dagger d}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{T}^{\dagger d}_{c]e} - \mathcal{R}^{\dagger d}_{[abc]} + \mathcal{H}^{\dagger}_{[ab} \delta^d_{c]} = 0, (\\ \mathcal{D}_{[a}^{\dagger} \mathcal{H}^{\dagger}_{bc]} - \mathcal{T}^{\dagger e}_{[ab} \mathcal{H}^{\dagger}_{c]e} = 0.)$
1510.06699_ EQ0115 (124)	022	0.692 N +0	$\nabla_{\sigma} g_{\mu\nu} = -2 B_{\sigma} g_{\mu\nu}$	$\nabla_{\sigma} g_{\mu\nu} = -2 B_{\sigma} g_{\mu\nu},$	
1510.06699_ EQ0226 (260)	042	0.694 N +1	$\mathcal{L}_{\mathrm{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi - \lambda \phi^4 - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}} \right]$	$\mathcal{L}_{\mathrm{M}} = h^{-1} \left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi - \lambda \phi^4 - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}} \right]$	$\mathcal{L}_{\mathrm{M}} = h^{-1} [\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi - \lambda \phi^4 - a \phi^2 \mathcal{R} + \phi^2 L_{\mathcal{T}^{*2}}],$
1510.06699_ EQ0218 (252)	040	0.699 C +4	$\begin{aligned} & \nu \mathcal{D}_a^{\dagger} (\mathcal{D}^{\dagger a} \phi) + 4 \lambda \phi^3 - \partial_{\phi} L_{\varphi} \\ & + \phi \left(a \mathcal{R}^{\dagger} - 2 \beta_1 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} - 2 \beta_2 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac} \right) = 0 \end{aligned}$	$\nu \mathcal{D}_a^{\dagger} (\mathcal{D}^{\dagger a} \phi) + 4 \lambda \phi^3 - \partial_{\phi} L_{\varphi} \\ + \phi \left(a \mathcal{R}^{\dagger} - 2 \beta_1 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} - 2 \beta_2 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac} \right) = 0$	$\nu \mathcal{D}_a^{\dagger} (\mathcal{D}^{\dagger a} \phi) + 4 \lambda \phi^3 - \partial_{\phi} L_{\varphi} \\ + \phi (a \mathcal{R}^{\dagger} - 2 \beta_1 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} - 2 \beta_2 \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}) = 0.$
1510.06699_ EQ0137 (151a)	026	0.704 W +2	$\begin{aligned} & \mathcal{R}^{\dagger}_{[a} \mathcal{D}^{\dagger}_{b]} - \mathcal{T}^{\dagger}_{[a} \mathcal{R}^{\dagger}_{b]} = 0, \\ & \mathcal{D}_{[a}^{\dagger} \mathcal{R}^{\dagger}_{b]} - \mathcal{T}^{\dagger}_{[a} \mathcal{D}^{\dagger}_{b]} = 0, \\ & \mathcal{D}_{[a}^{\dagger} \mathcal{H}^{\dagger}_{b]} - \mathcal{T}^{\dagger}_{[a} \mathcal{D}^{\dagger}_{b]} = 0. \end{aligned}$	$R^{\dagger ab}{}_{\mu\nu} \equiv \partial_{\mu} A^{\dagger ab}{}_{\nu} - \partial_{\nu} A^{\dagger ab}{}_{\mu} + A^{\dagger a}{}_{c\mu} A^{\dagger cb}{}_{\nu} - A^{\dagger a}{}_{c\nu} A^{\dagger cb}{}_{\mu} \\ = R^{ab}{}_{\mu\nu} + 4 b^{[b}{}_{[\nu} D_{\mu]} \mathcal{V}^{a]} + 4 \mathcal{V}^{[a} V_{[\mu} b^{b]}{}_{\nu]} - 2 \mathcal{V}^e \mathcal{V}_e b^{[a}{}_{\mu} b^{b]}{}_{\nu} + 2 \mathcal{V}^{[a} T^{b]}{}_{\mu\nu}$	$R^{\dagger ab}{}_{\mu\nu} \equiv \partial_{\mu} A^{\dagger ab}{}_{\nu} - \partial_{\nu} A^{\dagger ab}{}_{\mu} + A^{\dagger a}{}_{c\mu} A^{\dagger cb}{}_{\nu} - A^{\dagger a}{}_{c\nu} A^{\dagger cb}{}_{\mu}, \\ = R^{ab}{}_{\mu\nu} + 4 b^{[b}{}_{[\nu} D_{\mu]} \mathcal{V}^{a]} + 4 \mathcal{V}^{[a} V_{[\mu} b^{b]}{}_{\nu]} - 2 \mathcal{V}^e \mathcal{V}_e b^{[a}{}_{\mu} b^{b]}{}_{\nu} + 2 \mathcal{V}^{[a} T^{b]}{}_{\mu\nu},$
1510.06699_ EQ0022 (23)	009	0.723	$A^{\prime ab}{}_{\mu}(x') = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \left[\Lambda^a{}_c(x) \Lambda^b{}_d(x) A^{cd}{}_{\nu}(x) - \Lambda^{bc}(x) \partial_{\nu} \Lambda^a{}_c(x) \right],$	$A^{\prime ab}{}_{\mu}(x') = \frac{\partial x^{\nu}}{\partial x'^{\mu}} \left[\Lambda^a{}_c(x) \Lambda^b{}_d(x) A^{cd}{}_{\nu}(x) - \Lambda^{bc}(x) \partial_{\nu} \Lambda^a{}_c(x) \right],$	$A^{\prime ab}{}_{\mu}(x') = \frac{\partial x^{\nu}}{\partial x'^{\mu}} [\Lambda^a{}_c(x) \Lambda^b{}_d(x) A^{cd}{}_{\nu}(x) - \Lambda^{bc}(x) \partial_{\nu} \Lambda^a{}_c(x)],$
1510.06699_ EQ0013 (14)	008	0.734 N +0	$L_{\mathrm{M}} = \frac{1}{2} i \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi + \frac{1}{2} \partial_{\mu} \phi \partial^{\mu} \phi - \mu \phi \bar{\psi} \psi,$	$L_{\mathrm{M}} = \frac{1}{2} i \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi + \frac{1}{2} \partial_{\mu} \phi \partial^{\mu} \phi - \mu \phi \bar{\psi} \psi,$	$L_{\mathrm{M}} = \frac{1}{2} i \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi + \frac{1}{2} \partial_{\mu} \phi \partial^{\mu} \phi - \mu \phi \bar{\psi} \psi,$
Conf. MathPix confidence: 0.9 0.5-0.9 <0.5 no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C componentW weakS stableN noiseK cleannot measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0182 (198)	033	■ 0.744 ● W +0	$\begin{aligned} \mathrm{cal}_{L}_{\mathrm{T}}=& \mathrm{cal}_{L}_{\mathrm{G}}\left(\widehat{h}, \partial \widehat{h}, \partial^2 \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right. \\ & \left.+\mathrm{cal}_{L}_{\mathrm{M}}\left(\widehat{\varphi}, \partial \widehat{\varphi}, \phi_0, 0,, \widehat{h}, \partial \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) .\right. \\ & \left.+\mathrm{cal}_{L}_{\mathrm{M}}\left(\widehat{\varphi}, \partial \widehat{\varphi}, \phi_0, 0,, \widehat{h}, \partial \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) .\right. \end{aligned}$	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}\left(\widehat{h}, \partial \widehat{h}, \partial^2 \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) \\ +\mathcal{L}_{\mathrm{M}}\left(\widehat{\varphi}, \partial \widehat{\varphi}, \phi_0, 0,, \widehat{h}, \partial \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) .$	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}\left(\widehat{h}, \partial \widehat{h}, \partial^2 \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) \\ +\mathcal{L}_{\mathrm{M}}\left(\widehat{\varphi}, \partial \widehat{\varphi}, \phi_0, 0,, \widehat{h}, \partial \widehat{h}, \widehat{A}^{\dagger}, \partial \widehat{A}^{\dagger}\right) .$
1510.06699_ EQ0012 (13)	008	■ 0.749 ● N +2	$\begin{aligned} L_{\mathrm{D}}^{\prime} &= \frac{1}{2} i \bar{\psi}^{\prime} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu}^{\prime} \psi^{\prime}-m \bar{\psi}^{\prime} \psi^{\prime} \\ &= \frac{1}{2} i e^{(2 w-1) \rho} \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi-m e^{2 w \rho} \bar{\psi} \psi \end{aligned}$	$L_{\mathrm{D}}^{\prime}=\frac{1}{2} i \bar{\psi}^{\prime} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu}^{\prime} \psi^{\prime}-m \bar{\psi}^{\prime} \psi^{\prime} \\ =\frac{1}{2} i e^{(2 w-1) \rho} \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi-m e^{2 w \rho} \bar{\psi} \psi .$	$L_{\mathrm{D}}^{\prime}=\frac{1}{2} i \bar{\psi}^{\prime} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu}^{\prime} \psi^{\prime}-m \bar{\psi}^{\prime} \psi^{\prime} \\ =\frac{1}{2} i e^{(2 w-1) \rho} \bar{\psi} \gamma^{\mu} \overleftrightarrow{\partial}_{\mu} \psi-m e^{2 w \rho} \bar{\psi} \psi .$
1510.06699_ EQ0155 (169a)	029	■ 0.758 ● C +3	$\begin{aligned} & \& \mathrm{t}_{\mathrm{a} \mathrm{b}}^{\prime}=\mathrm{t}_{\mathrm{a} \mathrm{b}}+2 \theta \\\left(\mathrm{s}_{\mathrm{c}}\right\}^{\mathrm{c}} \mathrm{cal}_{\mathrm{P}}\}_{\mathrm{b}}-\mathrm{s}_{\mathrm{c} \mathrm{b} \mathrm{a}} \mathcal{P}^{\mathrm{c}} \\\mathrm{cal}_{\mathrm{P}}\}^{\mathrm{c}}\right) \& \& \mathrm{t}_{\mathrm{a} \mathrm{b}}^{\prime}=\mathrm{t}_{\mathrm{a} \mathrm{b}}+2 \theta \\\left(\mathrm{s}_{\mathrm{c}}\right\}^{\mathrm{c}} \mathrm{cal}_{\mathrm{P}}\}_{\mathrm{b}}-\mathrm{s}_{\mathrm{c} \mathrm{b} \mathrm{a}} \mathcal{P}^{\mathrm{c}} \\\mathrm{cal}_{\mathrm{P}}\}^{\mathrm{c}}\right) \end{aligned}$	$t_{a b}^{\prime}=t_{a b}+2 \theta\left(s_{a c}^c \mathcal{P}_b-s_{c b a} \mathcal{P}^c\right) \\ \tau_{a b}^{\prime}=\tau_{a b}+2 \theta\left(\sigma_{a c}^c \mathcal{P}_b-\sigma_{c b a} \mathcal{P}^c\right),$	$t_{a b}^{\prime}=t_{a b}+2 \theta\left(s_{a c}^c \mathcal{P}_b-s_{c b a} \mathcal{P}^c\right), \\ \tau_{a b}^{\prime}=\tau_{a b}+2 \theta\left(\sigma_{a c}^c \mathcal{P}_b-\sigma_{c b a} \mathcal{P}^c\right),$
1510.06699_ EQ0188 (116)	021	■ 0.766 ● W +0	$\hat{\boldsymbol{e}}_{\mathrm{a}} \cdot \hat{\boldsymbol{e}}_{\mathrm{b}}=\eta_{a b}=g_{\mu \nu} h_{\mathrm{a}}^{\mu} h_{\mathrm{b}}^{\nu} .$	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab} = g_{\mu \nu} h_a^{\mu} h_b^{\nu} .$	$\hat{e}_a \cdot \hat{e}_b = \eta_{ab} = g_{\mu \nu} h_a^{\mu} h_b^{\nu} .$
1510.06699_ EQ0123 (134)	023	■ 0.768 ● S +0	$\begin{aligned} & h_{\mathrm{a}}^{\prime \mu}(x)=e^{-\rho(x)} h_{\mathrm{a}}^{\mu}(x), \\ & A^{\prime a b}{}_{\mu}(x)=A^{a b}{}_{\mu}(x), \end{aligned}$	$h_a^{\prime \mu}(x)=e^{-\rho(x)} h_a^{\mu}(x), \\ A^{\prime a b}{}_{\mu}(x)=A^{a b}{}_{\mu}(x),$	$h_a^{\prime \mu}(x)=e^{-\rho(x)} h_a^{\mu}(x), \\ A^{\prime a b}{}_{\mu}(x)=A^{a b}{}_{\mu}(x),$
1510.06699_ EQ0222 (256)	041	■ 0.771 ● C +3	$\begin{aligned} \mathrm{cal}_{L}_{\mathrm{M}}=& h^{-1} \\\left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi-\mu \phi \bar{\psi} \psi+\frac{1}{2} \nu\left(\mathcal{D}_a \phi\right)\left(\mathcal{D}^a \phi\right)-\lambda \phi^4\right. \\ & \left.-a \phi^2 \mathcal{R}+\phi^2 L_{\mathcal{T}^2}\right], \end{aligned}$	$\mathcal{L}_{\mathrm{M}}=h^{-1}\left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi-\mu \phi \bar{\psi} \psi+\frac{1}{2} \nu\left(\mathcal{D}_a \phi\right)\left(\mathcal{D}^a \phi\right)-\lambda \phi^4\right. \\ \left.-a \phi^2 \mathcal{R}+\phi^2 L_{\mathcal{T}^2}\right],$	$\mathcal{L}_{\mathrm{M}}=h^{-1}\left[\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi-\mu \phi \bar{\psi} \psi+\frac{1}{2} \nu\left(\mathcal{D}_a \phi\right)\left(\mathcal{D}^a \phi\right)-\lambda \phi^4\right. \\ \left.-a \phi^2 \mathcal{R}+\phi^2 L_{\mathcal{T}^2}\right], \quad(256)$
1510.06699_ EQ0114 (123)	022	■ 0.786 ●	$\begin{aligned} & \Gamma_{\nu \mu}^{\lambda}=e_a^{\lambda}\left(\partial_{\mu}^* e_{\nu}^a+A_{b \mu}^a e_b^{\nu}\right), \\ & A_{b \mu}^a=e_a^{\lambda}\left(\partial_{\mu}^* e_b^{\lambda}+\Gamma_{\nu \mu}^{\lambda} e_b^{\nu}\right) . \end{aligned}$	$\Gamma_{\nu \mu}^{\lambda}=e_a^{\lambda}\left(\partial_{\mu}^* e_{\nu}^a+A_{b \mu}^a e_b^{\nu}\right), \\ A_{b \mu}^a=e_a^{\lambda}\left(\partial_{\mu}^* e_b^{\lambda}+\Gamma_{\nu \mu}^{\lambda} e_b^{\nu}\right) .$	$\Gamma_{\nu \mu}^{\lambda}=e_a^{\lambda}\left(\partial_{\mu}^* e_{\nu}^a+A_{b \mu}^a e_b^{\nu}\right), \\ A_{b \mu}^a=e_a^{\lambda}\left(\partial_{\mu}^* e_b^{\lambda}+\Gamma_{\nu \mu}^{\lambda} e_b^{\nu}\right) .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0168 (184c)	031	■ 0.814 ● C +25	$\begin{aligned} & \left(\frac{\delta \mathcal{L}_M}{\delta h_c^\mu}\right)_\dagger = \tau^c_\mu + 2\sigma_{ab}{}^c \mathcal{V}^a b_\mu^b - 2\sigma^{cb}{}_b V_\mu = \tau^{\dagger c}_\mu, (184a) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta A^{\dagger ab}{}_\mu}\right)_\dagger = \sigma_{ab}{}^\mu, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta V_\mu}\right)_\dagger = \zeta^\mu - 2h_a{}^\mu \sigma^{ab}{}_b = \zeta^{\dagger\mu} = 0 \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \varphi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \varphi}, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \phi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \phi}, \end{aligned}$	$\begin{aligned} & \left(\frac{\delta \mathcal{L}_M}{\delta h_c^\mu}\right)_\dagger = \tau^c_\mu + 2\sigma_{ab}{}^c \mathcal{V}^a b_\mu^b - 2\sigma^{cb}{}_b V_\mu = \tau^{\dagger c}_\mu, (184a) \\ & \left(\frac{\delta \mathcal{L}_M}{\delta A^{\dagger ab}{}_\mu}\right)_\dagger = \sigma_{ab}{}^\mu, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta V_\mu}\right)_\dagger = \zeta^\mu - 2h_a{}^\mu \sigma^{ab}{}_b = \zeta^{\dagger\mu} = 0 \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \varphi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \varphi}, \\ & \left(\frac{\delta \mathcal{L}_M}{\delta \phi}\right)_\dagger = \frac{\delta \mathcal{L}_M}{\delta \phi}, \end{aligned}$	
1510.06699_ EQ0103 (111)	020	■ 0.819 ● N +0	$L_{\{\mathrm{G}\}} = L_{\{0_{\{\mathrm{R}\}}\}} * 2 + L_{\{\mathrm{H}\}^{\{2\}}}$	$L_G = L_{0_{\mathcal{R}*2}} + L_{\mathcal{H}^2},$	$L_G = L_{0_{\mathcal{R}*2}} + L_{\mathcal{H}^2},$
1510.06699_ EQ0257 (C7)	050	■ 0.842 ● N +1	$\begin{aligned} & \mathcal{A}_{abc} = \frac{1}{2} (c_{abc} + c_{bca} - c_{cab}) \\ & + \kappa h (\sigma_{abc} - \sigma_{bca} - \sigma_{cab} + \eta_{ac} \sigma_{bd}{}^d - \eta_{bc} \sigma_{ad}{}^d), \end{aligned}$	$\begin{aligned} & \mathcal{A}_{abc} = \frac{1}{2} (c_{abc} + c_{bca} - c_{cab}) \\ & + \kappa h (\sigma_{abc} - \sigma_{bca} - \sigma_{cab} + \eta_{ac} \sigma_{bd}{}^d - \eta_{bc} \sigma_{ad}{}^d), \end{aligned}$	$\begin{aligned} & \mathcal{A}_{abc} = \frac{1}{2} (c_{abc} + c_{bca} - c_{cab}) \\ & + \kappa h (\sigma_{abc} - \sigma_{bca} - \sigma_{cab} + \eta_{ac} \sigma_{bd}{}^d - \eta_{bc} \sigma_{ad}{}^d), \end{aligned}$
1510.06699_ EQ0209 (.234)	039	■ 0.860 ● N +2	$\begin{aligned} & (s_{\mathcal{R}^2})_{ab}{}^c + (s_{\mathcal{H}^2})_{ab}{}^c \\ & + (\sigma_\varphi)_{ab}{}^c + (\sigma_\phi)_{ab}{}^c + (\sigma_{\mathcal{R}})_{ab}{}^c + (\sigma_{\mathcal{T}^2})_{ab}{}^c = 0, \end{aligned}$	$\begin{aligned} & (s_{\mathcal{R}^2})_{ab}{}^c + (s_{\mathcal{H}^2})_{ab}{}^c \\ & + (\sigma_\varphi)_{ab}{}^c + (\sigma_\phi)_{ab}{}^c + (\sigma_{\mathcal{R}})_{ab}{}^c + (\sigma_{\mathcal{T}^2})_{ab}{}^c = 0, \end{aligned}$	$\begin{aligned} & (s_{\mathcal{R}^2})_{ab}{}^c + (s_{\mathcal{H}^2})_{ab}{}^c \\ & + (\sigma_\varphi)_{ab}{}^c + (\sigma_\phi)_{ab}{}^c + (\sigma_{\mathcal{R}})_{ab}{}^c + (\sigma_{\mathcal{T}^2})_{ab}{}^c = 0, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0196 (213)	036	■ 0.861 ●	$\begin{aligned} R^{\dagger}{}_{\sigma\mu\nu} &= 2\left(\partial_{[\mu}\Gamma^{\rho}{}_{ \sigma \nu]} + \Gamma^{\rho}{}_{\lambda[\mu}\Gamma^{\lambda}{}_{ \sigma \nu]}\right) + H^{\dagger}{}_{\mu\nu}\delta^{\rho}_{\sigma}, \\ T^{\dagger\lambda}{}_{\mu\nu} &= 2\Gamma^{\lambda}{}_{[\nu\mu]}, \\ H^{\dagger}{}_{\mu\nu} &= \partial_{\mu}\left(V_{\nu} + \tfrac{1}{3}T_{\nu}\right) - \partial_{\nu}\left(V_{\mu} + \tfrac{1}{3}T_{\mu}\right). \end{aligned}$	$R^{\dagger\rho}{}_{\sigma\mu\nu} = 2\left(\partial_{[\mu}\Gamma^{\rho}{}_{ \sigma \nu]} + \Gamma^{\rho}{}_{\lambda[\mu}\Gamma^{\lambda}{}_{ \sigma \nu]}\right) + H^{\dagger}{}_{\mu\nu}\delta^{\rho}_{\sigma},$ $T^{\dagger\lambda}{}_{\mu\nu} = 2\Gamma^{\lambda}{}_{[\nu\mu]},$ $H^{\dagger}{}_{\mu\nu} = \partial_{\mu}\left(V_{\nu} + \tfrac{1}{3}T_{\nu}\right) - \partial_{\nu}\left(V_{\mu} + \tfrac{1}{3}T_{\mu}\right).$	$R^{\dagger\rho}{}_{\sigma\mu\nu} = 2(\partial_{[\mu}\Gamma^{\rho}{}_{ \sigma \nu]} + \Gamma^{\rho}{}_{\lambda[\mu}\Gamma^{\lambda}{}_{ \sigma \nu]}) + H^{\dagger}{}_{\mu\nu}\delta^{\rho}_{\sigma},$ $T^{\dagger\lambda}{}_{\mu\nu} = 2\Gamma^{\lambda}{}_{[\nu\mu]},$ $H^{\dagger}{}_{\mu\nu} = \partial_{\mu}(V_{\nu} + \tfrac{1}{3}T_{\nu}) - \partial_{\nu}(V_{\mu} + \tfrac{1}{3}T_{\mu}).$
1510.06699_ EQ0102 (110b)	020	■ 0.862 ● C +3	$\begin{aligned} \mathcal{R}^{ab}{}_{cd} &= {}^0\mathcal{R}^{*ab}{}_{cd} + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}{}_d - {}^0\mathcal{D}^*_d\mathcal{K}^{*ab}{}_c + \mathcal{K}^{*a}{}_{ec}\mathcal{K}^{*eb}{}_d - \mathcal{K}^{*a}{}_{ed}\mathcal{K}^{*eb}{}_c \\ \mathcal{R}^a{}_c &= {}^0\mathcal{R}^{*a}{}_c + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}{}_b - {}^0\mathcal{D}^*_b\mathcal{K}^{*ab}{}_c + \mathcal{K}^{*a}{}_{ec}\mathcal{K}^{*eb}{}_b - \mathcal{K}^{*a}{}_{eb}\mathcal{K}^{*eb}{}_c, \\ \mathcal{R} &= {}^0\mathcal{R}^* + \tfrac{1}{4}\mathcal{T}^{*abc}\mathcal{T}^*_{abc} + \tfrac{1}{2}\mathcal{T}^{*abc}\mathcal{T}^*_{bac} - \mathcal{T}^{*a}\mathcal{T}^*_a - 2{}^0\mathcal{D}^*_a\mathcal{T}^{*a}. \end{aligned}$	$\mathcal{R}^{ab}{}_{cd} = {}^0\mathcal{R}^{*ab}{}_{cd} + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}{}_d - {}^0\mathcal{D}^*_d\mathcal{K}^{*ab}{}_c + \mathcal{K}^{*a}{}_{ec}\mathcal{K}^{*eb}{}_d - \mathcal{K}^{*a}{}_{ed}\mathcal{K}^{*eb}{}_c$ $\mathcal{R}^a{}_c = {}^0\mathcal{R}^{*a}{}_c + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}{}_b - {}^0\mathcal{D}^*_b\mathcal{K}^{*ab}{}_c + \mathcal{K}^{*a}{}_{ec}\mathcal{K}^{*eb}{}_b - \mathcal{K}^{*a}{}_{eb}\mathcal{K}^{*eb}{}_c$ $\mathcal{R} = {}^0\mathcal{R}^* + \frac{1}{4}\mathcal{T}^{*abc}\mathcal{T}^*_{abc} + \frac{1}{2}\mathcal{T}^{*abc}\mathcal{T}^*_{bac} - \mathcal{T}^{*a}\mathcal{T}^*_a - 2{}^0\mathcal{D}^*_a\mathcal{T}^{*a}.$	$\mathcal{R}^{ab}{}_{cd} = {}^0\mathcal{R}^{*ab}{}_{cd} + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}{}_d - {}^0\mathcal{D}^*_d\mathcal{K}^{*ab}{}_c + \mathcal{K}^{*a}{}_{ec}\mathcal{K}^{*eb}{}_d - \mathcal{K}^{*a}{}_{ed}\mathcal{K}^{*eb}{}_c$ $\mathcal{R}^a{}_c = {}^0\mathcal{R}^{*a}{}_c + {}^0\mathcal{D}^*_c\mathcal{K}^{*ab}{}_b - {}^0\mathcal{D}^*_b\mathcal{K}^{*ab}{}_c + \mathcal{K}^{*a}{}_{ec}\mathcal{K}^{*eb}{}_b - \mathcal{K}^{*a}{}_{eb}\mathcal{K}^{*eb}{}_c,$ $\mathcal{R} = {}^0\mathcal{R}^* + \tfrac{1}{4}\mathcal{T}^{*abc}\mathcal{T}^*_{abc} + \tfrac{1}{2}\mathcal{T}^{*abc}\mathcal{T}^*_{bac} - \mathcal{T}^{*a}\mathcal{T}^*_a - 2{}^0\mathcal{D}^*_a\mathcal{T}^{*a}.$
1510.06699_ EQ0238 (A2)	047	■ 0.875 ● N +0	$\bar{\psi} \not{J} = J^{\mu} \bar{\psi} \gamma_{\mu} = \bar{\psi} \gamma^{\mu} \psi \bar{\psi} \gamma_{\mu}.$	$\bar{\psi} \not{J} = J^{\mu} \bar{\psi} \gamma_{\mu} = \bar{\psi} \gamma^{\mu} \psi \bar{\psi} \gamma_{\mu}.$	$\bar{\psi} \not{J} = J^{\mu} \bar{\psi} \gamma_{\mu} = \bar{\psi} \gamma^{\mu} \psi \bar{\psi} \gamma_{\mu}.$
1510.06699_ EQ0234 (268)	044	■ 0.882 ● C +5	$\begin{array}{l} \mathcal{L}_{\mathrm{M}} = h^{-1} \left[\tfrac{1}{2} i \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \tfrac{1}{2} \nu \left({}^0\mathcal{D}_a \phi \right) \left({}^0\mathcal{D}^a \phi \right) - \lambda \phi^4 \right. \\ \left. - a \phi^2 {}^0\mathcal{R} \right]. \end{array}$	$\mathcal{L}_{\mathrm{M}} = h^{-1} \left[\tfrac{1}{2} i \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \tfrac{1}{2} \nu \left({}^0\mathcal{D}_a \phi \right) \left({}^0\mathcal{D}^a \phi \right) - \lambda \phi^4 - a \phi^2 {}^0\mathcal{R} \right].$	$\mathcal{L}_{\mathrm{M}} = h^{-1} [\tfrac{1}{2} i \bar{\psi} \gamma^{a0} \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi + \tfrac{1}{2} \nu \left({}^0\mathcal{D}_a \phi \right) \left({}^0\mathcal{D}^a \phi \right) - \lambda \phi^4 - a \phi^2 {}^0\mathcal{R}]. \quad (268)$
1510.06699_ EQ0180 (196)	033	■ 0.900 ● C +5	$\widehat{\varphi} \equiv \left(\frac{\phi}{\phi_0} \right)^{-w} \varphi, \quad \widehat{h}_a{}^{\mu} \equiv \left(\frac{\phi}{\phi_0} \right)^{-1} h_a{}^{\mu}, \quad \widehat{A}^{\dagger ab}{}_{\mu} \equiv A^{\dagger ab}{}_{\mu},$	$\widehat{\varphi} \equiv \left(\frac{\phi}{\phi_0} \right)^{-w} \varphi, \quad \widehat{h}_a{}^{\mu} \equiv \left(\frac{\phi}{\phi_0} \right)^{-1} h_a{}^{\mu}, \quad \widehat{A}^{\dagger ab}{}_{\mu} \equiv A^{\dagger ab}{}_{\mu},$	$\widehat{\varphi} \equiv \left(\frac{\phi}{\phi_0} \right)^{-w} \varphi, \quad \widehat{h}_a{}^{\mu} \equiv \left(\frac{\phi}{\phi_0} \right)^{-1} h_a{}^{\mu}, \quad \widehat{A}^{\dagger ab}{}_{\mu} \equiv A^{\dagger ab}{}_{\mu},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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1510.06699_ E00190 (206)	035	■ 0.901 ● W +1	$\begin{aligned} \mathrm{L}_{\mathrm{T}} &= \mathrm{L}_{\mathrm{G}} \left(h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger \right) \\ &\quad + \mathcal{L}_{\mathrm{M}} \left(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger \right), \end{aligned}$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}} \left(h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger \right) + \mathcal{L}_{\mathrm{M}} \left(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger \right),$	
1510.06699_ E00145 (159)	027	■ 0.908 ●	$\{ \}^{\{0\}} D_{\{\mu\}}^{\{\dagger\}} \varphi \equiv \left(\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab} \right) \varphi,$	${}^0 D_{\mu}^{\dagger} \varphi \equiv \left(\partial_{\mu}^{\dagger} + \frac{1}{2} A^{\dagger ab}{}_{\mu} \Sigma_{ab} \right) \varphi,$	
1510.06699_ E00275 (D14)	053	■ 0.909 ● W +1	$\begin{aligned} \mathrm{D}_{\mathrm{a}} \mathrm{B}^{\mathrm{c}} &- \mathcal{R}^{ac} \mathcal{B}_{\mathrm{a}} + m^2 \mathcal{B}^c \\ &\quad + 2 \kappa \left[\left(\delta_b^c \mathcal{D}_{\mathrm{a}} - \kappa h \sigma_{ab}{}^c \right) \left(h \sigma^{abd} \mathcal{B}_{\mathrm{d}} \right) \right] = 0 \end{aligned}$	$\mathcal{D}_a \mathcal{D}^a \mathcal{B}^c - \mathcal{R}^{ac} \mathcal{B}_a + m^2 \mathcal{B}^c + 2 \kappa \left[\left(\delta_b^c \mathcal{D}_a - \kappa h \sigma_{ab}{}^c \right) \left(h \sigma^{abd} \mathcal{B}_d \right) \right] = 0,$	
1510.06699_ E00273 (D12)	053	■ 0.910 ● N +0	$L_{\psi} = \frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi,$	$L_{\psi} = \frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi,$	
1510.06699_ E00058 (62b)	014	■ 0.910 ● C +3	$\begin{aligned} t^a{}_b + \tau^a{}_b &= 0 \\ s_{ab}{}^c + \sigma_{ab}{}^c &= 0 \\ j^a + \zeta^a &= 0 \end{aligned}$	$\begin{aligned} t^a{}_b + \tau^a{}_b &= 0, \\ s_{ab}{}^c + \sigma_{ab}{}^c &= 0, \\ j^a + \zeta^a &= 0, \end{aligned}$	
1510.06699_ E00125 (138)	023	■ 0.912 ● N +1	$\mathcal{T}^{\prime a}{}_{bc} = e^{-\rho} \left(\mathcal{T}^a{}_{bc} + \mathcal{P}_b \delta_c^a - \mathcal{P}_c \delta_b^a \right),$	$\mathcal{T}^{ta}{}_{bc} = e^{-\rho} \left(\mathcal{T}^a{}_{bc} + \mathcal{P}_b \delta_c^a - \mathcal{P}_c \delta_b^a \right),$	
1510.06699_ E00161 (176)	030	■ 0.929 ● N +1	$\begin{aligned} D_c^\dagger \left(h j^{\dagger c} \right) &= 0, \\ h t^{\dagger c}{}_c &= 0. \end{aligned}$	$\begin{aligned} D_c^\dagger (h j^{\dagger c}) &= 0, \\ h t^{\dagger c}{}_c &= 0. \end{aligned}$	
1510.06699_ E00112 (120)	022	■ 0.934 ● K +0	$J^a + \delta J^a = (J^\mu + \delta J^\mu) e^a{}_\mu (x + \delta x).$	$J^a + \delta J^a = (J^\mu + \delta J^\mu) e^a{}_\mu (x + \delta x).$	
1510.06699_ E00240 (A4)	047	■ 0.943 ● N +0	$\bar{\psi} \not{J} = \bar{\psi} \left(\bar{\psi} \psi + \bar{\psi} i \gamma^5 \psi i \gamma^5 \right).$	$\bar{\psi} \not{J} = \bar{\psi} \left(\bar{\psi} \psi + \bar{\psi} i \gamma^5 \psi i \gamma^5 \right).$	
1510.06699_ E00264 (D3)	051	■ 0.943 ● W +0	$\begin{aligned} L_{\mathrm{T}} &= L_{\mathrm{M}} + L_{\mathrm{G}} \\ &= L_{\varphi} + \frac{1}{2} \nu \mathcal{D}_a^* \phi \mathcal{D}^{*a} \phi - \lambda \phi^4 - \frac{1}{2} a \phi^2 \mathcal{R} - \frac{1}{4} \xi \mathcal{H}_{ab} \mathcal{H}^{ab}. \end{aligned}$	$\begin{aligned} L_T &= L_M + L_G \\ &= L_\varphi + \frac{1}{2} \nu \mathcal{D}_a^* \phi \mathcal{D}^{*a} \phi - \lambda \phi^4 - \frac{1}{2} a \phi^2 \mathcal{R} - \frac{1}{4} \xi \mathcal{H}_{ab} \mathcal{H}^{ab}. \end{aligned}$	
1510.06699_ E00019 (20)	009	■ 0.944 ● W +2	$\begin{aligned} D_{\{\mu\}}^{\{*\}} \varphi &\equiv \left[\partial_{\mu} + \frac{1}{2} A^{ab}{}_{\mu} (x) \Sigma_{ab} + w B_{\mu} (x) \right] \varphi (x) \\ &= [D_{\mu} + w B_{\mu} (x)] \varphi (x) \end{aligned}$	$\begin{aligned} D_{\mu}^* \varphi (x) &\equiv \left[\partial_{\mu} + \frac{1}{2} A^{ab}{}_{\mu} (x) \Sigma_{ab} + w B_{\mu} (x) \right] \varphi (x) \\ &= [D_{\mu} + w B_{\mu} (x)] \varphi (x), \end{aligned}$	
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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1510.06699_ E00167 (183)	031	0.946 W +0	$\begin{aligned} \mathrm{L}_{\mathrm{T}} &= \mathrm{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger) \\ &\quad + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, A^\dagger, \partial A^\dagger), \end{aligned}$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A^\dagger, \partial A^\dagger) + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, A^\dagger, \partial A^\dagger),$	
1510.06699_ E00038 (39)	011	0.946	$\begin{aligned} R^{\mathrm{ab}}_{\mu\nu} &\equiv \partial_\mu A^{\mathrm{ab}}_\nu - \partial_\nu A^{\mathrm{ab}}_\mu + A^a_{c\mu} A^{\mathrm{cb}}_\nu - A^a_{c\nu} A^{\mathrm{cb}}_\mu, \\ H_{\mu\nu} &\equiv \partial_\mu B_\nu - \partial_\nu B_\mu. \end{aligned}$	$R^{\mathrm{ab}}_{\mu\nu} \equiv \partial_\mu A^{\mathrm{ab}}_\nu - \partial_\nu A^{\mathrm{ab}}_\mu + A^a_{c\mu} A^{\mathrm{cb}}_\nu - A^a_{c\nu} A^{\mathrm{cb}}_\mu, \\ H_{\mu\nu} \equiv \partial_\mu B_\nu - \partial_\nu B_\mu.$	
1510.06699_ E00124 (136)	023	0.952 W +1	$\begin{aligned} h_a^{\prime\mu} &= e^{-\rho} h_a^\mu \\ A^{\prime\mathrm{ab}}_\mu &= A^{\mathrm{ab}}_\mu + \theta (b^a_\mu \mathcal{P}^b - b^b_\mu \mathcal{P}^a), \end{aligned}$	$h_a^{\prime\mu} = e^{-\rho} h_a^\mu \\ A^{\prime\mathrm{ab}}_\mu = A^{\mathrm{ab}}_\mu + \theta (b^a_\mu \mathcal{P}^b - b^b_\mu \mathcal{P}^a),$	
1510.06699_ E00221 (255a)	041	0.954 N +1	$\begin{aligned} \mathcal{T}^{\mathrm{a}}_{bc} &= e^{-\rho} \left\{ \mathcal{T}^{\mathrm{a}}_{bc} + 2(1 - \theta) \mathcal{P}_{[b} \delta^{\mathrm{a}}_{c]} \right\}, \\ \mathcal{T}'_b &= e^{-\rho} \{ \mathcal{T}_b + 3(1 - \theta) \mathcal{P}_b \}. \end{aligned}$	$\mathcal{T}'^{\mathrm{a}}_{bc} = e^{-\rho} \{ \mathcal{T}^{\mathrm{a}}_{bc} + 2(1 - \theta) \mathcal{P}_{[b} \delta^{\mathrm{a}}_{c]} \}, \\ \mathcal{T}'_b = e^{-\rho} \{ \mathcal{T}_b + 3(1 - \theta) \mathcal{P}_b \}.$	
1510.06699_ E00086 (92)	018	0.955 N +2	$\begin{aligned} \mathcal{D}_{\mathrm{a}}^* \chi &\equiv (\mathcal{D}_{\mathrm{a}} + w \mathcal{B}_{\mathrm{a}}) \chi \\ &= \left(\frac{\phi}{\phi_0} \right)^{1-w} (\widehat{\mathcal{D}}_{\mathrm{a}} + w \widehat{\mathcal{B}}_{\mathrm{a}}) \widehat{\chi} \equiv \left(\frac{\phi}{\phi_0} \right)^{1-w} \widehat{\mathcal{D}}_{\mathrm{a}}^* \widehat{\chi} \end{aligned}$	$\mathcal{D}_{\mathrm{a}}^* \chi \equiv (\mathcal{D}_{\mathrm{a}} + w \mathcal{B}_{\mathrm{a}}) \chi \\ = \left(\frac{\phi}{\phi_0} \right)^{1-w} (\widehat{\mathcal{D}}_{\mathrm{a}} + w \widehat{\mathcal{B}}_{\mathrm{a}}) \widehat{\chi} \equiv \left(\frac{\phi}{\phi_0} \right)^{1-w} \widehat{\mathcal{D}}_{\mathrm{a}}^* \widehat{\chi},$	
1510.06699_ E00068 (72)	015	0.956 N +0	$\mathrm{L}_{\mathrm{D}} = h^{-1} \left(\frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi \right).$	$\mathcal{L}_{\mathrm{D}} = h^{-1} (\tfrac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_a \psi - \mu \phi \bar{\psi} \psi).$	
1510.06699_ E00093 (101)	019	0.959 N +0	$v^{\mathrm{c}0} \mathcal{D}_{\mathrm{c}}^* p_{\mathrm{a}} = e \mu^2 \phi^0 \mathcal{D}_{\mathrm{a}}^* \phi,$	$v^{\mathrm{c}0} \mathcal{D}_{\mathrm{c}}^* p_{\mathrm{a}} = e \mu^2 \phi^0 \mathcal{D}_{\mathrm{a}}^* \phi,$	
1510.06699_ E00026 (27)	010	0.964 N +0	$h_{\mathrm{a}}^{\prime\mu} (x') = \frac{\partial x^{\prime\mu}}{\partial x^\nu} e^{-\rho(x)} \Lambda_{\mathrm{a}}^{\mathrm{b}} (x) h_{\mathrm{b}}^\nu (x).$	$h_{\mathrm{a}}^{\prime\mu} (x') = \frac{\partial x^{\prime\mu}}{\partial x^\nu} e^{-\rho(x)} \Lambda_{\mathrm{a}}^{\mathrm{b}} (x) h_{\mathrm{b}}^\nu (x).$	
1510.06699_ E00008 (9)	008	0.965 K +0	$\mathrm{L}_{\mathrm{M}}(\varphi'(x'), \partial'_\mu \varphi'(x')) = e^{-4\rho} \mathrm{L}_{\mathrm{M}}(\varphi(x), \partial_\mu \varphi(x)),$	$L_{\mathrm{M}}(\varphi'(x'), \partial'_\mu \varphi'(x')) = e^{-4\rho} L_{\mathrm{M}}(\varphi(x), \partial_\mu \varphi(x)),$	
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00057 (61b)	014	■ 0.971 ● C +3	$\begin{aligned} t^a{}_\mu + \tau^a{}_\mu &= 0 \\ s_{ab}{}^\mu + \sigma_{ab}{}^\mu &= 0 \\ j^\mu + \zeta^\mu &= 0 \end{aligned}$	$t^a{}_\mu + \tau^a{}_\mu = 0$ $s_{ab}{}^\mu + \sigma_{ab}{}^\mu = 0$ $j^\mu + \zeta^\mu = 0$	$t^a{}_\mu + \tau^a{}_\mu = 0,$ $s_{ab}{}^\mu + \sigma_{ab}{}^\mu = 0,$ $j^\mu + \zeta^\mu = 0.$
1510.06699_ E00250 (B5)	049	■ 0.973 ● K +0	$L_{\mathrm{G}} = -\kappa^{-1} \left(\Lambda + a^0 \mathcal{R} \right) + \alpha_1{}^0 \mathcal{R}^2 + \alpha_2{}^0 \mathcal{R}_{ab}{}^0 \mathcal{R}^{ab}.$	$L_G = -\kappa^{-1} \left(\Lambda + a^0 \mathcal{R} \right) + \alpha_1{}^0 \mathcal{R}^2 + \alpha_2{}^0 \mathcal{R}_{ab}{}^0 \mathcal{R}^{ab}.$	$L_G = -\kappa^{-1} \left(\Lambda + a^0 \mathcal{R} \right) + \alpha_1{}^0 \mathcal{R}^2 + \alpha_2{}^0 \mathcal{R}_{ab}{}^0 \mathcal{R}^{ab}.$
1510.06699_ E00152 (166)	028	■ 0.974 ● K +0	$L_{\mathrm{G}} = L_{\mathrm{H}} + L_{\mathrm{H}^{\dagger 2}}$	$L_G = L_{\mathcal{R}^{\dagger 2}} + L_{\mathcal{H}^{\dagger 2}},$	$L_G = L_{\mathcal{R}^{\dagger 2}} + L_{\mathcal{H}^{\dagger 2}},$
1510.06699_ E00228 (262)	043	■ 0.976 ●	$\{ \}^{\{0\}} \mathcal{D}_{\{c\}}^{\{\prime\}} \varphi^{\prime} = e^{(w-1)\rho} \left[{}^0\mathcal{D}_c \varphi + w \mathcal{P}_c \varphi + \mathcal{P}^{[b} \delta_c^{a]} \Sigma_{ab} \varphi \right],$	${}^0\mathcal{D}'_c \varphi' = e^{(w-1)\rho} \left[{}^0\mathcal{D}_c \varphi + w \mathcal{P}_c \varphi + \mathcal{P}^{[b} \delta_c^{a]} \Sigma_{ab} \varphi \right],$	${}^0\mathcal{D}'_c \varphi' = e^{(w-1)\rho} \left[{}^0\mathcal{D}_c \varphi + w \mathcal{P}_c \varphi + \mathcal{P}^{[b} \delta_c^{a]} \Sigma_{ab} \varphi \right],$
1510.06699_ E00220 (254b)	041	■ 0.979 ● W +2	$\begin{aligned} & \mathcal{R}^{ab}{}_{cd} = e^{-2\rho} \left\{ \mathcal{R}^{ab}{}_{cd} + 2\theta \delta_d^{[a} (\mathcal{D}_c - \theta \mathcal{P}_c) \mathcal{P}^{b]} - 2\theta \delta_c^{[a} (\mathcal{D}_d - \theta \mathcal{P}_d) \mathcal{P}^{b]} - 2\theta \mathcal{P}^{[a} \mathcal{T}^{b]}{}_{cd} - 2\theta^2 \delta_c^{[a} \delta_d^{b]} \mathcal{P}^e \mathcal{P}_e \right\}, \\ & \mathcal{R}'^a{}_c = e^{-2\rho} \left\{ \mathcal{R}'^a{}_c - 2\theta \mathcal{D}_c \mathcal{P}^a - \theta \delta_c^a \mathcal{D}_b \mathcal{P}^b + 2\theta^2 \mathcal{P}^a \mathcal{P}_c - 2\theta^2 \delta_c^a \mathcal{P}^2 + \theta \mathcal{P}^b \mathcal{T}^a{}_{cb} - \theta \mathcal{P}^a \mathcal{T}_c \right\}, \\ & \mathcal{R}' = e^{-2\rho} \left\{ \mathcal{R} - 6\theta \left(\mathcal{D}_a + \frac{1}{3} \mathcal{T}_a \right) \mathcal{P}^a - 6\theta^2 \mathcal{P}^2 \right\}, \end{aligned}$	$\mathcal{R}^{ab}{}_{cd} = e^{-2\rho} \left\{ \mathcal{R}^{ab}{}_{cd} + 2\theta \delta_d^{[a} (\mathcal{D}_c - \theta \mathcal{P}_c) \mathcal{P}^{b]} - 2\theta \delta_c^{[a} (\mathcal{D}_d - \theta \mathcal{P}_d) \mathcal{P}^{b]} - 2\theta \mathcal{P}^{[a} \mathcal{T}^{b]}{}_{cd} - 2\theta^2 \delta_c^{[a} \delta_d^{b]} \mathcal{P}^e \mathcal{P}_e \right\},$ $\mathcal{R}'^a{}_c = e^{-2\rho} \left\{ \mathcal{R}'^a{}_c - 2\theta \mathcal{D}_c \mathcal{P}^a - \theta \delta_c^a \mathcal{D}_b \mathcal{P}^b + 2\theta^2 \mathcal{P}^a \mathcal{P}_c - 2\theta^2 \delta_c^a \mathcal{P}^2 + \theta \mathcal{P}^b \mathcal{T}^a{}_{cb} - \theta \mathcal{P}^a \mathcal{T}_c \right\},$ $\mathcal{R}' = e^{-2\rho} \left\{ \mathcal{R} - 6\theta \left(\mathcal{D}_a + \frac{1}{3} \mathcal{T}_a \right) \mathcal{P}^a - 6\theta^2 \mathcal{P}^2 \right\},$	$\mathcal{R}^{ab}{}_{cd} = e^{-2\rho} \left\{ \mathcal{R}^{ab}{}_{cd} + 2\theta \delta_d^{[a} (\mathcal{D}_c - \theta \mathcal{P}_c) \mathcal{P}^{b]} - 2\theta \delta_c^{[a} (\mathcal{D}_d - \theta \mathcal{P}_d) \mathcal{P}^{b]} - 2\theta \mathcal{P}^{[a} \mathcal{T}^{b]}{}_{cd} - 2\theta^2 \delta_c^{[a} \delta_d^{b]} \mathcal{P}^e \mathcal{P}_e \right\},$ $\mathcal{R}'^a{}_c = e^{-2\rho} \left\{ \mathcal{R}'^a{}_c - 2\theta \mathcal{D}_c \mathcal{P}^a - \theta \delta_c^a \mathcal{D}_b \mathcal{P}^b + 2\theta^2 \mathcal{P}^a \mathcal{P}_c - 2\theta^2 \delta_c^a \mathcal{P}^2 + \theta \mathcal{P}^b \mathcal{T}^a{}_{cb} - \theta \mathcal{P}^a \mathcal{T}_c \right\},$ $\mathcal{R}' = e^{-2\rho} \left\{ \mathcal{R} - 6\theta \left(\mathcal{D}_a + \frac{1}{3} \mathcal{T}_a \right) \mathcal{P}^a - 6\theta^2 \mathcal{P}^2 \right\},$
1510.06699_ E00120 (131)	023	■ 0.979 ● W +1	$\begin{aligned} & \{ \}^{\{0\}} \Gamma^{\{*} \lambda \} \{ \}^{\{ \mu \} \nu} \\ & \&= \frac{1}{2} g^{\lambda \rho} \left(\partial_\mu^* g_{\nu \rho} + \partial_\nu^* g_{\mu \rho} - \partial_\rho^* g_{\mu \nu} \right) \\ & g_{\{ \nu \} \rho} + \partial_{\{ \nu \} \rho}^* \{ \}^{\{ \mu \} \rho} - \partial_{\{ \mu \} \rho}^* \{ \}^{\{ \nu \} \rho} \} g_{\{ \mu \} \nu} \right) \&= \{ \}^{\{0\}} \Gamma^{\{*} \lambda \} \{ \}^{\{ \mu \} \nu} + \delta_{\nu}^{\lambda} B_{\mu} + \delta_{\mu}^{\lambda} B_{\nu} - g_{\mu \nu} B^{\lambda} \} \\ & \end{aligned}$	${}^0\Gamma^{*\lambda}{}_{\mu\nu} = \frac{1}{2} g^{\lambda\rho} \left(\partial_\mu^* g_{\nu\rho} + \partial_\nu^* g_{\mu\rho} - \partial_\rho^* g_{\mu\nu} \right)$ $= {}^0\Gamma^\lambda{}_{\mu\nu} + \delta_\nu^\lambda B_\mu + \delta_\mu^\lambda B_\nu - g_{\mu\nu} B^\lambda$	${}^0\Gamma^{*\lambda}{}_{\mu\nu} = \frac{1}{2} g^{\lambda\rho} \left(\partial_\mu^* g_{\nu\rho} + \partial_\nu^* g_{\mu\rho} - \partial_\rho^* g_{\mu\nu} \right)$ $= {}^0\Gamma^\lambda{}_{\mu\nu} + \delta_\nu^\lambda B_\mu + \delta_\mu^\lambda B_\nu - g_{\mu\nu} B^\lambda,$
1510.06699_ E00110 (118)	022	■ 0.979 ● N +1	$D_{\mu} J^a = \partial_{\mu} J^a + w B_{\mu} J^a + A^a{}_{b\mu} J^b = \partial_{\mu}^* J^a + A^a{}_{b\mu} J^b,$	$D_{\mu}^* J^a = \partial_{\mu} J^a + w B_{\mu} J^a + A^a{}_{b\mu} J^b = \partial_{\mu}^* J^a + A^a{}_{b\mu} J^b,$	$D_{\mu}^* J^a = \partial_{\mu} J^a + w B_{\mu} J^a + A^a{}_{b\mu} J^b = \partial_{\mu}^* J^a + A^a{}_{b\mu} J^b,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00233 (267)	044	0.982 C +5	$\begin{aligned} & \{ \}^{\{0\}} \mathcal{C}_{\{a\ b\ c\ d\}} = \{ \}^{\{0\}} \mathcal{R}_{\{a\ b\ c\ d\}} & - \frac{1}{2} \left(\eta_{ac} \mathcal{R}_{bd} - \eta_{ad} \mathcal{R}_{bc} - \eta_{bc} \mathcal{R}_{ad} + \eta_{bd} \mathcal{R}_{ac} \right) \\ & + \frac{1}{6} \left(\eta_{ac} \eta_{bd} - \eta_{ad} \eta_{bc} \right)^0 \mathcal{R}. \end{aligned}$	${}^0\mathcal{C}_{abcd} = {}^0\mathcal{R}_{abcd} - \frac{1}{2} \left(\eta_{ac} {}^0\mathcal{R}_{bd} - \eta_{ad} {}^0\mathcal{R}_{bc} - \eta_{bc} {}^0\mathcal{R}_{ad} + \eta_{bd} {}^0\mathcal{R}_{ac} \right) + \frac{1}{6} \left(\eta_{ac} \eta_{bd} - \eta_{ad} \eta_{bc} \right)^0 \mathcal{R}.$	${}^0\mathcal{C}_{abcd} = {}^0\mathcal{R}_{abcd} - \frac{1}{2} \left(\eta_{ac} {}^0\mathcal{R}_{bd} - \eta_{ad} {}^0\mathcal{R}_{bc} - \eta_{bc} {}^0\mathcal{R}_{ad} + \eta_{bd} {}^0\mathcal{R}_{ac} \right) + \frac{1}{6} \left(\eta_{ac} \eta_{bd} - \eta_{ad} \eta_{bc} \right)^0 \mathcal{R}. \quad (267)$
1510.06699_ E00184 (200)	034	0.982 N +0	$\phi\ v^{\{b\}\ \}\{0\}} \mathcal{D}_{\{b\}^{\{\dagger\}} v_a} = \left(\delta_{\{a\}^{\{b\}} - v_a v^{\{b\}} \right)^{\{0\}} \mathcal{D}_{\{b\}^{\{\dagger\}} \phi}$	$\phi v^{b\ 0} \mathcal{D}_b^\dagger v_a = \left(\delta_a^b - v_a v^b \right)^0 \mathcal{D}_b^\dagger \phi.$	$\phi v^{b\ 0} \mathcal{D}_b^\dagger v_a = \left(\delta_a^b - v_a v^b \right)^0 \mathcal{D}_b^\dagger \phi.$
1510.06699_ E00243 (A7)	047	0.983 K +0	$\nexists \partial \psi = J^\mu \partial_\mu \psi - i \sigma^{\mu \nu} J_\mu \partial_\nu \psi.$	$\nexists \partial \psi = J^\mu \partial_\mu \psi - i \sigma^{\mu \nu} J_\mu \partial_\nu \psi.$	$\nexists \partial \psi = J^\mu \partial_\mu \psi - i \sigma^{\mu \nu} J_\mu \partial_\nu \psi.$
1510.06699_ E00118 (129)	023	0.984 N +0	$\left[\nabla_{\mu}^*,\nabla_{\nu}^*\right]J^{\rho}=\widetilde{R}^{\rho}_{\sigma\mu\nu}J^{\sigma}+wH_{\mu\nu}J^{\rho}-T^{*\sigma}{}_{\mu\nu}\nabla_{\sigma}^*V^{\rho}.$	$[\nabla_{\mu}^*,\nabla_{\nu}^*]J^{\rho}=\widetilde{R}^{\rho}_{\sigma\mu\nu}J^{\sigma}+wH_{\mu\nu}J^{\rho}-T^{*\sigma}{}_{\mu\nu}\nabla_{\sigma}^*V^{\rho}.$	$[\nabla_{\mu}^*,\nabla_{\nu}^*]J^{\rho}=\widetilde{R}^{\rho}_{\sigma\mu\nu}J^{\sigma}+wH_{\mu\nu}J^{\rho}-T^{*\sigma}{}_{\mu\nu}\nabla_{\sigma}^*V^{\rho}.$
1510.06699_ E00111 (119)	022	0.984	$\Delta_{\mu}^*\equiv\partial_{\mu}^*+\Gamma_{\rho\mu}^{\sigma}X_{\sigma}^{\rho}+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab}=\nabla_{\mu}^*+D_{\mu}^*-\partial_{\mu}^*,$	$\Delta_{\mu}^*\equiv\partial_{\mu}^*+\Gamma_{\rho\mu}^{\sigma}X_{\sigma}^{\rho}+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab}=\nabla_{\mu}^*+D_{\mu}^*-\partial_{\mu}^*,$	$\Delta_{\mu}^*\equiv\partial_{\mu}^*+\Gamma_{\rho\mu}^{\sigma}X_{\sigma}^{\rho}+\frac{1}{2}A^{ab}{}_{\mu}\Sigma_{ab}=\nabla_{\mu}^*+D_{\mu}^*-\partial_{\mu}^*,$
1510.06699_ E00018 (19)	009	0.985 N +0	$\partial_{\mu}^{\prime}\varphi^{\prime}(x^{\prime})=\frac{\partial x^{\nu}}{\partial x^{\prime\mu}}e^{w\rho(x)}[S(\Lambda(x))\partial_{\nu}\varphi(x)+\partial_{\nu}S(\Lambda(x))\varphi(x)+w\partial_{\nu}\rho S(\Lambda(x))\varphi(x)].$	$\partial_{\mu}^{\prime}\varphi^{\prime}(x^{\prime})=\frac{\partial x^{\nu}}{\partial x^{\prime\mu}}e^{w\rho(x)}[S(\Lambda(x))\partial_{\nu}\varphi(x)+\partial_{\nu}S(\Lambda(x))\varphi(x)+w\partial_{\nu}\rho S(\Lambda(x))\varphi(x)].$	$\partial_{\mu}^{\prime}\varphi^{\prime}(x^{\prime})=\frac{\partial x^{\nu}}{\partial x^{\prime\mu}}e^{w\rho(x)}[S(\Lambda(x))\partial_{\nu}\varphi(x)+\partial_{\nu}S(\Lambda(x))\varphi(x)+w\partial_{\nu}\rho S(\Lambda(x))\varphi(x)].$
1510.06699_ E00033 (34)	010	0.985 N +0	$D_{\mu}^*b^a{}_{\nu}=\partial_{\mu}^*b^a{}_{\nu}+A^a{}_{b\mu}b^b{}_{\nu}.$	$D_{\mu}^*b^a{}_{\nu}=\partial_{\mu}^*b^a{}_{\nu}+A^a{}_{b\mu}b^b{}_{\nu}.$	$D_{\mu}^*b^a{}_{\nu}=\partial_{\mu}^*b^a{}_{\nu}+A^a{}_{b\mu}b^b{}_{\nu}.$
1510.06699_ E00175 (188)	032	0.985 N +0	$\tau^{\dagger a}{}_b=\tau^a{}_b+2\sigma_{cb}{}^a\mathcal{V}^c.$	$\tau^{\dagger a}{}_b=\tau^a{}_b+2\sigma_{cb}{}^a\mathcal{V}^c.$	$\tau^{\dagger a}{}_b=\tau^a{}_b+2\sigma_{cb}{}^a\mathcal{V}^c.$
1510.06699_ E00080 (85)	017	0.986 N +1	$\mathcal{D}_{[a}^*\mathcal{F}_{bc]}-\mathcal{T}^{*d}{}_{[ab}\mathcal{F}_{c]d}=0.$	$\mathcal{D}_{[a}^*\mathcal{F}_{bc]}-\mathcal{T}^{*d}{}_{[ab}\mathcal{F}_{c]d}=0.$	$\mathcal{D}_{[a}^*\mathcal{F}_{bc]}-\mathcal{T}^{*d}{}_{[ab}\mathcal{F}_{c]d}=0.$
1510.06699_ E00188 (204)	035	0.987 W +0	$\begin{array}{l} \mathcal{L}_{\{L\}_{\{\mathrm{M}\}}=h^{\{-1\}}\left[\frac{1}{2}i\bar{\psi}\gamma^{a0}\stackrel{\leftrightarrow}{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu\left({}^0\mathcal{D}_a^\dagger\phi\right)\left({}^0\mathcal{D}^{\dagger a}\phi\right)-\lambda\phi^4\right.\\ \left.-a\phi^{2\ 0}\mathcal{R}^\dagger\right].(204) \end{array}$	$\mathcal{L}_{\mathrm{M}}=h^{-1}\Big[\frac{1}{2}i\bar{\psi}\gamma^{a0}\stackrel{\leftrightarrow}{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu\left({}^0\mathcal{D}_a^\dagger\phi\right)\left({}^0\mathcal{D}^{\dagger a}\phi\right)-\lambda\phi^4-a\phi^{2\ 0}\mathcal{R}^\dagger\Big].(204)$	$\mathcal{L}_{\mathrm{M}}=h^{-1}[\frac{1}{2}i\bar{\psi}\gamma^{a\ 0}\stackrel{\leftrightarrow}{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu({}^0\mathcal{D}_a^\dagger\phi)({}^0\mathcal{D}^{\dagger a}\phi)-\lambda\phi^4-a\phi^{2\ 0}\mathcal{R}^\dagger].(204)$
1510.06699_ E00106 (114)	021	0.987 N +0	$\hat{\boldsymbol{e}}_{\{\boldsymbol{a}\}}=h_{\{a\}\ \}\{0\}} \boldsymbol{e}_{\{\mu\}}, \quad \boldsymbol{e}_{\{\mu\}}=\boldsymbol{b}^a{}_{\mu}\hat{\boldsymbol{e}}_a,$	$\hat{e}_a=h_a{}^\mu e_\mu, \quad e_\mu=b^a{}_\mu \hat{e}_a,$	$\hat{e}_a=h_a{}^\mu e_\mu, \quad e_\mu=b^a{}_\mu \hat{e}_a,$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0154 (168b)	029	■ 0.988 ● C +3	$\begin{aligned} t^a{}_b + \tau^a{}_b + \tau^a{}_b &= 0 \\ s_{ab}{}^c + \sigma_{ab}{}^c &= 0 \\ j^a + \zeta^a &= 0 \end{aligned}$	$t^a{}_b + \tau^a{}_b = 0$ $s_{ab}{}^c + \sigma_{ab}{}^c = 0$ $j^a + \zeta^a = 0$	$t^a{}_b + \tau^a{}_b = 0,$ $s_{ab}{}^c + \sigma_{ab}{}^c = 0,$ $j^a + \zeta^a = 0,$
1510.06699_ EQ0204 (224)	038	■ 0.988 ● C +6	$L_{\varphi} + L_{\phi} + \phi^2 L_{\mathcal{R}^\dagger} + \phi^2 L_{\mathcal{T}^\dagger 2}$	$L_{\varphi} + L_{\phi} + \phi^2 L_{\mathcal{R}^\dagger} + \phi^2 L_{\mathcal{T}^\dagger 2}$	$L_{\varphi} + L_{\phi} + \phi^2 L_{\mathcal{R}^\dagger} + \phi^2 L_{\mathcal{T}^\dagger 2}, \text{ where}$
1510.06699_ EQ0073 (77)	016	■ 0.990 ● N +2	$\begin{aligned} h \tau^a{}_b &= \frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_b \psi - \delta_b^a L_D \\ h \sigma_{abc} &= \frac{1}{4} i \bar{\psi} \gamma_{[a} \gamma_b \gamma_{c]} \psi \end{aligned}$	$h \tau^a{}_b = \frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_b \psi - \delta_b^a L_D$ $h \sigma_{abc} = \frac{1}{4} i \bar{\psi} \gamma_{[a} \gamma_b \gamma_{c]} \psi$	$h \tau^a{}_b = \frac{1}{2} i \bar{\psi} \gamma^a \overleftrightarrow{\mathcal{D}}_b \psi - \delta_b^a L_D,$ $h \sigma_{abc} = \frac{1}{4} i \bar{\psi} \gamma_{[a} \gamma_b \gamma_{c]} \psi,$
1510.06699_ EQ0163 (179)	030	■ 0.990 ● N +0	$\zeta^{\dagger a} \equiv \zeta^a - 2 \sigma^{ab}{}_b,$	$\zeta^{\dagger a} \equiv \zeta^a - 2 \sigma^{ab}{}_b,$	$\zeta^{\dagger a} \equiv \zeta^a - 2 \sigma^{ab}{}_b,$
1510.06699_ EQ0247 (B2)	048	■ 0.991 ● K +0	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, A, \partial A) + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, h, A),$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, A, \partial A) + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, h, A),$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, A, \partial A) + \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, h, A),$
1510.06699_ EQ0189 (205)	035	■ 0.991 ●	$\begin{aligned} \mathcal{L}_{\mathrm{T}} &= \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A, \partial A, V, \partial V) \\ &+ \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A, \partial A, V, \partial V). \end{aligned}$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$ $+ \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A, \partial A, V, \partial V).$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h, \partial h, \partial^2 h, A, \partial A, V, \partial V)$ $+ \mathcal{L}_{\mathrm{M}}(\varphi, \partial \varphi, \phi, \partial \phi, h, \partial h, \partial^2 h, A, \partial A, V, \partial V).$
1510.06699_ EQ0191 (207)	036	■ 0.992 ● N +0	$\delta J^a = - \left[A^{\dagger a}{}_{b\mu} - w \left(V_{\mu} + \frac{1}{3} T_{\mu} \right) \delta_b^a \right] J^b \delta x^{\mu}.$	$\delta J^a = - \left[A^{\dagger a}{}_{b\mu} - w \left(V_{\mu} + \frac{1}{3} T_{\mu} \right) \delta_b^a \right] J^b \delta x^{\mu}.$	$\delta J^a = - [A^{\dagger a}{}_{b\mu} - w (V_{\mu} + \frac{1}{3} T_{\mu}) \delta_b^a] J^b \delta x^{\mu}.$
1510.06699_ EQ0015 (16)	008	■ 0.993 ● N +0	$\begin{aligned} L'_{\mathrm{EM}} &= -\frac{1}{4} F'_{\mu\nu} F'^{\mu\nu} - J'^{\mu} A'_{\mu} \\ &= -\frac{1}{4} e^{2(w-1)\rho} F_{\mu\nu} F^{\mu\nu} - e^{(w+w_J)\rho} J^{\mu} A_{\mu}, \end{aligned}$	$L'_{\mathrm{EM}} = -\frac{1}{4} F'_{\mu\nu} F'^{\mu\nu} - J'^{\mu} A'_{\mu}$ $= -\frac{1}{4} e^{2(w-1)\rho} F_{\mu\nu} F^{\mu\nu} - e^{(w+w_J)\rho} J^{\mu} A_{\mu},$	$L'_{\mathrm{EM}} = -\frac{1}{4} F'_{\mu\nu} F'^{\mu\nu} - J'^{\mu} A'_{\mu}$ $= -\frac{1}{4} e^{2(w-1)\rho} F_{\mu\nu} F^{\mu\nu} - e^{(w+w_J)\rho} J^{\mu} A_{\mu},$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00065 (69b)	015	■0.993 ●W +1	$\begin{aligned} & \left(\mathrm{D}_{\mathrm{c}}^* \right) \left(\mathrm{T}_{\mathrm{c}}^* \right) \left(h \sigma_{ab}^c \right) + h \tau_{[ab]} + \frac{1}{2} \frac{\delta L_{\mathrm{M}}}{\delta \varphi} \Sigma_{ab} \varphi = 0, \\ & \left(\mathcal{D}_c^* + \mathcal{T}_c^*\right) \left(h \tau^c{}_d\right) - h \left(\sigma_{ab}{}^c \mathcal{R}^{ab}{}_{cd} - \tau^c{}_b \mathcal{T}^{*b}{}_{cd} + \zeta^c \mathcal{H}_{cd}\right) + \frac{\delta L_{\mathrm{M}}}{\delta \phi} \mathcal{D}_d^* \phi + \frac{\delta L_{\mathrm{M}}}{\delta \varphi} \mathcal{D}_d^* \varphi = 0, \\ & \left(\mathcal{D}_c^* + \mathcal{T}_c^*\right) \left(h \zeta^c\right) - h \tau^c{}_c - \frac{\delta L_{\mathrm{M}}}{\delta \phi} \phi + \frac{\delta L_{\mathrm{M}}}{\delta \varphi} w \varphi = 0. \end{aligned}$	$\begin{aligned} & \left(\mathcal{D}_c^* + \mathcal{T}_c^*\right) \left(h \sigma_{ab}^c \right) + h \tau_{[ab]} + \frac{1}{2} \frac{\delta L_{\mathrm{M}}}{\delta \varphi} \Sigma_{ab} \varphi = 0, \\ & \left(\mathcal{D}_c^* + \mathcal{T}_c^*\right) \left(h \tau^c{}_d\right) - h \left(\sigma_{ab}{}^c \mathcal{R}^{ab}{}_{cd} - \tau^c{}_b \mathcal{T}^{*b}{}_{cd} + \zeta^c \mathcal{H}_{cd}\right) + \frac{\delta L_{\mathrm{M}}}{\delta \phi} \mathcal{D}_d^* \phi + \frac{\delta L_{\mathrm{M}}}{\delta \varphi} \mathcal{D}_d^* \varphi = 0, \\ & \left(\mathcal{D}_c^* + \mathcal{T}_c^*\right) \left(h \zeta^c\right) - h \tau^c{}_c - \frac{\delta L_{\mathrm{M}}}{\delta \phi} \phi + \frac{\delta L_{\mathrm{M}}}{\delta \varphi} w \varphi = 0. \end{aligned}$	$\begin{aligned} & \left(\mathcal{D}_c^* + \mathcal{T}_c^*\right) \left(h \sigma_{ab}^c \right) + h \tau_{[ab]} + \frac{1}{2} \frac{\delta L_{\mathrm{M}}}{\delta \varphi} \Sigma_{ab} \varphi = 0, \\ & \left(\mathcal{D}_c^* + \mathcal{T}_c^*\right) \left(h \tau^c{}_d\right) - h \left(\sigma_{ab}{}^c \mathcal{R}^{ab}{}_{cd} - \tau^c{}_b \mathcal{T}^{*b}{}_{cd} + \zeta^c \mathcal{H}_{cd}\right) + \frac{\delta L_{\mathrm{M}}}{\delta \phi} \mathcal{D}_d^* \phi + \frac{\delta L_{\mathrm{M}}}{\delta \varphi} \mathcal{D}_d^* \varphi = 0, \\ & \left(\mathcal{D}_c^* + \mathcal{T}_c^*\right) \left(h \zeta^c\right) - h \tau^c{}_c - \frac{\delta L_{\mathrm{M}}}{\delta \phi} \phi + \frac{\delta L_{\mathrm{M}}}{\delta \varphi} w \varphi = 0. \end{aligned}$
1510.06699_ E00049 (52)	013	■0.993 ●C +3	$\begin{aligned} & \left(\mathrm{D}_{\mathrm{a}}^* \right) \left(\mathcal{R}^{de}{}_{bc}\right) - \mathcal{T}^{*f}{}_{[ab} \mathcal{R}^{de}{}_{c]f} = 0, \\ & \left(\mathcal{D}_{[a} \mathcal{T}^{*d}{}_{bc]}\right) - \mathcal{T}^{*e}{}_{[ab} \mathcal{T}^{*d}{}_{c]e} - \mathcal{R}^d{}_{[abc]} + \mathcal{H}_{[ab} \delta_{c]}^d = 0, \\ & \left(\mathcal{D}_{[a} \mathcal{H}_{bc]}\right) - \mathcal{T}^{*e}{}_{[ab} \mathcal{H}_{c]e} = 0. \end{aligned}$	$\begin{aligned} & \mathcal{D}_{[a}^* \mathcal{R}^{de}{}_{bc]} - \mathcal{T}^{*f}{}_{[ab} \mathcal{R}^{de}{}_{c]f} = 0, \\ & \mathcal{D}_{[a}^* \mathcal{T}^{*d}{}_{bc]} - \mathcal{T}^{*e}{}_{[ab} \mathcal{T}^{*d}{}_{c]e} - \mathcal{R}^d{}_{[abc]} + \mathcal{H}_{[ab} \delta_{c]}^d = 0, \\ & \mathcal{D}_{[a}^* \mathcal{H}_{bc]} - \mathcal{T}^{*e}{}_{[ab} \mathcal{H}_{c]e} = 0. \end{aligned}$	$\begin{aligned} & \mathcal{D}_{[a}^* \mathcal{R}^{de}{}_{bc]} - \mathcal{T}^{*f}{}_{[ab} \mathcal{R}^{de}{}_{c]f} = 0, \\ & \mathcal{D}_{[a}^* \mathcal{T}^{*d}{}_{bc]} - \mathcal{T}^{*e}{}_{[ab} \mathcal{T}^{*d}{}_{c]e} - \mathcal{R}^d{}_{[abc]} + \mathcal{H}_{[ab} \delta_{c]}^d = 0, \\ & \mathcal{D}_{[a}^* \mathcal{H}_{bc]} - \mathcal{T}^{*e}{}_{[ab} \mathcal{H}_{c]e} = 0. \end{aligned}$
1510.06699_ E00217 (250)	040	■0.995 ●	$\begin{aligned} & \left(\delta_{\phi} L_{\phi} = -\nu \mathcal{D}_a^{\dagger} \left(\mathcal{D}^{\dagger a} \phi\right) - 4 \lambda \phi^3\right) \\ & \left(\partial_{\phi} L_{\mathcal{R}} = -a \phi^2 \mathcal{R}^{\dagger},\right) \\ & \left(\partial_{\phi} L_{\mathcal{T}^2} = 2 \beta_1 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} + 2 \beta_2 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}.\right) \end{aligned}$	$\begin{aligned} & \delta_{\phi} L_{\phi} = -\nu \mathcal{D}_a^{\dagger} \left(\mathcal{D}^{\dagger a} \phi\right) - 4 \lambda \phi^3. \\ & \partial_{\phi} L_{\mathcal{R}} = -a \phi^2 \mathcal{R}^{\dagger}, \\ & \partial_{\phi} L_{\mathcal{T}^2} = 2 \beta_1 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} + 2 \beta_2 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}. \end{aligned}$	$\begin{aligned} & \delta_{\phi} L_{\phi} = -\nu \mathcal{D}_a^{\dagger} \left(\mathcal{D}^{\dagger a} \phi\right) - 4 \lambda \phi^3. \\ & \partial_{\phi} L_{\mathcal{R}} = -a \phi^2 \mathcal{R}^{\dagger}, \\ & \partial_{\phi} L_{\mathcal{T}^2} = 2 \beta_1 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger abc} + 2 \beta_2 \phi \mathcal{T}_{abc}^{\dagger} \mathcal{T}^{\dagger bac}. \end{aligned}$

Conf.

MathPix confidence:

■ ≥0.9

■ 0.5-0.9

■ <0.5

— no value

LaTeX source

MathPix's reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: basis]

beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0126 (139)	024	0.995 C +8	$\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\wedge\{\mathrm{\prime}\mathrm{a}\}\mathrm{b}\}\{\}_{\mathrm{c}}\mathrm{d}\} \& \mathrm{=}\mathrm{e}^{\wedge\{-2\}\mathrm{\rho}\}\backslash\mathrm{left}\backslash\{\backslash\mathrm{mathcal}\{\mathrm{R}\}^{\wedge\{\mathrm{a}\}\mathrm{b}\}\{\}_{\mathrm{c}}\mathrm{d}\}+2\backslash\mathrm{theta}\backslash\mathrm{delta}_{\mathrm{d}}\}^{\wedge\{[\mathrm{a}]\}\backslash\mathrm{left}\{\backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{c}}\}-\backslash\mathrm{theta}\backslash\mathrm{mathcal}\{\mathrm{P}\}_{\mathrm{c}}\}\backslash\mathrm{right}\}\backslash\mathrm{mathcal}\{\mathrm{P}\}^{\wedge\{\mathrm{b}\}\}-2\backslash\mathrm{theta}\backslash\mathrm{delta}_{\mathrm{c}}\}^{\wedge\{[\mathrm{a}]\}\backslash\mathrm{left}\{\backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{d}}\}-\backslash\mathrm{theta}\backslash\mathrm{mathcal}\{\mathrm{P}\}_{\mathrm{d}}\}\backslash\mathrm{right}\}\backslash\mathrm{mathcal}\{\mathrm{P}\}^{\wedge\{\mathrm{b}\}\}-2\backslash\mathrm{theta}\backslash\mathrm{mathcal}\{\mathrm{P}\}^{\wedge\{[\mathrm{a}]\}\backslash\mathrm{mathcal}\{\mathrm{T}\}^{\wedge\{\mathrm{b}\}\}\}\{\}_{\mathrm{c}}\mathrm{d}\}-2\backslash\mathrm{theta}^{\wedge\{2\}}\backslash\mathrm{delta}_{\mathrm{c}}\}^{\wedge\{[\mathrm{a}]\}\backslash\mathrm{delta}_{\mathrm{d}}\}^{\wedge\{\mathrm{b}\}\}\}\backslash\mathrm{mathcal}\{\mathrm{P}\}^{\wedge\{\mathrm{e}\}}\backslash\mathrm{mathcal}\{\mathrm{P}\}_{\mathrm{e}}\}\backslash\mathrm{right}\backslash\}, \backslash\backslash \backslash\mathrm{mathcal}\{\mathrm{T}\}^{\wedge\{\mathrm{\prime}\mathrm{a}\}\}\{\}_{\mathrm{b}}\mathrm{c}\} \& \mathrm{=}\mathrm{e}^{\wedge\{-\mathrm{\rho}\}\}\backslash\mathrm{left}\backslash\{\backslash\mathrm{mathcal}\{\mathrm{T}\}^{\wedge\{\mathrm{a}\}\}\{\}_{\mathrm{b}}\mathrm{c}\}+2(1-\backslash\mathrm{theta})\backslash\mathrm{mathcal}\{\mathrm{P}\}_{[\mathrm{b}]\backslash\mathrm{delta}_{\mathrm{c}}\}}^{\wedge\{\mathrm{a}\}}\}\backslash\mathrm{right}\backslash\} . \\ & \backslash\mathrm{end}\{\mathrm{aligned}\} \end{aligned}$	$\mathcal{R}^{\prime ab}_{cd} = e^{-2\rho} \Big\{ \mathcal{R}^{ab}_{cd} + 2\theta \delta^{[a}_d (\mathcal{D}_c - \theta \mathcal{P}_c) \mathcal{P}^{b]} - 2\theta \delta^{[a}_c (\mathcal{D}_d - \theta \mathcal{P}_d) \mathcal{P}^{b]} - 2\theta \mathcal{P}^{[a} \mathcal{T}^{b]}_{cd} - 2\theta^2 \delta^{[a}_c \delta^{b]}_d \mathcal{P}^e \mathcal{P}_e \Big\},$ $\mathcal{T}^{\prime a}_{bc} = e^{-\rho} \Big\{ \mathcal{T}^a_{bc} + 2(1-\theta) \mathcal{P}_{[b} \delta^a_{c]} \Big\}.$	$\mathcal{R}^{\prime ab}_{cd} = e^{-2\rho} \{ \mathcal{R}^{ab}_{cd} + 2\theta \delta^{[a}_d (\mathcal{D}_c - \theta \mathcal{P}_c) \mathcal{P}^{b]} - 2\theta \delta^{[a}_c (\mathcal{D}_d - \theta \mathcal{P}_d) \mathcal{P}^{b]} - 2\theta \mathcal{P}^{[a} \mathcal{T}^{b]}_{cd} - 2\theta^2 \delta^{[a}_c \delta^{b]}_d \mathcal{P}^e \mathcal{P}_e \},$ $\mathcal{T}^{\prime a}_{bc} = e^{-\rho} \{ \mathcal{T}^a_{bc} + 2(1-\theta) \mathcal{P}_{[b} \delta^a_{c]} \}.$
1510.06699_ EQ0021 (22)	009	0.996 N +0	$\mathrm{D}_{\mathrm{\mu}}^{\mathrm{\prime}\ast}\mathrm{\varphi}^{\mathrm{\prime}}(\mathrm{x}^{\mathrm{\prime}})\backslash\mathrm{varphi}^{\mathrm{\prime}\ast}\backslash\mathrm{left}(\mathrm{x}^{\mathrm{\prime}\ast}\backslash\mathrm{right})=\mathrm{\frac{\partial\mathrm{x}^{\mathrm{\nu}}}{\partial\mathrm{x}^{\mathrm{\prime}\mu}}}\mathrm{e}^{w\rho(\mathrm{x})}\mathrm{S}(\Lambda(\mathrm{x}))\mathrm{D}_{\mathrm{\nu}}^{\ast}\mathrm{\varphi}(\mathrm{x}).$	$D_{\mu}^{*\prime} \varphi'(x') = \frac{\partial x^{\nu}}{\partial x'^{\mu}} e^{w\rho(x)} S(\Lambda(x)) D_{\nu}^* \varphi(x).$	$D_{\mu}^{*\prime} \varphi'(x') = \frac{\partial x^{\nu}}{\partial x'^{\mu}} e^{w\rho(x)} S(\Lambda(x)) D_{\nu}^* \varphi(x).$
1510.06699_ EQ0165 (181)	030	0.996 N +0	$\backslash\mathrm{mathcal}\{\mathrm{T}\}^{\wedge\{\mathrm{\natural}\mathrm{a}\}\}\{\}_{\mathrm{b}}\mathrm{c}\} \backslash\mathrm{equiv}\mathrm{h}_{\mathrm{b}}^{\natural}\{\}_{\mathrm{c}}^{\natural}\{\mathrm{\mu}\}\mathrm{h}_{\mathrm{c}}^{\natural}\{\}_{\mathrm{b}}^{\natural}\{\}\backslash\mathrm{left}(\mathrm{D}_{\mathrm{\mu}}^{\wedge\{\mathrm{\natural}\}\}\mathrm{b}^{\wedge\{\mathrm{a}\}\}\{\}_{\mathrm{c}}^{\wedge\{\mathrm{\natural}\}\}\mathrm{D}_{\mathrm{\nu}}^{\wedge\{\mathrm{\natural}\}\}\mathrm{b}^{\wedge\{\mathrm{a}\}\}\{\}_{\mathrm{c}}^{\wedge\{\mathrm{\natural}\}\}\}\backslash\mathrm{right})$	$\mathcal{T}^{\natural a}_{bc} \equiv h_b{}^{\mu} h_c{}^{\nu} \left(D_{\mu}^{\natural} b^a{}_{\nu} - D_{\nu}^{\natural} b^a{}_{\mu} \right),$	$\mathcal{T}^{\natural a}_{bc} \equiv h_b{}^{\mu} h_c{}^{\nu} (D_{\mu}^{\natural} b^a{}_{\nu} - D_{\nu}^{\natural} b^a{}_{\mu}),$
1510.06699_ EQ0215 (246)	039	0.997 N -1	$\partial_{\phi} L_{\mathcal{R}^2} + \partial_{\phi} L_{\mathcal{H}^2} + \partial_{\phi} L_{\varphi} + \delta_{\phi} L_{\phi} + \partial_{\phi} L_{\mathcal{R}} + \partial_{\phi} L_{\mathcal{T}^2} = 0,$	$\partial_{\phi} L_{\mathcal{R}^2} + \partial_{\phi} L_{\mathcal{H}^2} + \partial_{\phi} L_{\varphi} + \delta_{\phi} L_{\phi} + \partial_{\phi} L_{\mathcal{R}} + \partial_{\phi} L_{\mathcal{T}^2} = 0,$	$\partial_{\phi} L_{\mathcal{R}^2} + \partial_{\phi} L_{\mathcal{H}^2} + \partial_{\phi} L_{\varphi} + \delta_{\phi} L_{\phi} + \partial_{\phi} L_{\mathcal{R}} + \partial_{\phi} L_{\mathcal{T}^2} = 0,$
1510.06699_ EQ0242 (A6)	047	0.997 N -1	$\mathrm{L}_{\mathrm{\mathrm{mathrm}\{D\}}}=\mathrm{\frac{\Re\left[i\bar{\psi}\left(\bar{\psi}\psi+\bar{\psi}i\gamma^5\psi i\gamma^5\right),\mathbb{J}\partial\psi\right]}{\left(\bar{\psi}\psi\right)^2+\left(\bar{\psi}i\gamma^5\psi\right)^2}}-m\bar{\psi}\psi.$	$L_{\mathrm{D}} = \frac{\Re\left[i\bar{\psi}\left(\bar{\psi}\psi+\bar{\psi}i\gamma^5\psi i\gamma^5\right)\not{\partial}\psi\right]}{(\bar{\psi}\psi)^2+(\bar{\psi}i\gamma^5\psi)^2} - m\bar{\psi}\psi.$	$L_{\mathrm{D}} = \frac{\Re[i\bar{\psi}(\bar{\psi}\psi+\bar{\psi}i\gamma^5\psi i\gamma^5)\not{\partial}\psi]}{(\bar{\psi}\psi)^2+(\bar{\psi}i\gamma^5\psi)^2} - m\bar{\psi}\psi.$
1510.06699_ EQ0239 (A3)	047	0.997 N +2	$\begin{aligned} \psi \bar{\psi} &= \frac{1}{4} \left(\bar{\psi} \psi + \bar{\psi} \gamma^{\nu} \psi \gamma_{\nu} - \bar{\psi} \gamma^{\nu} \gamma^5 \psi \gamma_{\nu} \gamma^5 + \frac{1}{2} \bar{\psi} \sigma^{\nu \rho} \psi \sigma_{\nu \rho} \right. \\ & \quad \left. + \bar{\psi} \gamma^5 \psi \gamma^5 \right) \end{aligned}$	$\psi \bar{\psi} = \frac{1}{4} \left(\bar{\psi} \psi + \bar{\psi} \gamma^{\nu} \psi \gamma_{\nu} - \bar{\psi} \gamma^{\nu} \gamma^5 \psi \gamma_{\nu} \gamma^5 + \frac{1}{2} \bar{\psi} \sigma^{\nu \rho} \psi \sigma_{\nu \rho} + \bar{\psi} \gamma^5 \psi \gamma^5 \right)$	$\psi \bar{\psi} = \frac{1}{4} (\bar{\psi} \psi + \bar{\psi} \gamma^{\nu} \psi \gamma_{\nu} - \bar{\psi} \gamma^{\nu} \gamma^5 \psi \gamma_{\nu} \gamma^5 + \frac{1}{2} \bar{\psi} \sigma^{\nu \rho} \psi \sigma_{\nu \rho} + \bar{\psi} \gamma^5 \psi \gamma^5), \quad ($
1510.06699_ EQ0056 (60)	013	0.997 N +0	$\backslash\mathrm{mathcal}\{\mathrm{L}\}_{\mathrm{\mathrm{mathrm}\{T\}}}=\backslash\mathrm{mathcal}\{\mathrm{L}\}_{\mathrm{\mathrm{mathrm}\{G\}}}\{\mathrm{h},\mathrm{A},\backslash\mathrm{partial}\mathrm{A},\backslash\mathrm{partial}\mathrm{B}\}+\backslash\mathrm{mathcal}\{\mathrm{L}\}_{\mathrm{\mathrm{mathrm}\{M\}}}\{\backslash\mathrm{varphi},\backslash\mathrm{partial}\backslash\mathrm{varphi},\backslash\mathrm{phi},\backslash\mathrm{partial}\backslash\mathrm{phi},\mathrm{h},\backslash\mathrm{partial}\mathrm{h},\mathrm{A},\backslash\mathrm{partial}\mathrm{A},\mathrm{B}\}$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(h,A,\partial A,\partial B) + \mathcal{L}_{\mathrm{M}}(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,A,\partial A,B),$	$\mathcal{L}_{\mathrm{T}}=\mathcal{L}_{\mathrm{G}}(h,A,\partial A,\partial B)+\mathcal{L}_{\mathrm{M}}(\varphi,\partial\varphi,\phi,\partial\phi,h,\partial h,A,\partial A,B),$
1510.06699_ EQ0077 (82)	017	0.997 K +0	$\widehat{\mathcal{F}}_{ab}^* = \mathcal{F}_{ab} - \mathcal{T}^{*c}{}_{ab} \mathcal{A}_c$	$\widehat{\mathcal{F}}_{ab}^* = \mathcal{F}_{ab} - \mathcal{T}^{*c}{}_{ab} \mathcal{A}_c,$	$\widehat{\mathcal{F}}_{ab}^* = \mathcal{F}_{ab} - \mathcal{T}^{*c}{}_{ab} \mathcal{A}_c,$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_E00162 (178b)	030	0.998	$\begin{aligned} & \mathcal{D}_c^\dagger(h\sigma_{ab}{}^c)+h\tau_{[ab]}^\dagger+\frac{1}{2}\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\Sigma_{ab}\varphi=0, \\ & \mathcal{D}_c^\dagger(h\tau^{\dagger c}{}_d)-h(\sigma_{ab}{}^c\mathcal{R}^{\dagger ab}{}_{cd}-\tau^{\dagger c}{}_b\mathcal{T}^{\dagger b}{}_{cd}+\zeta^{\dagger c}\mathcal{H}_{cd})+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\mathcal{D}_d^\dagger\phi+\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\mathcal{D}_d^\dagger\varphi=0, \\ & \mathcal{D}_c^\dagger(h\zeta^{\dagger c})=0, \\ & h\tau^{\dagger c}{}_c+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\phi-\frac{\delta L_{\mathrm{M}}}{\delta\varphi}w\varphi=0, \end{aligned}$	$\begin{aligned} & \mathcal{D}_c^\dagger(h\sigma_{ab}{}^c)+h\tau_{[ab]}^\dagger+\frac{1}{2}\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\Sigma_{ab}\varphi=0, \\ & \mathcal{D}_c^\dagger(h\tau^{\dagger c}{}_d)-h(\sigma_{ab}{}^c\mathcal{R}^{\dagger ab}{}_{cd}-\tau^{\dagger c}{}_b\mathcal{T}^{\dagger b}{}_{cd}+\zeta^{\dagger c}\mathcal{H}_{cd})+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\mathcal{D}_d^\dagger\phi+\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\mathcal{D}_d^\dagger\varphi=0, \\ & \mathcal{D}_c^\dagger(h\zeta^{\dagger c})=0, \\ & h\tau^{\dagger c}{}_c+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\phi-\frac{\delta L_{\mathrm{M}}}{\delta\varphi}w\varphi=0, \end{aligned}$	$\begin{aligned} & \mathcal{D}_c^\dagger(h\sigma_{ab}{}^c)+h\tau_{[ab]}^\dagger+\frac{1}{2}\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\Sigma_{ab}\varphi=0, \\ & \mathcal{D}_c^\dagger(h\tau^{\dagger c}{}_d)-h(\sigma_{ab}{}^c\mathcal{R}^{\dagger ab}{}_{cd}-\tau^{\dagger c}{}_b\mathcal{T}^{\dagger b}{}_{cd}+\zeta^{\dagger c}\mathcal{H}_{cd})+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\mathcal{D}_d^\dagger\phi+\frac{\delta L_{\mathrm{M}}}{\delta\varphi}\mathcal{D}_d^\dagger\varphi=0, \\ & \mathcal{D}_c^\dagger(h\zeta^{\dagger c})=0, \\ & h\tau^{\dagger c}{}_c+\frac{\delta L_{\mathrm{M}}}{\delta\phi}\phi-\frac{\delta L_{\mathrm{M}}}{\delta\varphi}w\varphi=0, \end{aligned}$
1510.06699_E00251 (B6)	049	0.998 K +0	$\begin{aligned} R=& {}^0R+\frac{1}{4}T^{\mu\nu\lambda}T_{\mu\nu\lambda}+\frac{1}{2}T^{\mu\nu\lambda}T_{\nu\mu\lambda}-T^\mu T_\mu \\ & -\frac{2}{\sqrt{-g}}\partial_\mu(\sqrt{-g}T^\mu), \end{aligned}$	$R={}^0R+\frac{1}{4}T^{\mu\nu\lambda}T_{\mu\nu\lambda}+\frac{1}{2}T^{\mu\nu\lambda}T_{\nu\mu\lambda}-T^\mu T_\mu-\frac{2}{\sqrt{-g}}\partial_\mu(\sqrt{-g}T^\mu),$	$R={}^0R+\frac{1}{4}T^{\mu\nu\lambda}T_{\mu\nu\lambda}+\frac{1}{2}T^{\mu\nu\lambda}T_{\nu\mu\lambda}-T^\mu T_\mu-\frac{2}{\sqrt{-g}}\partial_\mu(\sqrt{-g}T^\mu),$
1510.06699_E00104	021	0.999 W -1	$\begin{array}{l} \mathcal{L}_{\mathrm{M}}=\hbar^{-1}\left[\frac{1}{2}i\bar{\psi}\gamma^{a0}\overset{\leftrightarrow}{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu({}^0\mathcal{D}_a^*\phi)({}^0\mathcal{D}^{*a}\phi)-\lambda\phi^4\right. \\ \left.-a\phi^2{}^0\mathcal{R}^*\right],(112) \end{array}$	$\mathcal{L}_{\mathrm{M}}=h^{-1}\left[\frac{1}{2}i\bar{\psi}\gamma^{a0}\overset{\leftrightarrow}{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu({}^0\mathcal{D}_a^*\phi)({}^0\mathcal{D}^{*a}\phi)-\lambda\phi^4-a\phi^2{}^0\mathcal{R}^*\right],(112)$	$\mathcal{L}_{\mathrm{M}}=h^{-1}[\frac{1}{2}i\bar{\psi}\gamma^{a0}\overset{\leftrightarrow}{\mathcal{D}}_a\psi-\mu\phi\bar{\psi}\psi+\frac{1}{2}\nu({}^0\mathcal{D}_a^*\phi)({}^0\mathcal{D}^{*a}\phi)-\lambda\phi^4-a\phi^2{}^0\mathcal{R}^*],(112)$
1510.06699_E00171 (187)	031	0.999 N +0	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}\left(\frac{1}{2}i\bar{\psi}\gamma^a\overset{\leftrightarrow}{\mathcal{D}}_a^\dagger\psi-\mu\phi\bar{\psi}\psi\right),$	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}(\frac{1}{2}i\bar{\psi}\gamma^a\overset{\leftrightarrow}{\mathcal{D}}_a^\dagger\psi-\mu\phi\bar{\psi}\psi),$	$\mathcal{L}_{\mathrm{D}}=h^{-1}L_{\mathrm{D}}=h^{-1}(\frac{1}{2}i\bar{\psi}\gamma^a\overset{\leftrightarrow}{\mathcal{D}}_a^\dagger\psi-\mu\phi\bar{\psi}\psi),$
1510.06699_E00253 (C2)	050	0.999 N +0	$\begin{aligned} & \mathcal{R}^{ac}{}_{bc}-\frac{1}{2}\delta_b^a\mathcal{R}=\kappa h\tau^a{}_b, \\ & h b^c{}_\mu D_\nu[h^{-1}(h_a{}^\mu h_b{}^\nu-h_a{}^\nu h_b{}^\mu)]=2\kappa h\sigma_{ab}{}^c. \end{aligned}$	$\begin{aligned} & \mathcal{R}^{ac}{}_{bc}-\frac{1}{2}\delta_b^a\mathcal{R}=\kappa h\tau^a{}_b, \\ & h b^c{}_\mu D_\nu[h^{-1}(h_a{}^\mu h_b{}^\nu-h_a{}^\nu h_b{}^\mu)]=2\kappa h\sigma_{ab}{}^c. \end{aligned}$	$\begin{aligned} & \mathcal{R}^{ac}{}_{bc}-\frac{1}{2}\delta_b^a\mathcal{R}=\kappa h\tau^a{}_b, \\ & h b^c{}_\mu D_\nu[h^{-1}(h_a{}^\mu h_b{}^\nu-h_a{}^\nu h_b{}^\mu)]=2\kappa h\sigma_{ab}{}^c. \end{aligned}$
Conf. MathPix confidence: 0.998					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0265 (D4)	052	■ 1.000 ● C +3	$\begin{aligned} a\,\phi^2\left(\mathcal{R}^a{}_b-\frac{1}{2}\delta^a_b\mathcal{R}\right) =& h\left(\tau_\varphi\right)^a{}_b + \left[\nu\left(\mathcal{D}^{*a}\phi\right)\left(\mathcal{D}^*_b\phi\right) - \delta^a_bL_\phi\right] \\ & - \xi\left(\mathcal{H}^{ac}\mathcal{H}_{bc} - \frac{1}{4}\delta^a_b\mathcal{H}_{cd}\mathcal{H}^{cd}\right), \end{aligned}$	$a\phi^2\left(\mathcal{R}^a{}_b-\frac{1}{2}\delta^a_b\mathcal{R}\right) =h(\tau_\varphi)^a{}_b + [\nu(\mathcal{D}^{*a}\phi)(\mathcal{D}^*_b\phi) - \delta^a_bL_\phi] - \xi\left(\mathcal{H}^{ac}\mathcal{H}_{bc} - \frac{1}{4}\delta^a_b\mathcal{H}_{cd}\mathcal{H}^{cd}\right), \quad (\text{D4})$	
1510.06699_ EQ0001 (1)	002	■ 1.000 ● N +0	$L_{\{\mathrm{G}\}} = -\kappa^{-1}(\Lambda + a\,\mathrm{\mathcal{R}}) + L_{\mathcal{R}^2} + \kappa^{-1}L_{\mathcal{T}^2},$	$L_{\mathrm{G}} = -\kappa^{-1}(\Lambda + a\mathcal{R}) + L_{\mathcal{R}^2} + \kappa^{-1}L_{\mathcal{T}^2},$	
1510.06699_ EQ0003 (4)	005	■ 1.000 ● C +28	$L_{\{\mathrm{G}\}} = \alpha_1\mathcal{R}^2 + \alpha_2\mathcal{R}_{ab}\mathcal{R}^{ab} + \alpha_3\mathcal{R}_{ab}\mathcal{R}^{ba} + \alpha_4\mathcal{R}_{abcd}\mathcal{R}^{abcd} + \alpha_5\mathcal{R}_{abcd}\mathcal{R}^{acbd} + \alpha_6\mathcal{R}_{abcd}\mathcal{R}^{cdab} + \xi\mathcal{H}_{ab}\mathcal{H}^{ab} \equiv L_{\mathcal{R}^2} + L_{\mathcal{H}^2},$	$L_{\mathrm{G}} = \alpha_1\mathcal{R}^2 + \alpha_2\mathcal{R}_{ab}\mathcal{R}^{ab} + \alpha_3\mathcal{R}_{ab}\mathcal{R}^{ba} + \alpha_4\mathcal{R}_{abcd}\mathcal{R}^{abcd} + \alpha_5\mathcal{R}_{abcd}\mathcal{R}^{acbd} + \alpha_6\mathcal{R}_{abcd}\mathcal{R}^{cdab} + \xi\mathcal{H}_{ab}\mathcal{H}^{ab} \equiv L_{\mathcal{R}^2} + L_{\mathcal{H}^2},$	
1510.06699_ EQ0004 (5)	007	■ 1.000 ●	$S_{\{\mathrm{M}\}} = \int L_{\{\mathrm{M}\}}(\varphi, \partial_\mu \varphi) \, d^4x$	$S_{\mathrm{M}} = \int L_{\mathrm{M}}(\varphi, \partial_\mu \varphi) \, d^4x,$	
1510.06699_ EQ0006 (7)	007	■ 1.000 ● K +0	$\varphi'(\boldsymbol{x}') = e^{w\rho} S(\Lambda)\varphi(\boldsymbol{x}),$	$\varphi'(x') = e^{w\rho} \mathcal{S}(\Lambda)\varphi(x),$	
1510.06699_ EQ0007 (8)	008	■ 1.000 ● K +0	$\partial'_\mu \varphi'(\boldsymbol{x}') = e^{(w-1)\rho} \Lambda_\mu{}^\nu \mathcal{S}(\Lambda) \partial_\nu \varphi(\boldsymbol{x}).$	$\partial'_\mu \varphi'(x') = e^{(w-1)\rho} \Lambda_\mu{}^\nu \mathcal{S}(\Lambda) \partial_\nu \varphi(x).$	
1510.06699_ EQ0014 (15)	008	■ 1.000 ● N +0	$L_{\{\mathrm{EM}\}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu$	$L_{\mathrm{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu$	
1510.06699_ EQ0016 (17)	009	■ 1.000 ● K +0	$\varphi'(\boldsymbol{x}') = e^{w\rho(\boldsymbol{x})} \mathcal{S}(\Lambda(\boldsymbol{x}))\varphi(\boldsymbol{x}).$	$\varphi'(x') = e^{w\rho(x)} \mathcal{S}(\Lambda(x))\varphi(x).$	
1510.06699_ EQ0017 (18)	009	■ 1.000 ● N +0	$\gamma'_{\mu\nu}(x') = \frac{\partial x^\lambda}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} e^{2\rho(x)} \gamma_{\lambda\sigma}(x).$	$\gamma'_{\mu\nu}(x') = \frac{\partial x^\lambda}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\nu} e^{2\rho(x)} \gamma_{\lambda\sigma}(x).$	
1510.06699_ EQ0020 (21)	009	■ 1.000 ● N +0	$[\Sigma_{ab}, \Sigma_{cd}] = \eta_{ad}\Sigma_{bc} - \eta_{bd}\Sigma_{ac} + \eta_{bc}\Sigma_{ad} - \eta_{ac}\Sigma_{bd}.$	$[\Sigma_{ab}, \Sigma_{cd}] = \eta_{ad}\Sigma_{bc} - \eta_{bd}\Sigma_{ac} + \eta_{bc}\Sigma_{ad} - \eta_{ac}\Sigma_{bd}.$	
1510.06699_ EQ0023 (24)	009	■ 1.000 ● K +0	$B'_\mu(x') = \frac{\partial x^\nu}{\partial x'^\mu} [B_\nu(x) - \partial_\nu \rho(x)].$	$B'_\mu(x') = \frac{\partial x^\nu}{\partial x'^\mu} [B_\nu(x) - \partial_\nu \rho(x)].$	
1510.06699_ EQ0024 (25)	009	■ 1.000 ● N +0	$D_\mu^* \varphi(x) = \left[\partial_\mu^* + \frac{1}{2} A^{ab}{}_\mu(x) \Sigma_{ab} \right] \varphi(x),$	$D_\mu^* \varphi(x) = [\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu(x) \Sigma_{ab}] \varphi(x),$	
1510.06699_ EQ0025 (26)	009	■ 1.000 ● N +0	$\partial_\mu^* \equiv \partial_\mu + w B_\mu(x).$	$\partial_\mu^* \equiv \partial_\mu + w B_\mu(x).$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510..06699_ E00027 (28)	010	1.000 ● K +0	$\mathrm{h}_{\mathrm{a}}\{ \ }^{\wedge\{\mu\}}\mathrm{b}^{\mathrm{a}}\{ \ }_{\mathrm{\nu}}=\delta_{\mathrm{\nu}}^{\mathrm{a}}\{ \ }^{\wedge\{\mu\}}\quad \backslash\mathrm{text}\{ \ \mathrm{and}\ \}\quad \mathrm{h}_{\mathrm{a}}\{ \ }^{\wedge\{\mu\}}\mathrm{b}^{\mathrm{c}}\{ \ }_{\mathrm{\mu}}=\delta_{\mathrm{a}}^{\mathrm{c}}\{ \ }^{\wedge\{\mu\}}$	$h_a{}^\mu b^a{}_\nu = \delta_\nu^\mu \quad \text{and} \quad h_a{}^\mu b^c{}_\mu = \delta_a^c.$	$h_a{}^\mu b^a{}_\nu = \delta_\nu^\mu \quad \text{and} \quad h_a{}^\mu b^c{}_\mu = \delta_a^c.$
1510..06699_ E00020 (29)	010	1.000 ● N +0	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\ \backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{a}}^{\wedge\{*\}}\ \backslash\mathrm{varphi}(\mathrm{x})\ \&\ \backslash\mathrm{equiv}\ \mathrm{h}_{\mathrm{a}}\{ \ }^{\wedge\{\mu\}}(\mathrm{x})\ \mathrm{D}_{\mathrm{\mu}}^{\wedge\{*\}}\ \backslash\mathrm{varphi}(\mathrm{x})\ \backslash\backslash\ \&\ =\mathrm{h}_{\mathrm{a}}\{ \ }^{\wedge\{\mu\}}(\mathrm{x})\backslash\mathrm{left}[\backslash\mathrm{partial}_{\mathrm{\mu}}+\backslash\mathrm{frac}\{1\}{2}\ \mathrm{A}^{\mathrm{c}}\{\mathrm{d}\}\{ \ }_{\mathrm{\mu}}\}(\mathrm{x})\ \backslash\mathrm{Sigma}_{\mathrm{c}}\{\mathrm{d}\}+\mathrm{w}\ \mathrm{B}_{\mathrm{\mu}}(\mathrm{x})\backslash\mathrm{right}]\ \backslash\mathrm{varphi}(\mathrm{x}),\ \backslash\mathrm{end}\{\mathrm{aligned}\}$	$\mathcal{D}_a^*\varphi(x) \equiv h_a{}^\mu(x)D_\mu^*\varphi(x)$ $= h_a{}^\mu(x) \left[\partial_\mu + \frac{1}{2}A^{cd}{}_\mu(x)\Sigma_{cd} + wB_\mu(x) \right] \varphi(x),$	$\mathcal{D}_a^*\varphi(x) \equiv h_a{}^\mu(x)D_\mu^*\varphi(x)$ $= h_a{}^\mu(x)[\partial_\mu + \frac{1}{2}A^{cd}{}_\mu(x)\Sigma_{cd} + wB_\mu(x)]\varphi(x),$
1510..06699_ E00030 (31)	010	1.000 ● N +1	$\mathrm{L}_{\mathrm{\mathit{M}}}\backslash\mathrm{left}(\backslash\mathrm{varphi}^{\mathrm{\prime}}\backslash\mathrm{left}(\mathrm{x}^{\mathrm{\prime}}\backslash\mathrm{prime}\backslash\mathrm{right})\ ;\ \backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{a}}^{\wedge\{*\}}\ \backslash\mathrm{prime}\backslash\mathrm{varphi}^{\mathrm{\prime}}\backslash\mathrm{prime}\backslash\mathrm{left}(\mathrm{x}^{\mathrm{\prime}}\backslash\mathrm{prime}\backslash\mathrm{right})\backslash\mathrm{right})=\mathrm{e}^{\mathrm{-}4\ \mathrm{\rho}(\mathrm{x})}\ \mathrm{L}_{\mathrm{\mathit{M}}}\backslash\mathrm{left}(\backslash\mathrm{varphi}(\mathrm{x})\ ;\ \backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{a}}^{\wedge\{*\}}\ \backslash\mathrm{varphi}(\mathrm{x})\backslash\mathrm{right})$	$L_{\mathrm{M}}(\varphi'(x');\mathcal{D}_a^{*'}\varphi'(x')) = e^{-4\rho(x)}L_{\mathrm{M}}(\varphi(x);\mathcal{D}_a^*\varphi(x)).$	$L_{\mathrm{M}}(\varphi'(x');\mathcal{D}_a^{*'}\varphi'(x')) = e^{-4\rho(x)}L_{\mathrm{M}}(\varphi(x);\mathcal{D}_a^*\varphi(x)).$
1510..06699_ E00031 (32)	010	1.000 ● N +1	$\mathrm{S}_{\mathrm{\mathit{M}}}=\backslash\mathrm{int}\ \mathrm{h}^{\mathrm{-}1}\ \mathrm{L}_{\mathrm{\mathit{M}}}\backslash\mathrm{left}(\backslash\mathrm{varphi},\ \backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{a}}^{\wedge\{*\}}\ \backslash\mathrm{varphi}\backslash\mathrm{right})\ \mathrm{d}^{\mathrm{4}}\ \mathrm{x}$	$S_{\mathrm{M}} = \int h^{-1}L_{\mathrm{M}}(\varphi,\mathcal{D}_a^*\varphi)\,d^4x$	$S_{\mathrm{M}} = \int h^{-1}L_{\mathrm{M}}(\varphi,\mathcal{D}_a^*\varphi)\,d^4x,$
1510..06699_ E00034 (35)	010	1.000 ● C +4	$\mathrm{D}_{\mathrm{\mu}}^{\wedge\{*\}}\ \backslash\mathrm{equiv}\ \backslash\mathrm{partial}_{\mathrm{\mu}}^{\wedge\{*\}}+\{ \ }^{\wedge\{0\}}\ \backslash\mathrm{Gamma}^{\mathrm{a}}\{\backslash\mathrm{sigma}\}\{ \ }_{\mathrm{\rho}}\backslash\mathrm{rho}\ \mathrm{\mu}\backslash\mathrm{mathrm}\{\mathrm{X}\}^{\mathrm{a}}\{\backslash\mathrm{rho}\}\{ \ }_{\mathrm{\backslash\mathrm{sigma}}}\}+\backslash\mathrm{frac}\{1\}{2}\ \mathrm{A}^{\mathrm{a}}\{\mathrm{b}\}\{ \ }_{\mathrm{\mu}}\backslash\mathrm{Sigma}_{\mathrm{a}}\{\mathrm{b}\}=\{ \ }^{\wedge\{0\}}\ \backslash\mathrm{nabla}_{\mathrm{\mu}}^{\wedge\{*\}}+\backslash\mathrm{frac}\{1\}{2}\ \mathrm{A}^{\mathrm{a}}\{\mathrm{b}\}\{ \ }_{\mathrm{\mu}}\backslash\mathrm{Sigma}_{\mathrm{a}}\{\mathrm{b}\}$	$D_\mu^* \equiv \partial_\mu^* + {}^0\Gamma^\sigma{}_{\rho\mu}X^\rho{}_\sigma + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab} = {}^0\nabla_\mu^* + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab},$	$D_\mu^* \equiv \partial_\mu^* + {}^0\Gamma^\sigma{}_{\rho\mu}X^\rho{}_\sigma + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab} = {}^0\nabla_\mu^* + \frac{1}{2}A^{ab}{}_\mu\Sigma_{ab},$
1510..06699_ E00035 (36)	011	1.000 ●	$\mathrm{D}_{\mathrm{\mu}}^{\wedge\{*\}}\ \mathrm{h}_{\mathrm{a}}\{ \ }^{\wedge\{\nu\}}=\backslash\mathrm{partial}_{\mathrm{\mu}}^{\wedge\{*\}}\ \mathrm{h}_{\mathrm{a}}\{ \ }^{\wedge\{\nu\}}+\{ \ }^{\wedge\{0\}}\ \backslash\mathrm{Gamma}^{\mathrm{a}}\{\backslash\mathrm{nu}\}\{ \ }_{\mathrm{\backslash\mathrm{rho}}\ \mathrm{\mu}}\ \mathrm{h}_{\mathrm{a}}\{ \ }^{\wedge\{\backslash\mathrm{rho}\}\mathrm{-}\mathrm{A}\{\mathrm{b}\}\{ \ }_{\mathrm{a}}\ \mathrm{\mu}\}\ \mathrm{h}_{\mathrm{b}}\{ \ }^{\wedge\{\backslash\mathrm{nu}\}}$	$D_\mu^*h_a{}^\nu = \partial_\mu^*h_a{}^\nu + {}^0\Gamma^\nu{}_{\rho\mu}h_a{}^\rho - A^b{}_{a\mu}h_b{}^\nu.$	$D_\mu^*h_a{}^\nu = \partial_\mu^*h_a{}^\nu + {}^0\Gamma^\nu{}_{\rho\mu}h_a{}^\rho - A^b{}_{a\mu}h_b{}^\nu.$
1510..06699_ E00036 (37)	011	1.000 ● K +0	$\backslash\mathrm{left}[\{ \ }^{\wedge\{0\}}\ \backslash\mathrm{nabla}_{\mathrm{\mu}},\{ \ }^{\wedge\{0\}}\ \backslash\mathrm{nabla}_{\mathrm{\nu}}\backslash\mathrm{right}]\ \backslash\mathrm{varphi}=0$	$[{}^0\nabla_\mu, {}^0\nabla_\nu]\varphi = 0,$	$[{}^0\nabla_\mu, {}^0\nabla_\nu]\varphi = 0,$
1510..06699_ E00037 (38)	011	1.000 ● K +0	$\backslash\mathrm{left}[\mathrm{D}_{\mathrm{\mu}}^{\wedge\{*\}},\ \mathrm{D}_{\mathrm{\nu}}^{\wedge\{*\}}\backslash\mathrm{right}]\ \backslash\mathrm{varphi}=\backslash\mathrm{frac}\{1\}{2}\ \mathrm{R}^{\mathrm{a}}\{\mathrm{b}\}\{ \ }_{\mathrm{\mu}\ \mathrm{\nu}}\backslash\mathrm{Sigma}_{\mathrm{a}}\{\mathrm{b}\}\ \backslash\mathrm{varphi}+\mathrm{w}\ \mathrm{H}_{\mathrm{\mu}\ \mathrm{\nu}}\backslash\mathrm{varphi}$	$[D_\mu^*,D_\nu^*]\varphi = \frac{1}{2}R^{ab}{}_{\mu\nu}\Sigma_{ab}\varphi + wH_{\mu\nu}\varphi,$	$[D_\mu^*,D_\nu^*]\varphi = \frac{1}{2}R^{ab}{}_{\mu\nu}\Sigma_{ab}\varphi + wH_{\mu\nu}\varphi,$
1510..06699_ E00039 (41)	012	1.000 ● N +0	$\backslash\mathrm{left}[\backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{c}}^{\wedge\{*\}},\ \backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{d}}^{\wedge\{*\}}\backslash\mathrm{right}]\ \backslash\mathrm{varphi}=\backslash\mathrm{frac}\{1\}{2}\ \backslash\mathrm{mathcal}\{\mathrm{R}\}^{\mathrm{a}}\{\mathrm{b}\}\{ \ }_{\mathrm{c}\ \mathrm{d}}\ \backslash\mathrm{Sigma}_{\mathrm{a}}\{\mathrm{b}\}\ \backslash\mathrm{varphi}+\mathrm{w}\ \backslash\mathrm{mathcal}\{\mathrm{H}\}_{\mathrm{c}\ \mathrm{d}}\ \backslash\mathrm{varphi}-\backslash\mathrm{mathcal}\{\mathrm{T}\}^{\wedge\{*\}}\{\mathrm{a}\}\{ \ }_{\mathrm{c}\ \mathrm{d}}\ \backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{a}}^{\wedge\{*\}}\ \backslash\mathrm{varphi}$	$[\mathcal{D}_c^*,\mathcal{D}_d^*]\varphi = \frac{1}{2}\mathcal{R}^{ab}{}_{cd}\Sigma_{ab}\varphi + w\mathcal{H}_{cd}\varphi - \mathcal{T}^{*a}{}_{cd}\mathcal{D}_a^*\varphi,$	$[\mathcal{D}_c^*,\mathcal{D}_d^*]\varphi = \frac{1}{2}\mathcal{R}^{ab}{}_{cd}\Sigma_{ab}\varphi + w\mathcal{H}_{cd}\varphi - \mathcal{T}^{*a}{}_{cd}\mathcal{D}_a^*\varphi,$
1510..06699_ E00040 (42)	012	1.000 ● N +0	$\backslash\mathrm{mathcal}\{\mathrm{T}\}^{\wedge\{*\}}\{\mathrm{a}\}\{ \ }_{\mathrm{b}\ \mathrm{c}}\ \backslash\mathrm{equiv}\ \mathrm{h}_{\mathrm{b}}\{ \ }_{\mathrm{c}}\{ \ }^{\wedge\{\mu\}}\ \mathrm{h}_{\mathrm{c}}\{ \ }^{\wedge\{\backslash\mathrm{nu}\}}\ \mathrm{T}^{\mathrm{a}}\{\mathrm{a}\}\{ \ }_{\mathrm{\mu}\ \mathrm{\nu}}\backslash\mathrm{equiv}\ \mathrm{h}_{\mathrm{b}}\{ \ }_{\mathrm{c}}\{ \ }^{\wedge\{\mu\}}\ \mathrm{h}_{\mathrm{c}}\{ \ }^{\wedge\{\backslash\mathrm{nu}\}}\backslash\mathrm{left}(\mathrm{D}_{\mathrm{\mu}}^{\wedge\{*\}}\ \mathrm{b}^{\mathrm{a}}\{ \ }_{\mathrm{\nu}}-\mathrm{D}_{\mathrm{\nu}}^{\wedge\{*\}}\ \mathrm{b}^{\mathrm{a}}\{ \ }_{\mathrm{\mu}}\}\{ \ }_{\mathrm{\mu}}\backslash\mathrm{right})$	$\mathcal{T}^{*a}{}_{bc} \equiv h_b{}^\mu h_c{}^\nu T^{*a}{}_{\mu\nu} \equiv h_b{}^\mu h_c{}^\nu (D_\mu^*b^a{}_\nu - D_\nu^*b^a{}_\mu),$	$\mathcal{T}^{*a}{}_{bc} \equiv h_b{}^\mu h_c{}^\nu T^{*a}{}_{\mu\nu} \equiv h_b{}^\mu h_c{}^\nu (D_\mu^*b^a{}_\nu - D_\nu^*b^a{}_\mu),$
1510..06699_ E00041 (43)	012	1.000 ● N +0	$\backslash\mathrm{mathcal}\{\mathrm{T}\}^{\wedge\{*\}}\{\mathrm{a}\}\{ \ }_{\mathrm{b}\ \mathrm{c}}=-\mathrm{b}^{\mathrm{a}}\{\mathrm{a}\}\{ \ }_{\mathrm{\mu}}\backslash\mathrm{left}(\ \backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{b}}^{\wedge\{*\}}\ \mathrm{h}_{\mathrm{c}}\{ \ }^{\wedge\{\mu\}}-\backslash\mathrm{mathcal}\{\mathrm{D}\}_{\mathrm{c}}^{\wedge\{*\}}\ \mathrm{h}_{\mathrm{b}}\{ \ }^{\wedge\{\mu\}}\backslash\mathrm{right})$	$\mathcal{T}^{*a}{}_{bc} = -b^a{}_\mu (\mathcal{D}_b^*h_c{}^\mu - \mathcal{D}_c^*h_b{}^\mu).$	$\mathcal{T}^{*a}{}_{bc} = -b^a{}_\mu (\mathcal{D}_b^*h_c{}^\mu - \mathcal{D}_c^*h_b{}^\mu).$
1510..06699_ E00042 (44)	012	1.000 ● K +0	$\backslash\mathrm{mathcal}\{\mathrm{T}\}^{\wedge\{*\}}\{\mathrm{a}\}\{ \ }_{\mathrm{b}\ \mathrm{c}}=\backslash\mathrm{mathcal}\{\mathrm{T}\}^{\wedge\{*\}}\{\mathrm{a}\}\{ \ }_{\mathrm{b}\ \mathrm{c}}+\delta_{\mathrm{c}}^{\mathrm{a}}\{\mathrm{a}\}\ \backslash\mathrm{mathcal}\{\mathrm{B}\}_{\mathrm{b}}-\delta_{\mathrm{b}}^{\mathrm{a}}\{\mathrm{a}\}\ \backslash\mathrm{mathcal}\{\mathrm{B}\}_{\mathrm{c}}\}$	$\mathcal{T}^{*a}{}_{bc} = \mathcal{T}^a{}_{bc} + \delta_c^a\mathcal{B}_b - \delta_b^a\mathcal{B}_c,$	$\mathcal{T}^{*a}{}_{bc} = \mathcal{T}^a{}_{bc} + \delta_c^a\mathcal{B}_b - \delta_b^a\mathcal{B}_c,$
1510..06699_ E00043 (45)	012	1.000 ● N +0	$\mathrm{c}^{\mathrm{a}}\{\mathrm{a}\}\{ \ }_{\mathrm{b}\ \mathrm{c}}\ \backslash\mathrm{equiv}\ \mathrm{h}_{\mathrm{b}}\{ \ }_{\mathrm{c}}\{ \ }^{\wedge\{\mu\}}\ \mathrm{h}_{\mathrm{c}}\{ \ }^{\wedge\{\backslash\mathrm{nu}\}}\backslash\mathrm{left}(\backslash\mathrm{partial}_{\mathrm{\mu}}^{\wedge\{*\}}\ \mathrm{b}^{\mathrm{a}}\{ \ }_{\mathrm{\nu}}-\backslash\mathrm{partial}_{\mathrm{\nu}}^{\wedge\{*\}}\ \mathrm{b}^{\mathrm{a}}\{ \ }_{\mathrm{\mu}}\}\{ \ }_{\mathrm{\mu}}\backslash\mathrm{right})$	$c^{*a}{}_{bc} \equiv h_b{}^\mu h_c{}^\nu (\partial_\mu^*b^a{}_\nu - \partial_\nu^*b^a{}_\mu),$	$c^{*a}{}_{bc} \equiv h_b{}^\mu h_c{}^\nu (\partial_\mu^*b^a{}_\nu - \partial_\nu^*b^a{}_\mu),$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E09044 (46)	012	■ 1.000 ● N +0	$\mathrm{A}_{\{a\,b\,c\}}=\frac{1}{2}\left(c_{\{a\,b\,c\}}^*+c_{\{b\,c\,a\}}^*-c_{\{c\,a\,b\}}^*\right)-\frac{1}{2}\left(\mathrm{T}_{\{a\,b\,c\}}^*+\mathrm{T}_{\{b\,c\,a\}}^*-\mathrm{T}_{\{c\,a\,b\}}^*\right)$	$\mathcal{A}_{abc}=\frac{1}{2}(c_{abc}^*+c_{bca}^*-c_{cab}^*)-\frac{1}{2}(\mathcal{T}_{abc}^*+\mathcal{T}_{bca}^*-\mathcal{T}_{cab}^*).$	$\mathcal{A}_{abc}=\frac{1}{2}(c_{abc}^*+c_{bca}^*-c_{cab}^*)-\frac{1}{2}(\mathcal{T}_{abc}^*+\mathcal{T}_{bca}^*-\mathcal{T}_{cab}^*).$
1510.06699_ E09045 (47)	012	■ 1.000 ● N +0	$\mathrm{A}_{\{a\,b\,c\}}=\{ \}^{\{0\}}\,\mathrm{A}_{\{a\,b\,c\}}^*(h,\,\partial h,\,B)+\mathrm{K}_{\{a\,b\,c\}}^*(h,\,\partial h,\,A,\,B)$	$\mathcal{A}_{abc}={}^0\mathcal{A}_{abc}^*(h,\partial h,B)+\mathcal{K}_{abc}^*(h,\partial h,A,B),$	$\mathcal{A}_{abc}={}^0\mathcal{A}_{abc}^*(h,\partial h,B)+\mathcal{K}_{abc}^*(h,\partial h,A,B),$
1510.06699_ E09048 (50)	012	■ 1.000 ● W +2	$\left[\mathrm{D}_{\{a\}}^*,\mathrm{D}_{\{b\}}^*\right]\varphi+\left[\mathrm{D}_{\{c\}}^*,\mathrm{D}_{\{a\}}^*\right]\varphi+\left[\mathrm{D}_{\{b\}}^*,\mathrm{D}_{\{c\}}^*\right]\varphi+\left[\mathrm{D}_{\{a\}}^*,\mathrm{D}_{\{b\}}^*\right]\varphi+\left[\mathrm{D}_{\{c\}}^*,\mathrm{D}_{\{a\}}^*\right]\varphi+\left[\mathrm{D}_{\{b\}}^*,\mathrm{D}_{\{c\}}^*\right]\varphi=0$	$[\mathcal{D}_a^*,[\mathcal{D}_b^*,\mathcal{D}_c^*]]\varphi+[\mathcal{D}_c^*,[\mathcal{D}_a^*,\mathcal{D}_b^*]]\varphi+[\mathcal{D}_b^*,[\mathcal{D}_c^*,\mathcal{D}_a^*]]\varphi=0.$	$[\mathcal{D}_a^*,[\mathcal{D}_b^*,\mathcal{D}_c^*]]\varphi+[\mathcal{D}_c^*,[\mathcal{D}_a^*,\mathcal{D}_b^*]]\varphi+[\mathcal{D}_b^*,[\mathcal{D}_c^*,\mathcal{D}_a^*]]\varphi=0.$
1510.06699_ E09050 (54)	013	■ 1.000 ● N +0	$\mathrm{D}_{\{a\}}^*\mathrm{R}^{ae}_{\{b\,c\}}-2\mathrm{D}_{[b}^*\mathcal{R}^e_{\,c]}\mathrm{R}^{ae}_{\{a\,e\}}\}_{\{b\,c\}}-2\mathrm{D}_{\{[b\}}^*\mathrm{R}^{ae}_{\{e\}}\}_{\{c\}}-2\mathrm{T}_{\{a\,b\}}^*\mathrm{R}^{ae}_{\{c\}}\mathrm{R}^{ae}_{\{a\,e\}}\}_{\{c\}}-\mathrm{T}_{\{b\,c\}}^*\mathrm{R}^{ae}_{\{a\,e\}}\}_{\{b\,c\}}=0$	$\mathcal{D}_a^*\mathcal{R}^{ae}_{bc}-2\mathcal{D}_{[b}^*\mathcal{R}^e_{\,c]}\mathrm{R}^{ae}_{\,f}-\mathcal{T}_{a[b}^*\mathcal{R}^{ae}_{\,c]f}-\mathcal{T}_{bc}^*\mathcal{R}^e_{\,f}=0.$	$\mathcal{D}_a^*\mathcal{R}^{ae}_{bc}-2\mathcal{D}_{[b}^*\mathcal{R}^e_{\,c]}\mathrm{R}^{ae}_{\,f}-2\mathcal{T}_{a[b}^*\mathcal{R}^{ae}_{\,c]f}-\mathcal{T}_{bc}^*\mathcal{R}^e_{\,f}=0.$
1510.06699_ E09051 (55)	013	■ 1.000 ● N +0	$\mathrm{D}_{\{a\}}^*\left(\mathrm{R}^a_{\,c}-\frac{1}{2}\delta^a_c\mathcal{R}\right)+\mathcal{T}_{bc}^*\mathcal{R}^b_{\,f}+\frac{1}{2}\mathcal{T}_{ab}^*\mathcal{R}^{ab}_{\,cf}=0$	$\mathcal{D}_a^*\left(\mathcal{R}^a_{\,c}-\frac{1}{2}\delta^a_c\mathcal{R}\right)+\mathcal{T}_{bc}^*\mathcal{R}^b_{\,f}+\frac{1}{2}\mathcal{T}_{ab}^*\mathcal{R}^{ab}_{\,cf}=0.$	$\mathcal{D}_a^*(\mathcal{R}^a_{\,c}-\frac{1}{2}\delta^a_c\mathcal{R})+\mathcal{T}_{bc}^*\mathcal{R}^b_{\,f}+\frac{1}{2}\mathcal{T}_{ab}^*\mathcal{R}^{ab}_{\,cf}=0.$
1510.06699_ E09052 (56)	013	■ 1.000 ● N +0	$\mathrm{D}_{\{a\}}^*\mathrm{T}_{\{b\,c\}}^*+2\mathrm{D}_{[b}^*\mathcal{T}_{\,c]}^*+\mathcal{T}_{bc}^*\mathcal{T}_e^*+2\mathcal{R}_{[bc]}+2\mathcal{H}_{bc}=0$	$\mathcal{D}_a^*\mathcal{T}_{bc}^{*a}+2\mathcal{D}_{[b}^*\mathcal{T}_{\,c]}^*+\mathcal{T}_{bc}^{*e}\mathcal{T}_e^*+2\mathcal{R}_{[bc]}+2\mathcal{H}_{bc}=0.$	$\mathcal{D}_a^*\mathcal{T}_{bc}^{*a}+2\mathcal{D}_{[b}^*\mathcal{T}_{\,c]}^*+\mathcal{T}_{bc}^{*e}\mathcal{T}_e^*+2\mathcal{R}_{[bc]}+2\mathcal{H}_{bc}=0.$
1510.06699_ E09053 (57)	013	■ 1.000 ●	$S_{\{\mathrm{G}\}}=\int h^{-1}L_{\{\mathrm{G}\}}\left(\mathrm{R}_{\{a\,b\,c\,d\}},\,\mathrm{T}_{\{a\,b\,c\}}^*,\,\mathrm{H}_{\{a\,b\}}\right)d^4x$	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}\left(\mathcal{R}_{abcd},\mathcal{T}_{abc}^*,\mathcal{H}_{ab}\right)d^4x$	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}(\mathcal{R}_{abcd},\mathcal{T}_{abc}^*,\mathcal{H}_{ab})\,d^4x,$
1510.06699_ E09054 (58)	013	■ 1.000 ● N +1	$L_{\{\mathrm{G}\}}=L_{\{\mathrm{R}\}^2}+L_{\{\mathrm{H}\}^2}$	$L_{\mathrm{G}}=L_{\mathcal{R}^2}+L_{\mathcal{H}^2},$	$L_{\mathrm{G}}=L_{\mathcal{R}^2}+L_{\mathcal{H}^2},$
1510.06699_ E09055 (59)	013	■ 1.000 ● N +0	$\mathrm{R}^2-4\mathrm{R}_{ab}\mathrm{R}^{ba}+\mathrm{R}_{abcd}\mathrm{R}^{cdab}$	$\mathcal{R}^2-4\mathcal{R}_{ab}\mathcal{R}^{ba}+\mathcal{R}_{abcd}\mathcal{R}^{cdab}$	$\mathcal{R}^2-4\mathcal{R}_{ab}\mathcal{R}^{ba}+\mathcal{R}_{abcd}\mathcal{R}^{cdab}$
1510.06699_ E09059 (63)	014	■ 1.000 ● W +2	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-\partial_{\mu}\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial(\partial_{\mu}\varphi)}\right)=0$	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-\partial_{\mu}\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial(\partial_{\mu}\varphi)}\right)=0.$	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-\partial_{\mu}\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial(\partial_{\mu}\varphi)}\right)=0.$
1510.06699_ E09060 (64)	014	■ 1.000 ● N +1	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\bar{\partial}\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-D_{\mu}^*\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial(D_{\mu}^*\varphi)}\right)=0$	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\bar{\partial}\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-D_{\mu}^*\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial(D_{\mu}^*\varphi)}\right)=0$	$\frac{\delta\mathcal{L}_{\mathrm{M}}}{\delta\varphi}\equiv\frac{\bar{\partial}\mathcal{L}_{\mathrm{M}}}{\partial\varphi}-D_{\mu}^*\left(\frac{\partial\mathcal{L}_{\mathrm{M}}}{\partial(D_{\mu}^*\varphi)}\right)=0,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00061 (65)	014	■ 1.000 ● W +0	$\frac{\frac{\partial}{\partial \varphi} \left(\frac{\partial L_{\mathrm{M}}}{\partial \varphi} \right)}{\frac{\partial L_{\mathrm{M}}}{\partial \varphi}} - \mathcal{D}_a^* \left(\frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^* \varphi)} \right) = h D_{\mu}^* (h^{-1} h_a^{\mu}) \frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^* \varphi)},$	$\frac{\bar{\partial} L_{\mathrm{M}}}{\partial \varphi} - \mathcal{D}_a^* \left(\frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^* \varphi)} \right) = h D_{\mu}^* (h^{-1} h_a^{\mu}) \frac{\partial L_{\mathrm{M}}}{\partial (\mathcal{D}_a^* \varphi)},$	
1510.06699_ E00062 (66)	014	■ 1.000 ● N +2	$h D_{\mu}^* (h^{-1} h_a^{\mu}) \left(\frac{\partial L_{\mathrm{M}}}{\partial \varphi} \right) = b^b_{\mu} (\mathcal{D}_b^* h_a^{\mu} - \mathcal{D}_a^* h_b^{\mu}) = \mathcal{T}^{*b}_{ab} \equiv \mathcal{T}_a^*,$	$h D_{\mu}^* (h^{-1} h_a^{\mu}) = b^b_{\mu} (\mathcal{D}_b^* h_a^{\mu} - \mathcal{D}_a^* h_b^{\mu}) = \mathcal{T}^{*b}_{ab} \equiv \mathcal{T}_a^*,$	
1510.06699_ E00069 (73)	015	■ 1.000 ● N +0	$i \gamma^a \left(\mathcal{D}_a^* + \frac{1}{2} \mathcal{T}_a^* \right) \psi - \mu \phi \psi = 0.$	$i \gamma^a \left(\mathcal{D}_a^* + \frac{1}{2} \mathcal{T}_a^* \right) \psi - \mu \phi \psi = 0.$	
1510.06699_ E00070 (74)	015	■ 1.000 ● N +0	$i \gamma^a \left(\mathcal{D}_a + \frac{1}{2} \mathcal{T}_a \right) \psi - \mu \phi \psi = 0.$	$i \gamma^a \left(\mathcal{D}_a + \frac{1}{2} \mathcal{T}_a \right) \psi - \mu \phi \psi = 0.$	
1510.06699_ E00071 (75)	016	■ 1.000 ● N +0	$i \gamma^a \left({}^0 \mathcal{D}_a^* - \frac{1}{4} \mathcal{T}_{[abc]}^* \Sigma^{bc} \right) \psi - \mu \phi \psi = 0.$	$i \gamma^a \left({}^0 \mathcal{D}_a^* - \frac{1}{4} \mathcal{T}_{[abc]}^* \Sigma^{bc} \right) \psi - \mu \phi \psi = 0.$	
1510.06699_ E00072 (76)	016	■ 1.000 ● K +0	$i \gamma^a \left({}^0 \mathcal{D}_a - \frac{1}{4} \mathcal{T}_{[abc]} \Sigma^{bc} \right) \psi - \mu \phi \psi = 0,$	$i \gamma^a \left({}^0 \mathcal{D}_a - \frac{1}{4} \mathcal{T}_{[abc]} \Sigma^{bc} \right) \psi - \mu \phi \psi = 0,$	
1510.06699_ E00078 (83)	017	■ 1.000 ● K +0	$\mathcal{L}_{\mathrm{M}} = h^{-1} L_{\mathrm{M}} = -h^{-1} \left(\frac{1}{4} \mathcal{F}_{ab} \mathcal{F}^{ab} + \mathcal{J}^a \mathcal{A}_a \right),$	$\mathcal{L}_{\mathrm{M}} = h^{-1} L_{\mathrm{M}} = -h^{-1} \left(\frac{1}{4} \mathcal{F}_{ab} \mathcal{F}^{ab} + \mathcal{J}^a \mathcal{A}_a \right),$	
1510.06699_ E00084 (90)	017	■ 1.000 ● K +0	$h \tau^a{}_b = \frac{1}{4} \delta_b^a \mathcal{F}^{cd} \mathcal{F}_{cd} - \mathcal{F}^{ac} \mathcal{F}_{bc},$	$h \tau^a{}_b = \frac{1}{4} \delta_b^a \mathcal{F}^{cd} \mathcal{F}_{cd} - \mathcal{F}^{ac} \mathcal{F}_{bc},$	
1510.06699_ E00087 (93)	018	■ 1.000 ● N +0	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(\widehat{h}, \widehat{A}, \partial \widehat{A}, \partial \widehat{B}) + \mathcal{L}_{\mathrm{M}}(\widehat{\varphi}, \partial \widehat{\varphi}, \phi_0, 0, \widehat{h}, \partial \widehat{h}, \widehat{A}, \partial \widehat{A}, \widehat{B}).$	$\mathcal{L}_{\mathrm{T}} = \mathcal{L}_{\mathrm{G}}(\widehat{h}, \widehat{A}, \partial \widehat{A}, \partial \widehat{B}) + \mathcal{L}_{\mathrm{M}}(\widehat{\varphi}, \partial \widehat{\varphi}, \phi_0, 0, \widehat{h}, \partial \widehat{h}, \widehat{A}, \partial \widehat{A}, \widehat{B}).$	
1510.06699_ E00088 (91d)	018	■ 1.000 ●	$\left. \frac{\delta \mathcal{L}_{\mathrm{M}}}{\delta \phi} \right _{\phi=\phi_0} = -h_a{}^{\mu} \left. \frac{\delta \mathcal{L}_{\mathrm{M}}}{\delta h_a{}^{\mu}} \right _{\phi=\phi_0} + \partial_{\mu} \left(\frac{\delta \mathcal{L}_{\mathrm{M}}}{\delta B_{\mu}} \right)_{\phi=\phi_0},$	$\phi_0 \left. \frac{\delta \mathcal{L}_{\mathrm{M}}}{\delta \phi} \right _{\phi=\phi_0} = -h_a{}^{\mu} \left. \frac{\delta \mathcal{L}_{\mathrm{M}}}{\delta h_a{}^{\mu}} \right _{\phi=\phi_0} + \partial_{\mu} \left(\frac{\delta \mathcal{L}_{\mathrm{M}}}{\delta B_{\mu}} \right)_{\phi=\phi_0},$	
1510.06699_ E00090 (96)	019	■ 1.000 ● N +2	$S = - \int d\lambda \left[p_a v^a - \frac{1}{2} e (p_a p^a - \mu^2 \phi^2) \right]$	$S = - \int d\lambda [p_a v^a - \frac{1}{2} e (p_a p^a - \mu^2 \phi^2)],$	
1510.06699_ E00092 (100)	019	■ 1.000 ● K +0	$v^c (\mathcal{D}_c^* p_a - \mathcal{T}_{cab}^* p^b) = e \mu^2 \phi \mathcal{D}_a^* \phi.$	$v^c (\mathcal{D}_c^* p_a - \mathcal{T}_{cab}^* p^b) = e \mu^2 \phi \mathcal{D}_a^* \phi.$	
1510.06699_ E00096 (104)	019	■ 1.000 ● N +0	$v^b {}^0 \mathcal{D}_b v_a = 0.$	$v^b {}^0 \mathcal{D}_b v_a = 0.$	

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510..06699_ EQ0098 (106)	020	■ 1.000 ● N +1	$S=-\int d\widehat{\lambda}\left[\widehat{p}_a\widehat{v}^a-\frac{1}{2}\widehat{\epsilon}\left(\widehat{p}_a\widehat{p}^a-\mu^2\phi_0^2\right)\right]$	$S=-\int d\widehat{\lambda}\left[\widehat{p}_a\widehat{v}^a-\frac{1}{2}\widehat{\epsilon}\left(\widehat{p}_a\widehat{p}^a-\mu^2\phi_0^2\right)\right]$	$S=-\int d\widehat{\lambda}\left[\widehat{p}_a\widehat{v}^a-\frac{1}{2}\widehat{\epsilon}\left(\widehat{p}_a\widehat{p}^a-\mu^2\phi_0^2\right)\right],$
1510..06699_ EQ0099 (107)	020	■ 1.000 ● N +0	$\widehat{v}^b{}^0\widehat{\mathcal{D}}_b\widehat{v}_a=0,$	$\widehat{v}^{b\,0}\widehat{\mathcal{D}}_b\widehat{v}_a=0,$	$\widehat{v}^{b\,0}\widehat{\mathcal{D}}_b\widehat{v}_a=0,$
1510..06699_ EQ0100 (108)	020	■ 1.000 ● N +0	$\left[\begin{array}{c} \text{\texttt{D}}_{\text{\texttt{\mu}}}\text{\texttt{}}^{\text{\texttt{*}}},\text{\texttt{}}^{\text{\texttt{*}}} \text{\texttt{D}}_{\text{\texttt{\nu}}}\text{\texttt{}}^{\text{\texttt{*}}} \end{array}\right]\text{\texttt{\varphi}}=\frac{1}{2}\text{\texttt{R}}^{\text{\texttt{*}}\text{\texttt{a}}\text{\texttt{b}}}_{\text{\texttt{\mu}}\text{\texttt{\nu}}}\Sigma_{\text{\texttt{a}}\text{\texttt{b}}}\text{\texttt{\varphi}}+wH_{\text{\texttt{\mu}}\text{\texttt{\nu}}}\text{\texttt{\varphi}},$	$[{}^0D_{\mu}^*,{}^0D_{\nu}^*]\varphi=\frac{1}{2}{}^0R^{*ab}{}_{\mu\nu}\Sigma_{ab}\varphi+wH_{\mu\nu}\varphi,$	$[{}^0D_{\mu}^*,{}^0D_{\nu}^*]\varphi=\frac{1}{2}{}^0R^{*ab}{}_{\mu\nu}\Sigma_{ab}\varphi+wH_{\mu\nu}\varphi,$
1510..06699_ EQ0101 (109)	020	■ 1.000 ● N +0	$\left[\begin{array}{c} \text{\texttt{D}}_{\text{\texttt{c}}}\text{\texttt{}}^{\text{\texttt{*}}},\text{\texttt{}}^{\text{\texttt{*}}} \text{\texttt{D}}_{\text{\texttt{d}}}\text{\texttt{}}^{\text{\texttt{*}}} \end{array}\right]\text{\texttt{\varphi}}=\frac{1}{2}\text{\texttt{R}}^{\text{\texttt{*}}\text{\texttt{a}}\text{\texttt{b}}}_{\text{\texttt{c}}\text{\texttt{d}}}\Sigma_{\text{\texttt{a}}\text{\texttt{b}}}\text{\texttt{\varphi}}+w\mathcal{H}_{\text{\texttt{c}}\text{\texttt{d}}}\text{\texttt{\varphi}},$	$[{}^0\mathcal{D}_c^*,{}^0\mathcal{D}_d^*]\varphi=\frac{1}{2}{}^0\mathcal{R}^{*ab}{}_{cd}\Sigma_{ab}\varphi+w\mathcal{H}_{cd}\varphi,$	$[{}^0\mathcal{D}_c^*,{}^0\mathcal{D}_d^*]\varphi=\frac{1}{2}{}^0\mathcal{R}^{*ab}{}_{cd}\Sigma_{ab}\varphi+w\mathcal{H}_{cd}\varphi,$
1510..06699_ EQ0107 (115)	021	■ 1.000 ● N +0	$\boldsymbol{e}_{\text{\texttt{\mu}}}\cdot\boldsymbol{e}_{\text{\texttt{\nu}}}=\eta_{\text{\texttt{a}}\text{\texttt{b}}}b^{\text{\texttt{a}}}{}_{\text{\texttt{\mu}}}b^{\text{\texttt{b}}}{}_{\text{\texttt{\nu}}}\equiv g_{\text{\texttt{\mu}}\text{\texttt{\nu}}}.$	$\boldsymbol{e}_{\mu}\cdot\boldsymbol{e}_{\nu}=\eta_{ab}b^a{}_{\mu}b^b{}_{\nu}\equiv g_{\mu\nu}.$	$\boldsymbol{e}_{\mu}\cdot\boldsymbol{e}_{\nu}=\eta_{ab}b^a{}_{\mu}b^b{}_{\nu}\equiv g_{\mu\nu}.$
1510..06699_ EQ0113 (121)	022	■ 1.000 ● N +0	$\Delta_{\text{\texttt{\mu}}}^{\text{\texttt{*}}\text{\texttt{e}}^{\text{\texttt{a}}}\text{\texttt{\nu}}}\equiv\partial_{\text{\texttt{\mu}}}^{\text{\texttt{*}}\text{\texttt{e}}^{\text{\texttt{a}}}\text{\texttt{\nu}}}-\Gamma^{\sigma}{}_{\text{\texttt{\nu}}\text{\texttt{\mu}}}\text{\texttt{e}}^{\text{\texttt{a}}}\text{\texttt{\sigma}}+A^{\text{\texttt{a}}}{}_{\text{\texttt{b}}\text{\texttt{\mu}}}\text{\texttt{e}}^{\text{\texttt{a}}}\text{\texttt{\nu}}=0,$	$\Delta_{\mu}^{*\text{e}^a{}_{\nu}}\equiv\partial_{\mu}^{*\text{e}^a{}_{\nu}}-\Gamma^{\sigma}{}_{\nu\mu}\text{e}^a{}_{\sigma}+A^a{}_{b\mu}\text{e}^a{}_{\nu}=0,$	$\Delta_{\mu}^{*\text{e}^a{}_{\nu}}\equiv\partial_{\mu}^{*\text{e}^a{}_{\nu}}-\Gamma^{\sigma}{}_{\nu\mu}\text{e}^a{}_{\sigma}+A^a{}_{b\mu}\text{e}^a{}_{\nu}=0,$
1510..06699_ EQ0119 (130)	023	■ 1.000 ● N +0	$\Gamma^{\lambda}_{\text{\texttt{\mu}}\text{\texttt{\nu}}}={}^0\Gamma^{*\lambda}_{\text{\texttt{\mu}}\text{\texttt{\nu}}}+K^{*\lambda}_{\text{\texttt{\mu}}\text{\texttt{\nu}}},$	$\Gamma^{\lambda}_{\mu\nu}={}^0\Gamma^{*\lambda}_{\mu\nu}+K^{*\lambda}_{\mu\nu},$	$\Gamma^{\lambda}_{\mu\nu}={}^0\Gamma^{*\lambda}_{\mu\nu}+K^{*\lambda}_{\mu\nu},$
1510..06699_ EQ0121 (132)	023	■ 1.000 ● K +0	$K^{\text{\texttt{*}}\text{\texttt{\lambda}}}_{\text{\texttt{\mu}}\text{\texttt{\nu}}}=-\frac{1}{2}\left(T^{*\lambda}_{\text{\texttt{\mu}}\text{\texttt{\nu}}}-T^{*}{}_{\text{\texttt{\nu}}}{}^{\lambda}{}_{\text{\texttt{\mu}}}+T^{*}{}_{\text{\texttt{\mu}}\text{\texttt{\nu}}}{}^{\lambda}\right),$	$K^{*\lambda}_{\mu\nu}=-\frac{1}{2}\left(T^{*\lambda}_{\mu\nu}-T^{*}{}_{\nu}{}^{\lambda}{}_{\mu}+T^{*}{}_{\mu\nu}{}^{\lambda}\right),$	$K^{*\lambda}_{\mu\nu}=-\frac{1}{2}\left(T^{*\lambda}_{\mu\nu}-T^{*}{}_{\nu}{}^{\lambda}{}_{\mu}+T^{*}{}_{\mu\nu}{}^{\lambda}\right),$
1510..06699_ EQ0122 (133)	023	■ 1.000 ● N +0	$\left[\begin{array}{c} \text{\texttt{D}}_{\text{\texttt{\mu}}}\text{\texttt{}}^{\text{\texttt{*}}}\text{\texttt{e}}^{\text{\texttt{a}}}\text{\texttt{}}_{\text{\texttt{\nu}}}\text{\texttt{}}^{\text{\texttt{*}}} \end{array}\right]\text{\texttt{\varphi}}\equiv\left[\begin{array}{c} \text{\texttt{D}}_{\text{\texttt{\mu}}}\text{\texttt{}}^{\text{\texttt{*}}}\text{\texttt{e}}^{\text{\texttt{a}}}\text{\texttt{}}_{\text{\texttt{\nu}}}\text{\texttt{}}^{\text{\texttt{*}}} \end{array}\right]\text{\texttt{\varphi}}-\left[\begin{array}{c} \text{\texttt{D}}_{\text{\texttt{\mu}}}\text{\texttt{}}^{\text{\texttt{*}}}\text{\texttt{e}}^{\text{\texttt{a}}}\text{\texttt{}}_{\text{\texttt{\nu}}}\text{\texttt{}}^{\text{\texttt{*}}} \end{array}\right]\text{\texttt{\varphi}}+\left[\begin{array}{c} \text{\texttt{D}}_{\text{\texttt{\mu}}}\text{\texttt{}}^{\text{\texttt{*}}}\text{\texttt{e}}^{\text{\texttt{a}}}\text{\texttt{}}_{\text{\texttt{\nu}}}\text{\texttt{}}^{\text{\texttt{*}}} \end{array}\right]\text{\texttt{\varphi}}=0$	${}^0\Delta_{\mu}^{*\text{e}^a{}_{\nu}}\equiv\partial_{\mu}^{*\text{e}^a{}_{\nu}}-{}^0\Gamma^{*\sigma}{}_{\nu\mu}\text{e}^a{}_{\sigma}+{}^0A^{*a}{}_{b\mu}\text{e}^b{}_{\nu}=0.$	${}^0\Delta_{\mu}^{*\text{e}^a{}_{\nu}}\equiv\partial_{\mu}^{*\text{e}^a{}_{\nu}}-{}^0\Gamma^{*\sigma}{}_{\nu\mu}\text{e}^a{}_{\sigma}+{}^0A^{*a}{}_{b\mu}\text{e}^b{}_{\nu}=0.$
1510..06699_ EQ0127 (141)	024	■ 1.000 ● N +0	$A^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}{}_{\text{\texttt{\mu}}}=A^{\text{\texttt{a}}\text{\texttt{b}}}{}_{\text{\texttt{\mu}}}+(b^{\text{\texttt{a}}}{}_{\text{\texttt{\mu}}}\mathcal{P}^{\text{\texttt{b}}}-b^{\text{\texttt{b}}}{}_{\text{\texttt{\mu}}}\mathcal{P}^{\text{\texttt{a}}}),$	$A'^{ab}{}_{\mu}=A^{ab}{}_{\mu}+(b^a{}_{\mu}\mathcal{P}^b-b^b{}_{\mu}\mathcal{P}^a),$	$A'^{ab}{}_{\mu}=A^{ab}{}_{\mu}+(b^a{}_{\mu}\mathcal{P}^b-b^b{}_{\mu}\mathcal{P}^a),$
1510..06699_ EQ0128 (142)	024	■ 1.000 ● N +0	$A^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}{}_{\text{\texttt{\mu}}}\equiv A^{\text{\texttt{a}}\text{\texttt{b}}}{}_{\text{\texttt{\mu}}}+(\mathcal{V}^{\text{\texttt{a}}}b^{\text{\texttt{b}}}{}_{\text{\texttt{\mu}}}-\mathcal{V}^{\text{\texttt{b}}}b^{\text{\texttt{a}}}{}_{\text{\texttt{\mu}}}),$	$A^{\dagger ab}{}_{\mu}\equiv A^{ab}{}_{\mu}+(\mathcal{V}^ab^b{}_{\mu}-\mathcal{V}^bb^a{}_{\mu}),$	$A^{\dagger ab}{}_{\mu}\equiv A^{ab}{}_{\mu}+(\mathcal{V}^ab^b{}_{\mu}-\mathcal{V}^bb^a{}_{\mu}),$
1510..06699_ EQ0129 (143)	024	■ 1.000 ● N +0	$\text{\texttt{D}}_{\text{\texttt{\mu}}}\text{\texttt{}}^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}\text{\texttt{\varphi}}\equiv\left(\partial_{\text{\texttt{\mu}}}+\frac{1}{2}A^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}{}_{\text{\texttt{\mu}}}\Sigma_{\text{\texttt{a}}\text{\texttt{b}}}-wV_{\text{\texttt{\mu}}}-\frac{1}{3}wT_{\text{\texttt{\mu}}}\right)\text{\texttt{\varphi}},$	$D_{\mu}^{\dagger}\varphi\equiv\left(\partial_{\mu}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab}-wV_{\mu}-\frac{1}{3}wT_{\mu}\right)\varphi,$	$D_{\mu}^{\dagger}\varphi\equiv(\partial_{\mu}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab}-wV_{\mu}-\frac{1}{3}wT_{\mu})\varphi,$
1510..06699_ EQ0130 (144)	024	■ 1.000 ●	$V_{\text{\texttt{\mu}}}\text{\texttt{'}}=V_{\text{\texttt{\mu}}}+\theta P_{\text{\texttt{\mu}}},$	$V'_{\mu}=V_{\mu}+\theta P_{\mu},$	$V'_{\mu}=V_{\mu}+\theta P_{\mu},$
1510..06699_ EQ0131 (145)	025	■ 1.000 ● N +0	$\text{\texttt{D}}_{\text{\texttt{\mu}}}\text{\texttt{}}^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}\text{\texttt{\varphi}}=\left(\partial_{\text{\texttt{\mu}}}^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}+\frac{1}{2}A^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}{}_{\text{\texttt{\mu}}}\Sigma_{\text{\texttt{a}}\text{\texttt{b}}}\right)\text{\texttt{\varphi}},$	$D_{\mu}^{\dagger}\varphi=\left(\partial_{\mu}^{\dagger}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab}\right)\varphi,$	$D_{\mu}^{\dagger}\varphi=(\partial_{\mu}^{\dagger}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab})\varphi,$
1510..06699_ EQ0132 (146)	025	■ 1.000 ● N +0	$\partial_{\text{\texttt{\mu}}}^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}\equiv\partial_{\text{\texttt{\mu}}}-w\left(V_{\text{\texttt{\mu}}}+\frac{1}{3}T_{\text{\texttt{\mu}}}\right).$	$\partial_{\mu}^{\dagger}\equiv\partial_{\mu}-w\left(V_{\mu}+\frac{1}{3}T_{\mu}\right).$	$\partial_{\mu}^{\dagger}\equiv\partial_{\mu}-w(V_{\mu}+\frac{1}{3}T_{\mu}).$
1510..06699_ EQ0133 (147)	025	■ 1.000 ● N +0	$\text{\texttt{D}}_{\text{\texttt{\mu}}}\text{\texttt{}}^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}\text{\texttt{\varphi}}\equiv h_{\text{\texttt{a}}}{}^{\text{\texttt{\mu}}}\text{\texttt{D}}_{\text{\texttt{\mu}}}^{\text{\texttt{'}}\text{\texttt{a}}\text{\texttt{b}}}\text{\texttt{\varphi}},$	$\mathcal{D}_a^{\dagger}\varphi\equiv h_a{}^{\mu}D_{\mu}^{\dagger}\varphi,$	$\mathcal{D}_a^{\dagger}\varphi\equiv h_a{}^{\mu}D_{\mu}^{\dagger}\varphi,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00134 (148)	025	■ 1.000 ● N +2	$S_{\mathrm{M}}=\int h^{-1}L_{\mathrm{M}}(\varphi,\mathcal{D}_a^\dagger\varphi)\,d^4x$	$S_{\mathrm{M}}=\int h^{-1}L_{\mathrm{M}}(\varphi,\mathcal{D}_a^\dagger\varphi)\,d^4x$	$S_{\mathrm{M}}=\int h^{-1}L_{\mathrm{M}}(\varphi,\mathcal{D}_a^\dagger\varphi)\,d^4x.$
1510.06699_ E00135 (149)	026	■ 1.000 ● N +2	$D_{\mu}^{\dagger}=\partial_{\mu}+{}^0\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab}={}^0\nabla_{\mu}^{\dagger}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab},$	$D_{\mu}^{\dagger}=\partial_{\mu}+{}^0\Gamma^{\sigma}{}_{\rho\mu}X^{\rho}{}_{\sigma}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab}={}^0\nabla_{\mu}^{\dagger}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab},$	$D_{\mu}^{\dagger}=\partial_{\mu}^{\dagger}+{}^0\Gamma^{\sigma}{}_{\rho\mu}\mathbf{X}^{\rho}{}_{\sigma}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab}={}^0\nabla_{\mu}^{\dagger}+\frac{1}{2}A^{\dagger ab}{}_{\mu}\Sigma_{ab},$
1510.06699_ E00136 (150)	026	■ 1.000 ● K +0	$\left[D_{\mu}^{\dagger},D_{\nu}^{\dagger}\right]\varphi=\frac{1}{2}R^{\dagger ab}{}_{\mu\nu}\Sigma_{ab}\varphi-wH_{\mu\nu}^{\dagger}\varphi,$	$\left[D_{\mu}^{\dagger},D_{\nu}^{\dagger}\right]\varphi=\frac{1}{2}R^{\dagger ab}{}_{\mu\nu}\Sigma_{ab}\varphi-wH_{\mu\nu}^{\dagger}\varphi,$	$\left[D_{\mu}^{\dagger},D_{\nu}^{\dagger}\right]\varphi=\frac{1}{2}R^{\dagger ab}{}_{\mu\nu}\Sigma_{ab}\varphi-wH_{\mu\nu}^{\dagger}\varphi,$
1510.06699_ E00138 (152)	026	■ 1.000 ● N +0	$H_{\mu\nu}^{\dagger}=\partial_{\mu}\left(V_{\nu}+\frac{1}{3}T_{\nu}\right)-\partial_{\nu}\left(V_{\mu}+\frac{1}{3}T_{\mu}\right).$	$H_{\mu\nu}^{\dagger}=\partial_{\mu}\left(V_{\nu}+\frac{1}{3}T_{\nu}\right)-\partial_{\nu}\left(V_{\mu}+\frac{1}{3}T_{\mu}\right).$	$H_{\mu\nu}^{\dagger}=\partial_{\mu}(V_{\nu}+\frac{1}{3}T_{\nu})-\partial_{\nu}(V_{\mu}+\frac{1}{3}T_{\mu}).$
1510.06699_ E00139 (153)	026	■ 1.000 ● N +1	$\left[\mathcal{D}_c^{\dagger},\mathcal{D}_d^{\dagger}\right]\varphi=\frac{1}{2}\mathcal{R}^{\dagger ab}{}_{cd}\Sigma_{ab}\varphi-w\mathcal{H}_{cd}^{\dagger}\varphi-\mathcal{T}^{\dagger a}{}_{cd}\mathcal{D}_a^{\dagger}\varphi,$	$\left[\mathcal{D}_c^{\dagger},\mathcal{D}_d^{\dagger}\right]\varphi=\frac{1}{2}\mathcal{R}^{\dagger ab}{}_{cd}\Sigma_{ab}\varphi-w\mathcal{H}_{cd}^{\dagger}\varphi-\mathcal{T}^{\dagger a}{}_{cd}\mathcal{D}_a^{\dagger}\varphi,$	$[\mathcal{D}_c^{\dagger},\mathcal{D}_d^{\dagger}]\varphi=\frac{1}{2}\mathcal{R}^{\dagger ab}{}_{cd}\Sigma_{ab}\varphi-w\mathcal{H}_{cd}^{\dagger}\varphi-\mathcal{T}^{\dagger a}{}_{cd}\mathcal{D}_a^{\dagger}\varphi,$
1510.06699_ E00140 (154)	027	■ 1.000 ● N +1	$\mathcal{T}^{\dagger a}{}_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}\left(D_{\mu}^{\dagger}b^a{}_{\nu}-D_{\nu}^{\dagger}b^a{}_{\mu}\right)\equiv h_b{}^{\mu}h_c{}^{\nu}T^{\dagger a}{}_{\mu\nu}.$	$\mathcal{T}^{\dagger a}{}_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}\left(D_{\mu}^{\dagger}b^a{}_{\nu}-D_{\nu}^{\dagger}b^a{}_{\mu}\right)\equiv h_b{}^{\mu}h_c{}^{\nu}T^{\dagger a}{}_{\mu\nu}.$	$\mathcal{T}^{\dagger a}{}_{bc}\equiv h_b{}^{\mu}h_c{}^{\nu}(D_{\mu}^{\dagger}b^a{}_{\nu}-D_{\nu}^{\dagger}b^a{}_{\mu})\equiv h_b{}^{\mu}h_c{}^{\nu}T^{\dagger a}{}_{\mu\nu}.$
1510.06699_ E00142 (156)	027	■ 1.000 ● N +1	$\mathcal{T}^{\dagger a}{}_{bc}=\mathcal{T}^a{}_{bc}+\frac{1}{3}(\delta_b^a\mathcal{T}_c-\delta_c^a\mathcal{T}_b).$	$\mathcal{T}^{\dagger a}{}_{bc}=\mathcal{T}^a{}_{bc}+\frac{1}{3}(\delta_b^a\mathcal{T}_c-\delta_c^a\mathcal{T}_b).$	$\mathcal{T}^{\dagger a}{}_{bc}=\mathcal{T}^a{}_{bc}+\frac{1}{3}(\delta_b^a\mathcal{T}_c-\delta_c^a\mathcal{T}_b).$
1510.06699_ E00143 (157)	027	■ 1.000 ● N +1	$\mathcal{T}_b^{\dagger}\equiv\mathcal{T}^{\dagger a}{}_{ba}=0,$	$\mathcal{T}_b^{\dagger}\equiv\mathcal{T}^{\dagger a}{}_{ba}=0,$	$\mathcal{T}_b^{\dagger}\equiv\mathcal{T}^{\dagger a}{}_{ba}=0,$
1510.06699_ E00146 (160)	027	■ 1.000 ● N +0	$\mathcal{D}_a^{\dagger}\varphi=\left({}^0\mathcal{D}_a^{\dagger}+\frac{1}{2}\mathcal{K}^{\dagger bc}{}_a\Sigma_{bc}\right)\varphi,$	$\mathcal{D}_a^{\dagger}\varphi=\left({}^0\mathcal{D}_a^{\dagger}+\frac{1}{2}\mathcal{K}^{\dagger bc}{}_a\Sigma_{bc}\right)\varphi,$	$\mathcal{D}_a^{\dagger}\varphi=({}^0\mathcal{D}_a^{\dagger}+\frac{1}{2}\mathcal{K}^{\dagger bc}{}_a\Sigma_{bc})\varphi,$
1510.06699_ E00148 (162)	028	■ 1.000 ● N +0	$\mathcal{D}_a^{\dagger}\mathcal{R}^{\dagger ae}{}_{bc}-2\mathcal{D}_{[b}^{\dagger}\mathcal{R}^{\dagger e}{}_{c]}\!-\!2\mathcal{T}^{\dagger f}{}_{a[b}\mathcal{R}^{\dagger ae}{}_{c]f}\!-\!\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger e}{}_f=0.$	$\mathcal{D}_a^{\dagger}\mathcal{R}^{\dagger ae}{}_{bc}-2\mathcal{D}_{[b}^{\dagger}\mathcal{R}^{\dagger e}{}_{c]}\!-\!2\mathcal{T}^{\dagger f}{}_{a[b}\mathcal{R}^{\dagger ae}{}_{c]f}\!-\!\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger e}{}_f=0.$	$\mathcal{D}_a^{\dagger}\mathcal{R}^{\dagger ae}{}_{bc}-2\mathcal{D}_{[b}^{\dagger}\mathcal{R}^{\dagger e}{}_{c]}\!-\!2\mathcal{T}^{\dagger f}{}_{a[b}\mathcal{R}^{\dagger ae}{}_{c]f}\!-\!\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger e}{}_f=0.$
1510.06699_ E00149 (163)	028	■ 1.000 ● W +2	$\mathcal{D}_a^{\dagger}\left(\mathcal{R}^{\dagger a}{}_c-\frac{1}{2}\delta_c^a\mathcal{R}^{\dagger}\right)+\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger b}{}_f+\frac{1}{2}\mathcal{T}^{\dagger f}{}_{ab}\mathcal{R}^{\dagger ab}{}_{cf}=0.$	$\mathcal{D}_a^{\dagger}\left(\mathcal{R}^{\dagger a}{}_c-\frac{1}{2}\delta_c^a\mathcal{R}^{\dagger}\right)+\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger b}{}_f+\frac{1}{2}\mathcal{T}^{\dagger f}{}_{ab}\mathcal{R}^{\dagger ab}{}_{cf}=0.$	$\mathcal{D}_a^{\dagger}(\mathcal{R}^{\dagger a}{}_c-\frac{1}{2}\delta_c^a\mathcal{R}^{\dagger})+\mathcal{T}^{\dagger f}{}_{bc}\mathcal{R}^{\dagger b}{}_f+\frac{1}{2}\mathcal{T}^{\dagger f}{}_{ab}\mathcal{R}^{\dagger ab}{}_{cf}=0.$
1510.06699_ E00150 (164)	028	■ 1.000 ●	$\mathcal{D}_a^{\dagger}\mathcal{T}^{\dagger a}{}_{bc}+2\mathcal{R}^{\dagger}{}_{[bc]}+2\mathcal{H}_{bc}^{\dagger}=0.$	$\mathcal{D}_a^{\dagger}\mathcal{T}^{\dagger a}{}_{bc}+2\mathcal{R}^{\dagger}{}_{[bc]}+2\mathcal{H}_{bc}^{\dagger}=0.$	$\mathcal{D}_a^{\dagger}\mathcal{T}^{\dagger a}{}_{bc}+2\mathcal{R}^{\dagger}{}_{[bc]}+2\mathcal{H}_{bc}^{\dagger}=0.$
1510.06699_ E00151 (165)	028	■ 1.000 ● N +1	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}\left(\mathcal{R}_{abcd}^{\dagger},\mathcal{T}_{abc}^{\dagger},\mathcal{H}_{ab}^{\dagger}\right)d^4x$	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}\left(\mathcal{R}_{abcd}^{\dagger},\mathcal{T}_{abc}^{\dagger},\mathcal{H}_{ab}^{\dagger}\right)d^4x$	$S_{\mathrm{G}}=\int h^{-1}L_{\mathrm{G}}(\mathcal{R}_{abcd}^{\dagger},\mathcal{T}_{abc}^{\dagger},\mathcal{H}_{ab}^{\dagger})d^4x,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ EQ0156 (170)	029	1.000 • N +0	$\tau_{a\,b}^{\dagger}\equiv\tau_{a\,b}+2\,\sigma_{c\,b\,a}\,\mathcal{V}^c-2\sigma_{a\,c}^{\,\,c}\mathcal{V}_b,$	$\tau_{ab}^{\dagger}\equiv\tau_{ab}+2\sigma_{cba}\mathcal{V}^c-2\sigma_{ac}^{\,\,c}\mathcal{V}_b,$	$\tau_{ab}^{\dagger}\equiv\tau_{ab}+2\sigma_{cba}\mathcal{V}^c-2\sigma_{ac}^{\,\,c}\mathcal{V}_b,$
1510.06699_ EQ0157 (171)	029	1.000 • N +0	$t^{\dagger a}{}_b+\tau^{\dagger a}{}_b=0.$	$t^{\dagger a}{}_b+\tau^{\dagger a}{}_b=0.$	$t^{\dagger a}{}_b+\tau^{\dagger a}{}_b=0.$
1510.06699_ EQ0158 (168a)	029	1.000 • W +0	$\begin{aligned} & \, \& \frac{\delta L_{\mathrm{M}}}{\delta\varphi}=\frac{\bar{\partial} L_{\mathrm{M}}}{\partial\varphi}-\mathcal{D}_a^{\dagger}\left(\frac{\partial L_{\mathrm{M}}}{\partial(\mathcal{D}_a^{\dagger}\varphi)}\right)=0, \\ & \, \& \frac{\delta L_{\mathrm{M}}}{\delta\phi}=\frac{\bar{\partial} L_{\mathrm{M}}}{\partial\phi}-\mathcal{D}_a^{\dagger}\left(\frac{\partial L_{\mathrm{M}}}{\partial(\mathcal{D}_a^*\phi)}\right)=0, \end{aligned}$	$\frac{\delta L_{\mathrm{M}}}{\delta\varphi}=\frac{\bar{\partial} L_{\mathrm{M}}}{\partial\varphi}-\mathcal{D}_a^{\dagger}\left(\frac{\partial L_{\mathrm{M}}}{\partial\left(\mathcal{D}_a^{\dagger}\varphi\right)}\right)=0,$ $\frac{\delta L_{\mathrm{M}}}{\delta\phi}=\frac{\bar{\partial} L_{\mathrm{M}}}{\partial\phi}-\mathcal{D}_a^{\dagger}\left(\frac{\partial L_{\mathrm{M}}}{\partial\left(\mathcal{D}_a^*\phi\right)}\right)=0,$	$\frac{\delta L_{\mathrm{M}}}{\delta\varphi}=\frac{\bar{\partial} L_{\mathrm{M}}}{\partial\varphi}-\mathcal{D}_a^{\dagger}\left(\frac{\partial L_{\mathrm{M}}}{\partial(\mathcal{D}_a^{\dagger}\varphi)}\right)=0,$ $\frac{\delta L_{\mathrm{M}}}{\delta\phi}=\frac{\bar{\partial} L_{\mathrm{M}}}{\partial\phi}-\mathcal{D}_a^{\dagger}\left(\frac{\partial L_{\mathrm{M}}}{\partial(\mathcal{D}_a^*\phi)}\right)=0,$
1510.06699_ EQ0160 (175)	029	1.000 • N +0	$j^{\dagger a}\equiv j^a-2\,s^{ab}{}_b$	$j^{\dagger a}\equiv j^a-2s^{ab}{}_b.$	$j^{\dagger a}\equiv j^a-2s^{ab}{}_b.$
1510.06699_ EQ0164 (180)	030	1.000 • N +0	$\mathcal{D}_a^{\natural}\varphi\equiv h_a^{\,\,\mu}D_{\mu}^{\natural}\varphi\equiv h_a^{\,\,\mu}\left(\partial_{\mu}+\frac{1}{2}A^{\dagger bc}{}_{\mu}\Sigma_{bc}\right)\varphi.$	$\mathcal{D}_a^{\natural}\varphi\equiv h_a^{\,\,\mu}D_{\mu}^{\natural}\varphi\equiv h_a^{\,\,\mu}\left(\partial_{\mu}+\frac{1}{2}A^{\dagger bc}{}_{\mu}\Sigma_{bc}\right)\varphi.$	$\mathcal{D}_a^{\natural}\varphi\equiv h_a^{\,\,\mu}D_{\mu}^{\natural}\varphi\equiv h_a^{\,\,\mu}(\partial_{\mu}+\frac{1}{2}A^{\dagger bc}{}_{\mu}\Sigma_{bc})\varphi.$
1510.06699_ EQ0166 (182)	031	1.000 • K +0	$\mathcal{D}_a^{\dagger}\varphi=\left(\mathcal{D}_a^{\natural}-\frac{1}{3}w\mathcal{T}_a^{\natural}\right)\varphi,$	$\mathcal{D}_a^{\dagger}\varphi=\left(\mathcal{D}_a^{\natural}-\frac{1}{3}w\mathcal{T}_a^{\natural}\right)\varphi,$	$\mathcal{D}_a^{\dagger}\varphi=(\mathcal{D}_a^{\natural}-\frac{1}{3}w\mathcal{T}_a^{\natural})\varphi,$
1510.06699_ EQ0169 (185)	031	1.000 • N +0	$\zeta^a=2\sigma^{ab}{}_b,$	$\zeta^a=2\sigma^{ab}{}_b,$	$\zeta^a=2\sigma^{ab}{}_b,$
1510.06699_ EQ0172 (191)	032	1.000 • N +0	$i\,\gamma^a\mathcal{D}_a^{\dagger}\psi-\mu\phi\psi=0.$	$i\gamma^a\mathcal{D}_a^{\dagger}\psi-\mu\phi\psi=0.$	$i\gamma^a\mathcal{D}_a^{\dagger}\psi-\mu\phi\psi=0.$
1510.06699_ EQ0173 (189)	032	1.000 • N +0	$\tau^a{}_b=\frac{1}{2}i\hbar^{-1}\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}_b\psi-\delta_b^a\mathcal{L}_{\mathrm{D}}-2\sigma_{cb}^{\,\,a}\mathcal{V}^c,$	$\tau^a{}_b=\frac{1}{2}i\hbar^{-1}\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}_b\psi-\delta_b^a\mathcal{L}_{\mathrm{D}}-2\sigma_{cb}^{\,\,a}\mathcal{V}^c,$	$\tau^a{}_b=\frac{1}{2}i\hbar^{-1}\bar{\psi}\gamma^a\overleftrightarrow{\mathcal{D}}_b^{\dagger}\psi-\delta_b^a\mathcal{L}_{\mathrm{D}}-2\sigma_{cb}^{\,\,a}\mathcal{V}^c,$
1510.06699_ EQ0174 (190)	032	1.000 • N +1	$\tau'^a{}_b=\tau^a{}_b-2\theta\sigma_{cd}^{\,\,a}\mathcal{P}^c\delta_b^d$	$\tau'^a{}_b=\tau^a{}_b-2\theta\sigma_{cd}^{\,\,a}\mathcal{P}^c\delta_b^d$	$\tau'^a{}_b=\tau^a{}_b-2\theta\sigma_{cd}^{\,\,a}\mathcal{P}^c\delta_b^d.$
1510.06699_ EQ0177 (193)	032	1.000 • K +0	$\mathcal{L}_{\mathrm{M}}=h^{-1}L_{\mathrm{M}}=-h^{-1}\left(\frac{1}{4}\mathcal{F}_{ab}\mathcal{F}^{ab}+\mathcal{J}^a\mathcal{A}_a\right),$	$\mathcal{L}_{\mathrm{M}}=h^{-1}L_{\mathrm{M}}=-h^{-1}\left(\frac{1}{4}\mathcal{F}_{ab}\mathcal{F}^{ab}+\mathcal{J}^a\mathcal{A}_a\right),$	$\mathcal{L}_{\mathrm{M}}=h^{-1}L_{\mathrm{M}}=-h^{-1}(\frac{1}{4}\mathcal{F}_{ab}\mathcal{F}^{ab}+\mathcal{J}^a\mathcal{A}_a),$
1510.06699_ EQ0179 (195)	033	1.000 • N +1	$\mathcal{D}_{[a}^{\dagger}\mathcal{F}_{bc]}-\mathcal{T}^{\dagger d}{}_{[ab}\mathcal{F}_{c]d}=0.$	$\mathcal{D}_{[a}^{\dagger}\mathcal{F}_{bc]}-\mathcal{T}^{\dagger d}{}_{[ab}\mathcal{F}_{c]d}=0.$	$\mathcal{D}_{[a}^{\dagger}\mathcal{F}_{bc]}-\mathcal{T}^{\dagger d}{}_{[ab}\mathcal{F}_{c]d}=0.$
1510.06699_ EQ0181 (197)	033	1.000 • N +0	$\mathcal{D}_a^{\dagger}\chi=\left(\frac{\phi}{\phi_0}\right)^{1-w}\widehat{\mathcal{D}}_a^{\dagger}\widehat{\chi},$	$\mathcal{D}_a^{\dagger}\chi=\left(\frac{\phi}{\phi_0}\right)^{1-w}\widehat{\mathcal{D}}_a^{\dagger}\widehat{\chi},$	$\mathcal{D}_a^{\dagger}\chi=\left(\frac{\phi}{\phi_0}\right)^{1-w}\widehat{\mathcal{D}}_a^{\dagger}\widehat{\chi},$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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1510.06699_ E00186 (202)	035	1.000	$\left[{\ }^{\{0\}}\mathrm{D}_{\mathrm{c}}^{\dagger},{ }^{\{0\}}\mathrm{D}_{\mathrm{d}}^{\dagger}\right]\varphi=\frac{1}{2}\{{}^0\mathcal{R}^{\dagger ab}_{cd}\Sigma_{ab}\varphi+w\mathcal{H}^{\dagger}_{cd}\varphi,$	$\left[{}^0\mathcal{D}^{\dagger}_c,{}^0\mathcal{D}^{\dagger}_d\right]\varphi=\frac{1}{2}{}^0\mathcal{R}^{\dagger ab}_{cd}\Sigma_{ab}\varphi+w\mathcal{H}^{\dagger}_{cd}\varphi,$	
1510.06699_ E00193 (209)	036	1.000 N +0	$\Delta_{\mu}^{\dagger}\equiv\partial_{\mu}^{\dagger}+\Gamma^{\sigma}_{\rho\mu}\mathbf{X}^{\rho}_{\sigma}+\frac{1}{2}A^{ab}_{\mu}\Sigma_{ab}=\nabla_{\mu}^{\dagger}+D_{\mu}^{\dagger}-\partial_{\mu}^{\dagger},$	$\Delta_{\mu}^{\dagger}\equiv\partial_{\mu}^{\dagger}+\Gamma^{\sigma}_{\rho\mu}\mathbf{X}^{\rho}_{\sigma}+\frac{1}{2}A^{ab}_{\mu}\Sigma_{ab}=\nabla_{\mu}^{\dagger}+D_{\mu}^{\dagger}-\partial_{\mu}^{\dagger},$	
1510.06699_ E00194 (210)	036	1.000 N +0	$\Delta_{\mu}^{\dagger}\mathrm{e}^a_{\nu}\equiv\partial_{\mu}^{\dagger}\mathrm{e}^a_{\nu}-\Gamma^{\sigma}_{\nu\mu}e^a_{\sigma}+A^{\dagger a}_{b\mu}e^b_{\nu}=0,$	$\Delta_{\mu}^{\dagger}e^a_{\nu}\equiv\partial_{\mu}^{\dagger}e^a_{\nu}-\Gamma^{\sigma}_{\nu\mu}e^a_{\sigma}+A^{\dagger a}_{b\mu}e^b_{\nu}=0,$	
1510.06699_ E00195 (211)	036	1.000 K +0	$\nabla_{\sigma}g_{\mu\nu}=2\left(V_{\sigma}+\frac{1}{3}T_{\sigma}\right)g_{\mu\nu}.$	$\nabla_{\sigma}g_{\mu\nu}=2(V_{\sigma}+\frac{1}{3}T_{\sigma})g_{\mu\nu}.$	
1510.06699_ E00197 (215)	037	1.000 K +0	$\Gamma^{\lambda}_{\mu\lambda}=\Gamma^{\lambda}_{\lambda\mu}.$	$\Gamma^{\lambda}_{\mu\lambda}=\Gamma^{\lambda}_{\lambda\mu}.$	
1510.06699_ E00198 (216)	037	1.000 N +0	$\widehat{R}^{\rho}_{\sigma\mu\nu}\equiv R^{\dagger\rho}_{\sigma\mu\nu}-H^{\dagger}_{\mu\nu}\delta^{\rho}_{\sigma}.$	$\widehat{R}^{\rho}_{\sigma\mu\nu}\equiv R^{\dagger\rho}_{\sigma\mu\nu}-H^{\dagger}_{\mu\nu}\delta^{\rho}_{\sigma}.$	
1510.06699_ E00199 (217)	037	1.000 N +0	$\left[\nabla_{\mu}^{\dagger},\nabla_{\nu}^{\dagger}\right]J^{\rho}=\widehat{R}^{\rho}_{\sigma\mu\nu}J^{\sigma}-wH^{\dagger}_{\mu\nu}J^{\rho}-T^{\dagger\sigma}_{\mu\nu}\nabla_{\sigma}^{\dagger}V^{\rho}.$	$[\nabla_{\mu}^{\dagger},\nabla_{\nu}^{\dagger}]J^{\rho}=\widehat{R}^{\rho}_{\sigma\mu\nu}J^{\sigma}-wH^{\dagger}_{\mu\nu}J^{\rho}-T^{\dagger\sigma}_{\mu\nu}\nabla_{\sigma}^{\dagger}V^{\rho}.$	
1510.06699_ E00200 (218)	037	1.000 N +0	$\Gamma^{\lambda}_{\mu\nu}={}^0\Gamma^{\dagger\lambda}_{\mu\nu}+K^{\dagger\lambda}_{\mu\nu},$	$\Gamma^{\lambda}_{\mu\nu}={}^0\Gamma^{\dagger\lambda}_{\mu\nu}+K^{\dagger\lambda}_{\mu\nu},$	
1510.06699_ E00202 (220)	037	1.000 K +0	$K^{\dagger\lambda}_{\mu\nu}=-\frac{1}{2}\left(T^{\dagger\lambda}_{\mu\nu}-T^{\dagger}_{\nu}{}^{\lambda}_{\mu}+T^{\dagger}_{\mu\nu}{}^{\lambda}_{}\right).$	$K^{\dagger\lambda}_{\mu\nu}=-\frac{1}{2}(T^{\dagger\lambda}_{\mu\nu}-T^{\dagger}_{\nu}{}^{\lambda}_{\mu}+T^{\dagger}_{\mu\nu}{}^{\lambda}_{}).$	
1510.06699_ E00206 (228)	038	1.000 K +0	$\left(t_{\mathrm{R}}^{\dagger}\right)^a{}_b+\left(t_{\mathcal{H}^2}^{\dagger}\right)^a{}_b+\left(\tau_{\varphi}^{\dagger}\right)^a{}_b+\left(\tau_{\phi}^{\dagger}\right)^a{}_b+\left(\tau_{\mathcal{R}}^{\dagger}\right)^a{}_b+\left(\tau_{\mathcal{T}^2}^{\dagger}\right)^a{}_b=0,$	$(t_{\mathcal{R}^2}^{\dagger})^a{}_b+(t_{\mathcal{H}^2}^{\dagger})^a{}_b+(\tau_{\varphi}^{\dagger})^a{}_b+(\tau_{\phi}^{\dagger})^a{}_b+(\tau_{\mathcal{R}}^{\dagger})^a{}_b+(\tau_{\mathcal{T}^2}^{\dagger})^a{}_b=0,$	
1510.06699_ E00212 (240)	039	1.000 N +0	$(j_{\mathcal{R}^2})_a+(j_{\mathcal{H}^2})_a+(\zeta_{\varphi})_a+(\zeta_{\phi})_a+(\zeta_{\mathcal{R}})_a+(\zeta_{\mathcal{T}^2})_a=0,$	$(j_{\mathcal{R}^2})_a+(j_{\mathcal{H}^2})_a+(\zeta_{\varphi})_a+(\zeta_{\phi})_a+(\zeta_{\mathcal{R}})_a+(\zeta_{\mathcal{T}^2})_a=0,$	
1510.06699_ E00219 (253)	041	1.000 N +0	$\mathcal{D}'_c\varphi'=e^{(w-1)\rho}\left[\mathcal{D}_c\varphi+w\mathcal{P}_c\varphi+\theta\mathcal{P}^{[b}\delta^{a]}_c\Sigma_{ab}\varphi\right],$	$\mathcal{D}'_c\varphi'=e^{(w-1)\rho}[\mathcal{D}_c\varphi+w\mathcal{P}_c\varphi+\theta\mathcal{P}^{[b}\delta^{a]}_c\Sigma_{ab}\varphi],$	
1510.06699_ E00224 (258)	042	1.000 N +0	$\frac{1}{2}\nu=-6a\theta=3(\theta-1)(2\beta_1+\beta_2+3\beta_3),$	$\frac{1}{2}\nu=-6a\theta=3(\theta-1)(2\beta_1+\beta_2+3\beta_3),$	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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1510.06699_ EQ0225 (259)	042	■ 1.000 ● K +0	$\mathrm{L}_{\mathrm{M}}^{\prime}=\mathrm{L}_{\mathrm{M}}+6a\theta h^{-1}\left(\mathcal{D}_a+\mathcal{T}_a\right)\left(\phi^2\mathcal{P}^a\right).$	$\mathcal{L}_{\mathrm{M}}'=\mathcal{L}_{\mathrm{M}}+6a\theta h^{-1}(\mathcal{D}_a+\mathcal{T}_a)(\phi^2\mathcal{P}^a).$	
1510.06699_ EQ0232 (266)	043	■ 1.000 ●	$L_{\mathrm{G}}=2\alpha\left({}^0\mathcal{R}_{ab}{}^0\mathcal{R}^{ab}-\frac{1}{3}{}^0\mathcal{R}^2\right),$	$L_{\mathrm{G}}=2\alpha({}^0\mathcal{R}_{ab}{}^0\mathcal{R}^{ab}-\tfrac{1}{3}{}^0\mathcal{R}^2),$	
1510.06699_ EQ0236 (270)	044	■ 1.000 ● N +0	$\mathrm{L}_{\mathrm{M}}^{\prime}=\mathrm{L}_{\mathrm{M}}-h^{-1}\left[\frac{1}{2}\nu^0\mathcal{D}_a\left(\phi^2\mathcal{P}^a\right)\right],$	$\mathcal{L}_{\mathrm{M}}'=\mathcal{L}_{\mathrm{M}}-h^{-1}[\tfrac{1}{2}\nu^0\mathcal{D}_a(\phi^2\mathcal{P}^a)],$	
1510.06699_ EQ0237 (A1)	047	■ 1.000 ● C +4	$L_{\mathrm{D}}=\frac{1}{2}i\left[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-\left(\partial_{\mu}\bar{\psi}\right)\gamma^{\mu}\psi\right]-m\bar{\psi}\psi\equiv\Re(i\bar{\psi}\not{\partial}\psi)-m\bar{\psi}\psi,$	$L_{\mathrm{D}}=\tfrac{1}{2}i[\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi-(\partial_{\mu}\bar{\psi})\gamma^{\mu}\psi]-m\bar{\psi}\psi\equiv\Re(i\bar{\psi}\not{\partial}\psi)-m\bar{\psi}\psi,$	
1510.06699_ EQ0245 (A9)	047	■ 1.000 ● K +0	$S=-\int d\lambda\left[p_{\mu}\dot{x}^{\mu}-\frac{1}{2}e\left(p_{\mu}p^{\mu}-m^2\right)\right]$	$S=-\int d\lambda\left[p_{\mu}\dot{x}^{\mu}-\tfrac{1}{2}e(p_{\mu}p^{\mu}-m^2)\right],$	
1510.06699_ EQ0246 (B1)	048	■ 1.000 ● K +0	$L_{\mathrm{G}}=-\kappa^{-1}(\Lambda+a\mathcal{R})+L_{\mathcal{R}^2}+\kappa^{-1}L_{\mathcal{T}^2},$	$L_{\mathrm{G}}=-\kappa^{-1}(\Lambda+a\mathcal{R})+L_{\mathcal{R}^2}+\kappa^{-1}L_{\mathcal{T}^2},$	
1510.06699_ EQ0252 (C1)	049	■ 1.000 ● K +0	$\mathcal{L}_{\mathrm{G}}=-\frac{1}{2\kappa}h^{-1}\mathcal{R},$	$\mathcal{L}_{\mathrm{G}}=-\frac{1}{2\kappa}h^{-1}\mathcal{R},$	
1510.06699_ EQ0254 (C4)	050	■ 1.000 ● N +2	$\mathcal{T}_{ab}^c+\delta_a^c\mathcal{T}_b-\delta_b^c\mathcal{T}_a=2\kappa h\sigma_{ab}^c$	$\mathcal{T}_{ab}^c+\delta_a^c\mathcal{T}_b-\delta_b^c\mathcal{T}_a=2\kappa h\sigma_{ab}^c.$	
1510.06699_ EQ0255 (C5)	050	■ 1.000 ● N +0	$\mathcal{T}_a=-\kappa h\sigma_{ab}^b.$	$\mathcal{T}_a=-\kappa h\sigma_{ab}^b.$	
1510.06699_ EQ0256 (C6)	050	■ 1.000 ● K +0	$\mathcal{T}_{ab}^c=\kappa h\left(2\sigma_{ab}^c+\delta_a^c\sigma_{bd}^d-\delta_b^c\sigma_{ad}^d\right).$	$\mathcal{T}_{ab}^c=\kappa h(2\sigma_{ab}^c+\delta_a^c\sigma_{bd}^d-\delta_b^c\sigma_{ad}^d).$	
1510.06699_ EQ0259 (C9)	050	■ 1.000 ● K +0	$\mathcal{T}_a=-\kappa h\sigma_{ab}^b=0,$	$\mathcal{T}_a=-\kappa h\sigma_{ab}^b=0,$	
1510.06699_ EQ0260 (C10)	050	■ 1.000 ● K +0	$i\gamma^a\mathcal{D}_a\psi-m\psi=0,$	$i\gamma^a\mathcal{D}_a\psi-m\psi=0,$	
1510.06699_ EQ0262 (D1)	051	■ 1.000 ● K +0	$L_{\mathrm{G}}=-\frac{1}{4}\xi\mathcal{H}_{ab}\mathcal{H}^{ab},$	$L_{\mathrm{G}}=-\tfrac{1}{4}\xi\mathcal{H}_{ab}\mathcal{H}^{ab},$	
1510.06699_ EQ0267 (D6)	052	■ 1.000 ● N +0	$\xi\left[\frac{1}{2}\mathcal{T}_{ab}^{*c}\mathcal{H}^{ab}-\left(\mathcal{D}_a^*+\mathcal{T}_a^*\right)\mathcal{H}^{ac}\right]-\nu\phi\mathcal{D}^{*c}\phi=h\left(\zeta_{\varphi}\right)^c$	$\xi[\tfrac{1}{2}\mathcal{T}_{ab}^{*c}\mathcal{H}^{ab}-(\mathcal{D}_a^*+\mathcal{T}_a^*)\mathcal{H}^{ac}]-\nu\phi\mathcal{D}^{*c}\phi=h(\zeta_{\varphi})^c$	
1510.06699_ EQ0268 (D7)	052	■ 1.000 ● N +1	$\frac{\partial L_{\varphi}}{\partial \phi}-\nu\left(\mathcal{D}_a^*+\mathcal{T}_a^*\right)\mathcal{D}^{*a}\phi-4\lambda\phi^3-a\phi\mathcal{R}=0$	$\frac{\partial L_{\varphi}}{\partial \phi}-\nu(\mathcal{D}_a^*+\mathcal{T}_a^*)\mathcal{D}^{*a}\phi-4\lambda\phi^3-a\phi\mathcal{R}=0.$	
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_ E00269 (D8)	052	■ 1.000 ● N +0	$\mathcal{R}^a{}_b - \frac{1}{2} \delta^a_b \mathcal{R} - \frac{1}{2} \delta^a_b \Lambda = \kappa \left[h(\tau_\varphi)^a{}_b _{\phi=\phi_0} + \nu \left(\mathcal{B}^a \mathcal{B}_b - \frac{1}{2} \delta^a_b \mathcal{B}^c \mathcal{B}_c \right) - \xi \left(\mathcal{H}^{ac} \mathcal{H}_{bc} - \frac{1}{4} \delta^a_b \mathcal{H}_{cd} \mathcal{H}^{cd} \right) \right].$	$\mathcal{R}^a{}_b - \frac{1}{2} \delta^a_b \mathcal{R} - \delta^a_b \Lambda = \kappa \left[h(\tau_\varphi)^a{}_b _{\phi=\phi_0} + \nu (\mathcal{B}^a \mathcal{B}_b - \frac{1}{2} \delta^a_b \mathcal{B}^c \mathcal{B}_c) - \xi (\mathcal{H}^{ac} \mathcal{H}_{bc} - \frac{1}{4} \delta^a_b \mathcal{H}_{cd} \mathcal{H}^{cd}) \right].$	
1510.06699_ E00270 (D9)	052	■ 1.000 ●	$\mathcal{T}^c{}_{ab} + \delta^c_a \mathcal{T}_b - \delta^c_b \mathcal{T}_a = 2\kappa h(\sigma_\varphi)_{ab}{}^c _{\phi=\phi_0},$	$\mathcal{T}^c{}_{ab} + \delta^c_a \mathcal{T}_b - \delta^c_b \mathcal{T}_a = 2\kappa h(\sigma_\varphi)_{ab}{}^c _{\phi=\phi_0},$	
1510.06699_ E00271 (D10)	052	■ 1.000 ● N +0	$\xi \left[\frac{1}{2} \mathcal{T}^c{}_{ab} \mathcal{H}^{ab} - (\mathcal{D}_a + \mathcal{T}_a) \mathcal{H}^{ac} \right] + \nu \phi_0^2 \mathcal{B}^c = h(\zeta_\varphi)^c _{\phi=\phi_0},$	$\xi \left[\frac{1}{2} \mathcal{T}^c{}_{ab} \mathcal{H}^{ab} - (\mathcal{D}_a + \mathcal{T}_a) \mathcal{H}^{ac} \right] + \nu \phi_0^2 \mathcal{B}^c = h(\zeta_\varphi)^c _{\phi=\phi_0},$	
1510.06699_ E00272 (D11)	052	■ 1.000 ● N +0	$\mathcal{R} + 4\Lambda = \kappa \left[\phi_0 \partial_\phi L_\varphi _{\phi=\phi_0} + \nu \phi_0^2 (\mathcal{D}_a + \mathcal{T}_a + \mathcal{B}_a) \mathcal{B}^a \right]$	$\mathcal{R} + 4\Lambda = \kappa \left[\phi_0 \partial_\phi L_\varphi _{\phi=\phi_0} + \nu \phi_0^2 (\mathcal{D}_a + \mathcal{T}_a + \mathcal{B}_a) \mathcal{B}^a \right]$	
1510.06699_ E00274 (D13)	053	■ 1.000 ● N +0	$i\gamma^a \mathcal{D}_a \psi - \mu \phi \psi = 0,$	$i\gamma^a \mathcal{D}_a \psi - \mu \phi \psi = 0,$	
1510.06699_ E00276 (E1)	053	■ 1.000 ● K +0	$A^{\dagger ab}{}_\mu = A^{ab}{}_\mu + (\mathcal{V}^a b^b{}_\mu - \mathcal{V}^b b^a{}_\mu),$	$A^{\dagger ab}{}_\mu = A^{ab}{}_\mu + (\mathcal{V}^a b^b{}_\mu - \mathcal{V}^b b^a{}_\mu),$	
1510.06699_ E00277 (E2)	053	■ 1.000 ● N +0	$\mathcal{D}^\natural_a \varphi \equiv h_a{}^\mu D_\mu^\natural \varphi \equiv h_a{}^\mu \left(\partial_\mu + \frac{1}{2} A^{\dagger bc}{}_\mu \Sigma_{bc} \right) \varphi.$	$\mathcal{D}^\natural_a \varphi \equiv h_a{}^\mu D_\mu^\natural \varphi \equiv h_a{}^\mu (\partial_\mu + \frac{1}{2} A^{\dagger bc}{}_\mu \Sigma_{bc}) \varphi.$	
1510.06699_ E00278 (E3)	054	■ 1.000 ● K +0	$[D^\natural_\mu, D^\natural_\nu] \varphi = \frac{1}{2} R^{\dagger ab}{}_{\mu\nu} \Sigma_{ab} \varphi,$	$[D^\natural_\mu, D^\natural_\nu] \varphi = \frac{1}{2} R^{\dagger ab}{}_{\mu\nu} \Sigma_{ab} \varphi,$	
1510.06699_ E00279 (E4)	054	■ 1.000 ● K +0	$V'_\mu = V_\mu + \theta P_\mu,$	$V'_\mu = V_\mu + \theta P_\mu,$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Tables

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image						
1510.06699_TAB_004	055	—	$\begin{tabular}[t]{l l} \hline \mathcal{R}^{\mathbf{a}\mathbf{b}}_{\mathbf{cd}}=\mathcal{R}^{*\mathbf{a}\mathbf{b}}_{\mathbf{cd}}=2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]}+A^a{}_{c[\mu} A^{cb}{}_{\nu]}) \\ \mathcal{R}^{\dagger\mathbf{a}\mathbf{b}}_{\mathbf{cd}}=2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger\mathbf{a}\mathbf{b}}{}_{\nu]}+A^{\dagger a}{}_{c[\mu} A^{\dagger c\mathbf{b}}{}_{\nu]}) \\ \mathcal{T}^a{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]} \\ \mathcal{T}^{*a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]} \\ \mathcal{T}^{\dagger a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\dagger} b^a{}_{\nu]} \\ \mathcal{T}^{\mathfrak{a}}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]} \\ \mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]} \\ \mathcal{H}^{\dagger}{}_{ab}=2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_{\nu}+\tfrac{1}{3}T_{\nu]}) \end{tabular}$	$\mathcal{R}^{\mathbf{a}\mathbf{b}}_{\mathbf{cd}}=\mathcal{R}^{*\mathbf{a}\mathbf{b}}_{\mathbf{cd}}=2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]}+A^a{}_{c[\mu} A^{cb}{}_{\nu]}) \\ \mathcal{R}^{\dagger\mathbf{a}\mathbf{b}}_{\mathbf{cd}}=2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger\mathbf{a}\mathbf{b}}{}_{\nu]}+A^{\dagger a}{}_{c[\mu} A^{\dagger c\mathbf{b}}{}_{\nu]}) \\ \mathcal{T}^a{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]} \\ \mathcal{T}^{*a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]} \\ \mathcal{T}^{\dagger a}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\dagger} b^a{}_{\nu]} \\ \mathcal{T}^{\mathfrak{a}}{}_{bc}=2h_b{}^\mu h_c{}^\nu D_{[\mu}^{\mathfrak{a}} b^a{}_{\nu]} \\ \mathcal{H}_{ab}=2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]} \\ \mathcal{H}^{\dagger}{}_{ab}=2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_{\nu}+\tfrac{1}{3}T_{\nu]})$							
1510.06699_TAB_005	055	—	$\begin{tabular}[t]{l l} \hline (c^{\mathbf{a}}_{\mathbf{bc}})_{\mathbf{}}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]} \\ (c^{*\mathbf{a}}_{\mathbf{bc}})_{\mathbf{}}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]} \\ (c^{\dagger\mathbf{a}}_{\mathbf{bc}})_{\mathbf{}}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger} b^a{}_{\nu]} \\ {}^0A^{ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{ab}{}_c+c^b{}_c{}^a-c_c{}^{ab}) \\ {}^0A^{*ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c^{*ab}) \\ {}^0A^{\dagger ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger ab}) \end{tabular}$	<table><tr><td>$c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$</td></tr><tr><td>$c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$</td></tr><tr><td>$c^{\dagger a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$</td></tr><tr><td>${}^0A^{ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{ab}{}_c+c^b{}_c{}^a-c_c{}^{ab})$</td></tr><tr><td>${}^0A^{*ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c^{*ab})$</td></tr><tr><td>${}^0A^{\dagger ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger ab})$</td></tr></table>	$c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$	$c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$	$c^{\dagger a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$	${}^0A^{ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{ab}{}_c+c^b{}_c{}^a-c_c{}^{ab})$	${}^0A^{*ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c^{*ab})$	${}^0A^{\dagger ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger ab})$	$\begin{array}{l} c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]} \\ c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]} \\ c^{\dagger a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger} b^a{}_{\nu]} \\ {}^0A^{ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{ab}{}_c+c^b{}_c{}^a-c_c{}^{ab}) \\ {}^0A^{*ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c^{*ab}) \\ {}^0A^{\dagger ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger ab}) \end{array}$
$c^a{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a{}_{\nu]}$											
$c^{*a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a{}_{\nu]}$											
$c^{\dagger a}{}_{bc}\equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger} b^a{}_{\nu]}$											
${}^0A^{ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{ab}{}_c+c^b{}_c{}^a-c_c{}^{ab})$											
${}^0A^{*ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{*ab}{}_c+c^{*b}{}_c{}^a-c_c^{*ab})$											
${}^0A^{\dagger ab}{}_{\mu}\equiv \tfrac{1}{2}b^c{}_{\mu} (c^{\dagger ab}{}_c+c^{\dagger b}{}_c{}^a-c_c^{\dagger ab})$											

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_TAB_006	055	—	$\begin{tabular}[t]{l} \hline \\(\\) \\ \hline \\(\mathrm{L}\equiv h^{-1} L)\ \\ \hline \\(t^{\{a\}}{}_{{}_\mu} h_{\{b\}}{}^{\{\mu\}} \equiv h_{\{b\}}{}^{\{\mu\}} \delta \mathrm{L}_{\mathrm{G}} / \delta h_a{}^\mu\}\\ \hline \\(s_{ab}){}_{{}_\mu} s^{\{c\}}{}_{{}_\mu} \equiv s_{ab}{}^\mu s^c{}_{{}_\mu} \equiv s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)\ \\ \hline \\(j^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)) \\ \hline \\(\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_{\mathrm{M}} / \delta h_a{}^\mu) \\ \hline \\(\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_{{}_\mu} \equiv b^c{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta A^{ab}{}_{{}_\mu}) \\ \hline \\(\zeta^a \equiv \zeta^\mu b^a{}_{{}_\mu} \equiv b^a{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta B_\mu) \\ \hline \\(\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)) \\ \hline \\(\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b)\end{tabular}$	L $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_{\mathrm{G}} / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_{{}_\mu} \equiv b^c{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta A^{ab}{}_{{}_\mu}$ $j^a \equiv j^\mu b^a{}_{{}_\mu} \equiv b^a{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_{\mathrm{M}} / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_{{}_\mu} \equiv b^c{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta A^{ab}{}_{{}_\mu}$ $\zeta^a \equiv \zeta^\mu b^a{}_{{}_\mu} \equiv b^a{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	L $\mathcal{L} \equiv h^{-1} L$ $t^a{}_b \equiv t^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_{\mathrm{G}} / \delta h_a{}^\mu$ $s_{ab}{}^c \equiv s_{ab}{}^\mu b^c{}_{{}_\mu} \equiv b^c{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta A^{ab}{}_{{}_\mu}$ $j^a \equiv j^\mu b^a{}_{{}_\mu} \equiv b^a{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{G}} / \delta B_\mu$ $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$ $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ $\tau^a{}_b \equiv \tau^a{}_\mu h_b{}^\mu \equiv h_b{}^\mu \delta \mathcal{L}_{\mathrm{M}} / \delta h_a{}^\mu$ $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^\mu b^c{}_{{}_\mu} \equiv b^c{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta A^{ab}{}_{{}_\mu}$ $\zeta^a \equiv \zeta^\mu b^a{}_{{}_\mu} \equiv b^a{}_{{}_\mu} \delta \mathcal{L}_{\mathrm{M}} / \delta B_\mu$ $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$ $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$
Conf. MathPix confidence: ≥ 0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1510.06699_TAB_007	000	—	\begin{tabular}{ll} {\em Gravitational gauge field variables} & \\\ \$h_a^{\mu}\$ & translational gauge field (\$h\$-field) \\ \\\ \$h\$ & determinant of \$h_a^\mu\$ \\ \\\ \$b^a_\mu\$ & inverse \$h\$-field \\ \\\ \$A^{ab}_\mu\$ & rotational gauge field (\$A\$-field) \\ \\\ \$B_\mu\$ & WGT dilational gauge field (\$B\$-field) \\ \\\ \$V_\mu\$ & eWGT dilational gauge field (\$V\$-field) \\ \\\ \$\mathcal{A}^{\dagger ab}_\mu\$ & \$\mu \equiv A^{ab}_\mu + 2\eta^{ca} b^b_\mu\$ \\ \\\ \$h_c^\nu V_\nu\$ & eWGT extended rotational gauge field (\$\mathcal{A}^{\dagger\dagger}\$-field) \\ \\\ & \\ {\em Derivative operators} & \\ \\\ \$\mathcal{D}_a \equiv h_a^\mu D_\mu \equiv h_a^\mu (\partial_\mu + \frac{1}{2}\{A^{ab}\}_\mu \Sigma_{ab})\$ & PGT (generalised) covariant derivative\footnotemark[1] \\ \\\ \$\partial^{\ast}_\mu \equiv \partial_\mu + w B_\mu\$ & WGT `augmented' partial derivative \\ \\\ \$\mathcal{D}^{\ast}_a \equiv h_a^\mu D^{\ast}_\mu \equiv h_a^\mu (\partial^{\ast}_\mu + \frac{1}{2}\{A^{ab}\}_\mu \Sigma_{ab})\$ & WGT (generalised) covariant derivative\footnotemark[1] \\ \\\ \$\partial^{\dagger}_\mu \equiv \partial_\mu - w(V_\mu + \frac{1}{3}T_\mu)\$ & eWGT `augmented' partial derivative \\ \\\ \$\mathcal{D}^{\dagger}_a \equiv h_a^\mu D^{\dagger}_\mu \equiv h_a^\mu (\partial^{\dagger}_\mu + \frac{1}{2}\{A^{\dagger\dagger}_{ab}\}_\mu \Sigma_{ab})\$ & eWGT (generalised) covariant derivative\footnotemark[1] \\ \\\ \$\mathcal{D}^{\natural}_a \equiv h_a^\mu D^{\natural}_\mu \equiv h_a^\mu (\partial_\mu + \frac{1}{2}\{A^{\dagger\dagger}_{ab}\}_\mu \Sigma_{ab})\$ & eWGT (generalised) semi-covariant derivative\footnotemark[1] \\ \\\ & \\ {\em Gauge field strengths} & \\ \\\ \$\mathcal{R}^{ab}_{cd} = \mathcal{R}^{\dagger\dagger}_{ab}_{cd} = 2h_a^\mu h_b^\nu (\partial_\mu \{A^{cb}\}_\nu - \{A^{cb}\}_\mu \partial_\nu)\$ & PGT and WGT rotational gauge field strength\footnotemark[1], \footnotemark[2] \\ \\\ \$\mathcal{R}^{\dagger\dagger}_{ab}_{cd} = 2h_a^\mu h_b^\nu (\partial_\mu \{A^{\dagger\dagger}_{ab}\}_\nu - \{A^{\dagger\dagger}_{ab}\}_\mu \partial_\nu + \{A^{\dagger\dagger}_{ab}\}_c \{A^{\dagger\dagger}_{cb}\}_\nu - \{A^{\dagger\dagger}_{cb}\}_c \{A^{\dagger\dagger}_{ab}\}_\nu)\$ & eWGT rotational gauge field strength\footnotemark[1], \footnotemark[2] \\ \\\ \$\mathcal{T}^a_{bc} = 2h_b^\mu h_c^\nu D_\mu \{b^a_\nu\}\$ & PGT translational gauge field strength\footnotemark[2] \\ \\\ \$\{\mathcal{T}^{\ast}_a\}_{bc} = 2h_b^\mu h_c^\nu D^{\ast}_\mu \{b^a_\nu\}\$ & WGT translational gauge field strength\footnotemark[2] \\ \\\ \$\{\mathcal{T}^{\dagger}_a\}_{bc} = 2h_b^\mu h_c^\nu D^{\dagger}_\mu \{b^a_\nu\}\$ & eWGT translational gauge field strength\footnotemark[2] \\ \\\ \$\{\mathcal{T}^{\natural}_a\}_{bc} = 2h_b^\mu h_c^\nu D^{\natural}_\mu \{b^a_\nu\}\$ & eWGT translational gauge `semi' field strength\footnotemark[1], \footnotemark[2] \\ \\\ \$\mathcal{H}_{ab} = 2h_a^\mu h_b^\nu \partial_\mu \{b^a_\nu\}\$ & WGT dilational gauge field strength \\ \\\ \$\mathcal{H}^{\dagger}_{ab} = 2h_a^\mu h_b^\nu \partial_\mu (V_\nu + \frac{1}{3}T_\nu)\$ & eWGT dilational gauge field strength \\ \\\ & \\ {\em Reduced \$A\$-fields and related quantities} & \\ \\\ \$c^a_{bc} \equiv 2h_b^\mu h_c^\nu \partial_\mu \{b^a_\nu\}\$ & PGT Ricci rotation coefficients \\ \\\ \$c^{\ast}_{abc} \equiv 2h_b^\mu h_c^\nu \partial_\mu \{b^a_\nu\}\$ & WGT Ricci rotation coefficients \\ \\\ \$c^{\dagger}_{abc} \equiv 2h_b^\mu h_c^\nu \partial_\mu \{b^a_\nu\}\$ & eWGT Ricci rotation coefficients \\ \\\ \$\{^0A^{ab}\}_\mu \equiv \frac{1}{2}\{b^c_\mu (c^{ab}_c + \{c^b_c\}^a - \{c_c\}^{ab})\$ & PGT reduced \$A\$-field (when \$\{\mathcal{T}^a\}_{bc} \equiv 0\$) \\ \\\ \$\mu \equiv \frac{1}{2}\{b^c_\mu (c^{\ast ab}_c + \{c^{\ast b}_c\}^a - \{c^{\ast c}\}^{ab})\$ & WGT reduced \$A\$-field (when \$\{\mathcal{T}^{\ast}_a\}_{bc} \equiv 0\$) \\ \\\ \$\{^0A^{\dagger ab}\}_\mu \equiv \frac{1}{2}\{b^c_\mu\$	(not rendered)	—

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Image regions — rendered against scan

Two comparable cells per row. **Rendered:** the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan:** the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
1510.06699_PIC_0001	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$	$h_a{}^\mu$ h $b^a{}_\mu$ $A^{ab}{}_\mu$ B_μ V_μ $A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$
1510.06699_PIC_0002	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	translational gauge field (<i>h</i> -field) determinant of $h_a{}^\mu$ inverse <i>h</i> -field rotational gauge field (<i>A</i> -field) WGT dilational gauge field (<i>B</i> -field) eWGT dilational gauge field (<i>V</i> -field) eWGT extended rotational gauge field ($A^\dagger$ -field)	translational gauge field (<i>h</i> -field) determinant of $h_a{}^\mu$ inverse <i>h</i> -field rotational gauge field (<i>A</i> -field) WGT dilational gauge field (<i>B</i> -field) eWGT dilational gauge field (<i>V</i> -field) eWGT extended rotational gauge field ($A^\dagger$ -field)
1510.06699_PIC_0003	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$h_a{}^\mu$ translational gauge field (<i>h</i> -field)	$h_a{}^\mu$ translational gauge field (<i>h</i> -field)
1510.06699_PIC_0004	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	h determinant of $h_a{}^\mu$	h determinant of $h_a{}^\mu$
1510.06699_PIC_0005	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$b^a{}_\mu$ inverse <i>h</i> -field	$b^a{}_\mu$ inverse <i>h</i> -field
1510.06699_PIC_0006	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$A^{ab}{}_\mu$ rotational gauge field (<i>A</i> -field)	$A^{ab}{}_\mu$ rotational gauge field (<i>A</i> -field)
1510.06699_PIC_0007	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	B_μ WGT dilational gauge field (<i>B</i> -field)	B_μ WGT dilational gauge field (<i>B</i> -field)
1510.06699_PIC_0008	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	V_μ eWGT dilational gauge field (<i>V</i> -field)	V_μ eWGT dilational gauge field (<i>V</i> -field)
1510.06699_PIC_0009	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ eWGT extended rotational gauge field ($A^\dagger$ -field)	$A^{\dagger ab}{}_\mu \equiv A^{ab}{}_\mu + 2\eta^{c[a}b^{b]}{}_\mu h_c{}^\nu V_\nu$ eWGT extended rotational gauge field ($A^\dagger$ -field)
1510.06699_PIC_0010	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_a^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w (V_\mu + \tfrac{1}{3} T_\mu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\natural \equiv h_a{}^\mu D_\mu^\natural \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$	$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^* \equiv \partial_\mu + w B_\mu$ $\mathcal{D}_a^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$ $\partial_\mu^\dagger \equiv \partial_\mu - w (V_\mu + \tfrac{1}{3} T_\mu)$ $\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$ $\mathcal{D}_a^\natural \equiv h_a{}^\mu D_\mu^\natural \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$
1510.06699_PIC_0011	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$	$\mathcal{D}_a \equiv h_a{}^\mu D_\mu \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$
1510.06699_PIC_0012	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$\partial_\mu^* \equiv \partial_\mu + w B_\mu$	$\partial_\mu^* \equiv \partial_\mu + w B_\mu$
1510.06699_PIC_0013	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$\mathcal{D}_a^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$	$\mathcal{D}_a^* \equiv h_a{}^\mu D_\mu^* \equiv h_a{}^\mu (\partial_\mu^* + \tfrac{1}{2} A^{ab}{}_\mu \Sigma_{ab})$
1510.06699_PIC_0014	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$\partial_\mu^\dagger \equiv \partial_\mu - w (V_\mu + \tfrac{1}{3} T_\mu)$	$\partial_\mu^\dagger \equiv \partial_\mu - w (V_\mu + \tfrac{1}{3} T_\mu)$
1510.06699_PIC_0015	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$	$\mathcal{D}_a^\dagger \equiv h_a{}^\mu D_\mu^\dagger \equiv h_a{}^\mu (\partial_\mu^\dagger + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$
1510.06699_PIC_0016	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$\mathcal{D}_a^\natural \equiv h_a{}^\mu D_\mu^\natural \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$	$\mathcal{D}_a^\natural \equiv h_a{}^\mu D_\mu^\natural \equiv h_a{}^\mu (\partial_\mu + \tfrac{1}{2} A^{\dagger ab}{}_\mu \Sigma_{ab})$
1510.06699_PIC_0017	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$, $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^\natural a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\natural b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_\nu)$	$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$, $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^\natural a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\natural b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_\nu)$
1510.06699_PIC_0018	055	— (dup)	tex.zip holds no images	(no tex.zip crop for this row)	$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^\natural a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\natural b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_\nu)$	$\mathcal{R}^{ab}{}_{cd} = \mathcal{R}^{*ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}{}_{\nu]} + A^a{}_{c[\mu} A^{cb}{}_{\nu]})$ $\mathcal{R}^{\dagger ab}{}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}{}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}{}_{\nu]})$ $\mathcal{T}^a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a{}_{\nu]}$ $\mathcal{T}^{*a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^* b^a{}_{\nu]}$ $\mathcal{T}^{\dagger a}{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\dagger b^a{}_{\nu]}$ $\mathcal{T}^\natural a{}_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu}^\natural b^a{}_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}_{ab}^\dagger = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \tfrac{1}{3} T_\nu)$

